

West Visayas State University
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
La Paz, Iloilo City, Philippines

ZenLens: Image-Based Stress Detection and Tailored
Recommendations for Enhanced Student Stress Management

An Undergraduate Thesis

Presented to the Faculty of the
College of Information and Communications Technology
West Visayas State University
La Paz, Iloilo City

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Computer Science

by

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Approval Sheet

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Abstract

College students face academic stress on a regular basis. Stressors such as severe workloads, course discomfort, low motivation, and even high family expectations cause mild stress among students. This study reinforces the general idea of using facial expressions to reveal stress levels by analyzing multiple facial expressions among students and implementing Computer Vision Facial Expression Recognition using RetinaFace and the Deep Learning Model ZFNet to categorize facial expressions to stress emotions. The ZFNET model achieved 87.14% accuracy in emotion recognition, which is consistent with the Depression Anxiety Stress Scale-21 evaluation measures. Meanwhile, the software Quality Standard Evaluation achieved 3.99, which corresponds to a descriptive category of "Good" standard. The research data shows that Zenlens uses deep learning algorithms and ZFNet for emotion classification, as well as RetinaFace for facial landmark detection, to measure students' stress levels through facial

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expression analysis. The advantage of Zenlens is that it examines various facial expressions over time rather than using one-time facial assessments, which is useful for monitoring the stress activities of each class period. It is advisable to consider changing the environmental conditions to better examine the external elements influencing classroom stress.

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CHAPTER 1 INTRODUCTION TO THE STUDY

Background of the Study and Conceptual Framework

Stress is one's condition of worry or mental tension caused by a difficult situation; it is a natural response that makes humans address challenges and threats [1]. Everyone experiences stress to some extent, and how a person responds to it makes a big difference to their overall well-being [2]. Thus, it is important to recognize stress symptoms early to help people figure out healthy coping methods [3].

Among college students, academic stress is one of the top factors that affect students' stress levels [4]-[7]; self-report assessment tools measured these as the basis for determining the causes. Academic overloads, course awkwardness, inadequate time to study, workload every semester, exam awkwardness, low motivation, and high family expectations drive moderate stress among students. However, the significant source of stress among undergraduate students is the fear of failure. Stress became prevalent among college students; various issues appear when stressful experiences are higher than coping capacity. Furthermore, academic stress among students with different levels of study and specializations is the same. With academic-related workloads

as major stressors among students, the importance of the role of teachers may significantly impact students' perceptions and emotions as they are the managers of the academic program of their subjects [8]. Having a piece of equipment that can help instructors gauge or assess their students' mental stress is vital in promoting a healthy and conducive learning environment.

Concerningly, research conducted by Lerner et al. [9] reveals that facial expressions of emotions are linked with cortisol levels, cardiovascular reactivity, and facial muscle activity, which are physiological stress responses. Also, a multimodal dataset by Hosseini et al. [10] reveals that facial expressions of emotions such as anger, sadness, and surprise are highly correlated with stress. This reinforces the general idea that a person's facial expressions reveal his or her stress levels.

Computer vision is a field in artificial intelligence (AI) that enables computers to identify and understand objects and people in videos and images by combining techniques that process raw inputs into usable data and insights [11]. Specifically, facial expression recognition (FER), under computer vision, is relevant in identifying and categorizing

emotions depicted on a human face, thus detecting the presence of stress in a person.

Studies have utilized deep learning to predict mental stress by analyzing a human's facial expressions. However, these studies have one thing in common: their capabilities are limited to only assessing one facial expression in a single image or video frame. This situation creates an opportunity to create a model that can assess mental stress levels by analyzing multiple facial expressions in images and applying it in a classroom setup, which can benefit educational institutions. As such, the researchers proposed a study of analyzing multiple facial expressions to assess stress levels among students by implementing deep learning models and techniques.

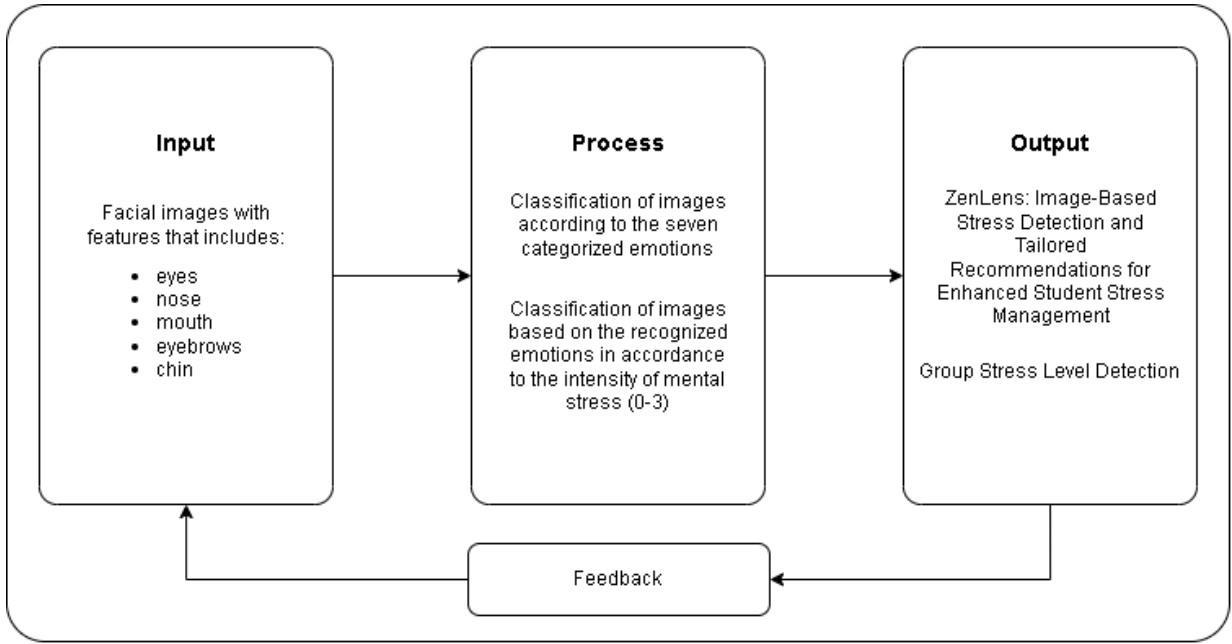


Figure 1. Conceptual Framework

Figure 1 shows the conceptual framework of the proposed application. The application needs the face, so the primary input will be facial images containing features like eyes, nose, mouth, eyebrows, and chin.

These facial images will be processed using the ZFNet model and classified into seven categories of emotions: anger, disgust, fear, happiness, sadness, surprise, and neutral. Classified images will be categorized again between two categories: "stressed", "neutral", and "non-stressed".

Afterwards, the frequency of "stressed" images will be measured, which will output a stress level assessment and a

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graphical representation of the frequency of emotions in the classified images. A corresponding stress management recommendation will follow.

Objectives of the Study

The study aimed to develop ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management, a system for monitoring students' mental stress.

Specifically, this study aims to:

1. develop and train a ZFNet model to estimate students' stress levels by analyzing their emotions through facial features;
2. evaluate the system's performance in determining the stress level using accuracy, precision, recall, and a confusion matrix;
3. integrate the model into a web application for monitoring the stress level of students; and
4. evaluate the usability and effectiveness of the system using ISO 25010 Software Quality Standard.

Significance of the Study

The outcomes of the proposed study on ZenLens aimed to provide stress detection and recommendations. The findings may will offer benefits to the following:

Teachers. ZenLens enables teachers to detect students' stress through image-based facial expression analysis. This helps educators offer personalized support and adapt their teaching methods to create a more conducive learning environment. Early detection of stress can prevent negative impacts on academic performance and promote a healthier atmosphere in classrooms.

Counselors. By using ZenLens, counselors can receive data on students' stress levels, allowing them to develop tailored stress management strategies. This enhances their ability to provide focused, effective support for students, improving mental well-being.

Facilitators. ZenLens aids facilitators in identifying student stress through facial expression recognition, allowing for timely and accurate interventions. This helps foster a more supportive environment, addressing emotional and mental needs proactively.

Students. ZenLens offers students personalized recommendations for managing stress, leading to improved

mental health, academic performance, and emotional resilience. With tailored coping strategies, students can better navigate stress and enhance their overall well-being.

Parents. Parents benefit from ZenLens by being informed about their child's stress levels. This information enables them to offer appropriate support at home, enhancing family communication and contributing to a positive, stress-free environment.

Schools and Institutions. They can make use of ZenLens as it contributes to creating a healthier and more supportive environment within schools and institutions by addressing student stress proactively, promoting mental wellness, and supporting the implementation of proactive mental health strategies for students, staff, and faculty.

Future Researchers. The results and analysis of the study may provide valuable insights for future researchers to advance stress detection and management technologies, benefiting the field of mental health research and innovation.

Delimitation of the Study

This study focused on analyzing multiple facial expressions, including happiness, surprise, neutral, sadness, anger, fear, and disgust, to measure stress levels among college students at West Visayas State University. It specifically examined the correlation between these facial expressions and stress levels, excluding other physiological or psychological indicators of stress, such as heart rate variability or self-reported stress levels.

Images will serve as the data for stress assessment and are taken at 90-second intervals during specified observation periods. This study was limited to frame-by-frame image processing for stress assessment and does not incorporate real-time or video-based data collection methods.

The recommendations for stress management provided by this study are tailored specifically for the demographic of college students based on the results of the stress level assessments. The system developed does not function as a chatbot or provide in-depth, interactive support for stress management. The specific stress levels, severe and extremely severe, are not included in the tailored recommendations, as it is advised to consult directly with the domain expert or

licensed professionals. They are the appropriate people to give proper recommendations.

Image capture is performed using a camera setup that can only detect front-facing student faces. It is acknowledged that this setup may not capture all students' faces in a single snapshot, potentially limiting the comprehensiveness of the data collection.

The accuracy of emotion detection is further assessed through the DASS stress scale and manual annotation by a domain expert, such as a psychiatrist, to ensure the reliability of the automated system's assessments.

The study focuses on the development of a deep learning model and a ZFNet algorithm, both implemented into a web application. This system is designed to function as a unified tool for assessing stress levels through facial expression analysis.

The model is adept at multi-face detection using RetinaFace, limited to identifying five facial landmarks (the eyes, nose, and mouth corners). This limitation is by design, as these landmarks are critical for recognizing the targeted facial expressions associated with stress levels.

Definition of Terms

For better understanding, the following terms were defined conceptually and operationally:

System. A system is a cohesive arrangement of interrelated components working together, united to achieve a specific set of tasks or solve a problem [12].

System. In this study, it referred to ZenLens Web Application.

Model. Visual, mathematical, or simply a logical representation of a system, entity, phenomenon, or process. A model is a system to follow or imitate a representation of a person or a proposed structure, typically on a smaller scale than the original [13].

Model. In this study, it refers to the ZFNet model to be used in the system.

Algorithm. An algorithm is a method used for solving a problem or accomplishing a task that follows several steps or a sequence of instructions [14].

Algorithm. In this study, takes ZFNet as its approach or method that guarantees to solve problems or to perform a function in a particular system.

Facial Landmarks. Facial landmarks refer to the specific points or features of the human face determined to identify and analyze facial expressions, shapes, and structures. These particular patterns of minor muscle movements in the face convey a person's distinct emotional state [15].

Facial Landmarks. In this study, facial landmarks were defined as reference points used to track and measure facial movements as a basis for evaluating changes in expression.

Stress. According to the World Health Organization, Stress can be defined as a natural human response that prompts people to address challenges and threats. Additionally, stress is a feeling that affects both the mind and the body. The physiological stress reaction is caused by how a person's traits and the environment interact [16].

Stress. In this study, stress referred to the response state described numerically by the average amount of negative Stress using the developed application.

Emotion. Emotions are a complex process that includes automatic evaluation, which is impacted by our evolutionary and personal history. They are triggered when we see something significant that affects our well-being, leading to

psychological changes and emotional responses to address the problem [17].

Emotion. In this study, emotion is detected via a deep learning algorithm and classified as angry, disgust, fear, happy, neutral, sad, and surprise, then it would be labeled as two types: positive and negative emotions. These types are labeled to identify stress-related emotions [18].

Facial Emotion Recognition. Facial Emotion Recognition (FER) is the technology that automates recognizing facial expressions from the faces. This measurement requires both emotional images and videos to show that emotional state, which is enhanced by isolating the face from other people, detecting that the face exists, and establishing features like eyes, nose, and mouth. The machine gets going by categorizing the emotions here [19].

Facial Emotion Recognition. In this study, FER measures emotional state via pictures focused on healthcare to explore non-invasive stress assessment to identify users in need of health assistance.

CHAPTER 2 REVIEW OF RELATED LITERATURE

Review of Existing and Related Studies

Emotion, Stress, and Rumination

Previous studies found that the release of stress hormones like adrenaline and serotonin in the brain is triggered by negative emotions such as anger, frustration, and fear. These biochemical reactions influence memory and emotional regulation. Without proper management, chronic stress stemming from these negative emotions can profoundly impact overall health [20].

Emotion

Emotions are reactions that human beings experience in response to events or situations. The emotion humans feel depends entirely on the circumstance, with good news bringing joy and threats inciting fear. These emotions significantly impact daily lives, influencing decisions, activity choices, and stress levels [21].

Identifying stress through facial expressions involves understanding the nuanced movements of facial muscles and their relation to emotions. This section delves into how

specific emotions, evidenced through particular facial regions' movements, serve as indicators of stress levels.

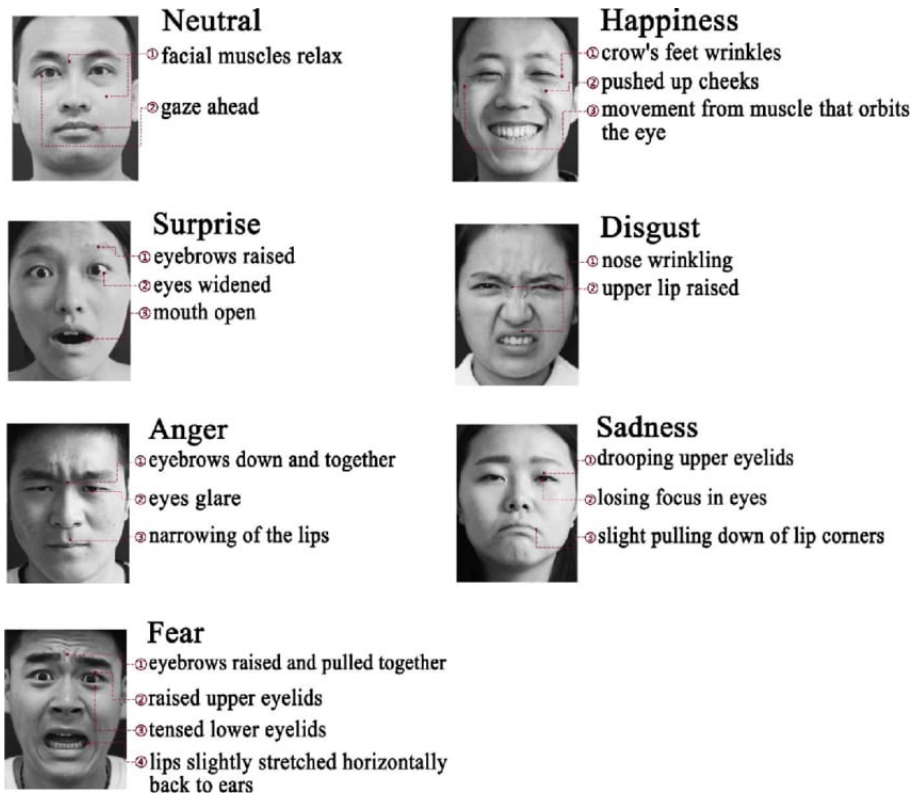


Figure 2. Seven Universal Emotions [22]

Figure 2 depicts the seven universal emotions—Neutral, Happiness, Surprise, Disgust, Anger, Sadness, and Fear—along with their distinct facial expressions. These cues, such as relaxed muscles for Happiness or narrowed lips for Anger, are crucial for accurate emotion detection. As noted by Ma et al. [22], cultural and ethnic diversity, as seen in their facial-expression database for Chinese, Han, Hui, and Tibetan

populations, can influence emotion expression and interpretation. This highlights the need for culturally inclusive approaches in emotion recognition research, ensuring more accurate and personalized systems for improving emotional well-being and communication.

Table 1. Anatomy, Head and Neck: Facial Muscles [23-28]

Emotion	Muscle Involved
Happiness	Zygomatic major (raises the corners of the mouth into a smile)
Surprise	Frontalis (raises eyebrows), Orbicularis oculi (widens eyes), Masseter (drops jaw)
Neutral	Relaxed face, no movement
Sadness	Corrugator supercilii (draws eyebrows together), Depressor anguli oris (pulls corners of the mouth down)
Fear	Frontalis (raises eyebrows), Platysma (tenses neck), Orbicularis oculi (widens eyes)
Anger	Corrugator supercilii (draws eyebrows together), Levator labii superioris (raises upper lip) Orbicularis oris (tightens lips)
Disgust	Levator labii superioris (raises upper lip), Nasalis (wrinkles nose), Orbicularis oris (tightens lips)

Negative Emotions

Anger

Anger is an intense emotion that occurs when something goes wrong or when someone is wronged. Feelings of stress, frustration, and irritation typically characterize it. It is a perfectly normal response to frustrating or difficult situations. It only becomes a problem when it is excessively displayed and affects daily functioning and relationships with others [29].

Anger is often expressed through distinctive facial movements involving the corrugator supercilii, the orbicularis oculi, and the levator labii superioris [30]. Moore and Zimmermann [31] highlight that these universal expressions of anger are recognized cross-culturally, serving as important indicators of stress.

Fear

Fear arises with the threat of harm, either physical, emotional, or psychological, real or imagined. While traditionally considered a "negative" emotion, fear is vital in keeping humans safe as it mobilizes them to cope with potential danger [32].

Fear, characterized by widened eyes and tensed facial muscles, signals heightened alertness to threats and is a potent indicator of stress. This emotion primarily involves the levator palpebrae superioris and the corrugator supercilii [33]. LaBar et al.'s [34] findings on amygdala activation during fear conditioning underscores the primal connection between fear and stress.

Sadness

Sadness is one of the seven universal emotions experienced by everyone worldwide resulting from the loss of someone or something important. What causes us sadness varies considerably based on personal and cultural notions of loss. While sadness is often considered a "negative" emotion, it is essential in signaling a need for help or comfort [35].

Sadness, often linked with stress, is characterized by downturned lips and slightly lowered eyebrows, involving the corrugator supercilii, mentalis, and depressor anguli oris muscles [36]. Research by Tomarken and Davidson [37] shows that it correlates with increased right frontal brain activity, indicating its role in stress and emotional challenges.

Disgust

Disgust is one of the seven universal emotions and arises as an aversion towards something offensive. Humans can feel disgusted by something they perceive with their physical senses (sight, smell, touch, sound, taste), by the actions or appearances of people, and even by ideas [38].

Disgust, characterized by a distinct facial expression involving the wrinkling of the nose, elevation of the upper lip, and sometimes a slight opening of the mouth. This emotion primarily activates the levator labii superioris, and the corrugator supercilii. These facial movements are often a reaction to repulsive stimuli but can also be indicative of stress [39].

Neutral Emotion

Neutral is applying balanced emotions and inner awareness. Having a neutral state does not mean having no emotions [40].

Neutral expressions are typically identified by a relaxed facial muscle state without the pronounced movements seen in expressions of joy, sadness, anger, or fear [41]. Noelle-

Neumann's spiral of silence theory provides a socio-psychological perspective on why individuals might hide stress behind a neutral facade, emphasizing the importance of sensitivity to minor cues in stress detection [42].

Non-negative emotions

Happiness

Happiness is an emotional state characterized by joy, satisfaction, contentment, and fulfillment. While happiness has many definitions, it often involves positive emotions and life satisfaction [43].

Happiness, often associated with well-being, involves the zygomaticus major and orbicularis oculi muscles [44]. Frank and Ekman's work further differentiates between genuine (Duchenne) smiles, involving both muscles, and forced smiles, adding depth to understanding happiness's impact on stress interpretation [45].

Surprise

Surprise arises when humans encounter sudden and unexpected sounds or movements. As the briefest of universal

emotions, it focuses on determining what is happening and whether it is dangerous [46].

Surprise is marked by raised eyebrows and wide-open eyes, which can indicate sudden stress. This expression involves the frontalis muscle [47]. Olofsson et al. [48] review on affective picture processing elucidates the neural mechanisms behind processing surprise, offering insights into its potential as a stress indicator.

Stress

Stress is the result of the interaction between individuals and their environment when faced with challenges that they perceive as personally significant and beyond their ability to cope [49]. The Stress and Coping theory [50] offer a framework for studying how stress affects physical and mental health. It highlights that emotions like anger, shame, and anxiety typically emerge in response to stressful situations marked by harm, threat, or difficulty. This demonstrates the link between stress and emotions.

Academic Stress

Academic Stress is a type of stress where students' experiences in response to the demands of academic work. Students have always complained about feeling stressed in school more so while studying for exams, trying to succeed in a competitive system, and coping with large amounts of work within a fixed time span. Many students exhibit stress during college transition due to changes in education and sociocultural environment strategies. This stress can be even more an issue for international students depending on other stressors for instance, moving into a new culture and new language apart from the class work stresses [51]. When stressors accumulate in an individual's life, the individual is likely to reach the extent of their ability to cope or handle the stress and adapt and may neglect his or her physical and psychological reservoirs. This leads to increased probabilities of physical disorders or psychological problems [51].

Academic Stress in Classroom Environment

Many college students report dealing with varying stress levels throughout college for several reasons. Stress affects everyone differently and for different reasons.

There are several sources of academic stress among college students, and among the primary ones are workload, time management difficulties, pressure, and financial issues. The amount and type of studies and tasks create a rather significant level of pressure. The daily activities that college students have to go through include several assignments, constant exams, and complicated studying, which can all be stressful [52]. Studying load piles and pressures accompanying academic education cause learners to experience inadequacy and anxiety.

Moreover, time management is one of the most significant sources of stress affecting students in terms of their academics. It is common for students to have issues with time management; thus, body and mind might need to be fresher for a lecture or examination. This will likely lead to failure in promptly completing tasks, combined with increased load and stress levels. Britton and Tesser [53] reasoned that time management is critical towards achieving academic success and diminishing stress; however, it continues to be a significant area of difficulty for learners in college.

Also, pressure comes with the need to perform well academically to obtain good grades and occupy future

professions. This pressure can be either internal or external in that it could originate from within the individual due to personal goals or be influenced by factors such as parents and society, among others. In an article, Parker, Jimmieson, and Amiot [54] explain how families and students associate academic excellence with much pressure when students set high standards for themselves for their academic work.

Rumination and negative emotion

Rumination can be defined as the process of dwelling on distressing experiences, searching for meaning in negative events, and thinking about potential ways to change the situation related to those negative events [55]. Rumination has been characterized as having both state and trait components in previous research [56, 57]. In everyday life, people can face different types of stressful events. Previous studies have found that psychological stress responses often involve negative emotions. In simpler terms, individuals tend to experience a multitude of negative emotions, such as depression, anxiety, anger, and distress, during real-life stressful situations [50],[58],[59]. Furthermore, Muris et al. [60] discovered a positive association between rumination and both anxiety and depression. This finding suggests that

rumination may serve as a potential mechanism in the development of negative emotions.

Rumination and stress

Nolen-Hoeksema et al. [61] provided evidence for this relationship. Numerous researchers have suggested that there is a concurrent association between rumination in response to stress and depressive symptoms. In an experimental study by Smith and Alloy [62] observed that individuals' level of rumination was responsive to their perceived levels of stress.

Furthermore, the research found that people's transient ruminative thoughts were also positively correlated with their experiences of negative emotions in the following period. Primarily, it showed that rumination helped to significantly mediate the tie between stress and negative feelings in the dynamic two-point process.

Current Systems

In the measurement of human stress, there are two general ways to estimate the stress level of a human being: (1) self-report measures and (2) biomarkers [63]. Self-report measures consist of questions about the respondents' experience and are

often needed in clinical psychology to identify signs of disturbances. The standard format of most self-report questionnaires is Likert-style responses to the items present in terms of subjective experience [64].

Several questionnaires being utilized in psychological research concerning Filipino adolescents are Depression, Anxiety, Stress Scale (DASS) [65-67] and Perceived Stress Scale (PSS) [68-70]. Meanwhile, as stressors elicit various biological responses, biomarkers are used to measure these physiological reactions from the hypothalamic-pituitary-adrenal (HPA) axis, the autonomic nervous system (ANS), and the immune system [63]. Some biochemical tests to measure stress levels are sweat analysis (e.g., chloride), salivary analysis (e.g., cortisol), blood cortisol, blood prolactin, and plasma metanephrines [71].

In a study titled "Frequency of depression, anxiety and stress among university students" [72], the researchers used DASS-21 (Depression, Anxiety Stress Scale) to measure the level of depression, anxiety, and stress among university students. The study found that 75% of university students experience depression, anxiety, and stress, with prevalence ranging from normal to extremely severe.

DASS-21 is a 21-item set of three self-report scales measuring emotional states of depression, anxiety, and stress. It is based on a dimensional conception of psychological disorder, allowing for a more accurate assessment of patients' emotional states [73]. The DASS 21 scale assigns items D (Depression), A (Anxiety), and S (Stress) and sums scores. The final score is multiplied by two, allowing comparisons and percentile rankings. The score is then transferred to the DASS profile sheet. Making it possible to compare the scales and provide severity labels and percentile rankings [74]. This instrument utilizes the screening severity of symptoms of depression, anxiety, and stress together for monitoring the actual changes in the severity of different categories of symptoms across time among Filipino adolescents. The measure can be helpful to psychiatrists, mental health clinicians, and counsellors in the early screening of students with potential mental health concerns. The evidence of DASS-21 reliability and longitudinal invariance to non-native English speakers shows adequate reliability across several categories, which means that the instrument can aid in accurately assigning individuals to groups based on the severity of their symptoms. Additionally, this tool is essential for university students

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in resource-constrained settings where mental health is primarily ignored relative to the general population [75].

DASS₂₁		<i>Name:</i>	<i>Date:</i>
Please read each statement and circle a number 0, 1, 2 or 3 which indicates how much the statement applied to you <i>over the past week</i>. There are no right or wrong answers. Do not spend too much time on any statement. <i>The rating scale is as follows:</i> 0 Did not apply to me at all 1 Applied to me to some degree, or some of the time 2 Applied to me to a considerable degree, or a good part of time 3 Applied to me very much, or most of the time			
1	I found it hard to wind down	0	1 2 3
2	I was aware of dryness of my mouth	0	1 2 3
3	I couldn't seem to experience any positive feeling at all	0	1 2 3
4	I experienced breathing difficulty (eg, excessively rapid breathing, breathlessness in the absence of physical exertion)	0	1 2 3
5	I found it difficult to work up the initiative to do things	0	1 2 3
6	I tended to over-react to situations	0	1 2 3
7	I experienced trembling (eg, in the hands)	0	1 2 3
8	I felt that I was using a lot of nervous energy	0	1 2 3
9	I was worried about situations in which I might panic and make a fool of myself	0	1 2 3
10	I felt that I had nothing to look forward to	0	1 2 3
11	I found myself getting agitated	0	1 2 3
12	I found it difficult to relax	0	1 2 3
13	I felt down-hearted and blue	0	1 2 3
14	I was intolerant of anything that kept me from getting on with what I was doing	0	1 2 3
15	I felt I was close to panic	0	1 2 3
16	I was unable to become enthusiastic about anything	0	1 2 3
17	I felt I wasn't worth much as a person	0	1 2 3
18	I felt that I was rather touchy	0	1 2 3
19	I was aware of the action of my heart in the absence of physical exertion (eg, sense of heart rate increase, heart missing a beat)	0	1 2 3
20	I felt scared without any good reason	0	1 2 3
21	I felt that life was meaningless	0	1 2 3

Figure 3. A DASS-21 Sample Form [76]

Table 2. DASS-21 Scoring [76,77]

DASS-21 Scoring	Stress
Normal	0-7
Mild	8-9
Moderate	10-12
Severe	13-16
Extremely Severe	17+

Another way to measure stress is the Perceived Stress Scale. The Perceived Stress Scale (PSS) measures an individual's stress level and helps determine whether such levels are normal. Although it can be used alongside other stress management techniques, such as the Vital Wellbeing coaching program, it is not a diagnostic tool for stress-related disorders [78]. The survey measures feelings and thoughts over the last month, scoring responses on a five-point scale from "never" to "very often," wherein Never = 0, Almost never = 1, Sometimes = 2, Fairly often = 3, and Very often = 4. Responses to the four favorably stated items (items 4, 5, 7, and 8) must first be reversed to determine the overall PSS score. Next, the PSS score is calculated by adding together each item. Higher scores correspond to higher reported stress levels [79].

Related Systems or Solutions

The facial emotion recognition (FER) technique analyzes facial expressions in images or videos and categorizes the displayed emotions. Universal emotions are recognized among humans, including happiness, anger, sadness, surprise, fear, and disgust. In FER, the primary facial landmarks, such as the eyes, brows, and mouth corners, are delineated and employed in the identified features to categorize emotional facial expressions [53].

Related Studies

Deep Learning Model

Artificial intelligence (AI) deep learning trains machines to process information like the human brain. To generate precise insights and predictions, deep learning models can identify complicated patterns in images, text, sounds, and other data types [80 -81].

Convolution Neural Network in FER Mental Stress

The study with Facial Emotion Recognition (FER) carries out similar experiments, but instead of tracking the participants' emotions in the classroom, they infer the

recorded emotions. Not only some of the students created the FER models to enhance and support the education system.

The study of Zahara et al. [82] proposes designing a system that can predict and recognize the classification of facial emotions based on feature extraction using the Convolution Neural Network (CNN) algorithm in real-time with the OpenCV library. The prediction of microexpressions is observed and distinguished because it is one of the communication methods in social psychology. The ability to recognize this facilitates the intended communication. The design recognition of microexpressions on faces was carried out in this study in several testing stages on the system. The results showed that using the CNN architectural model in the facial expression detection system could be done optimally and in real-time. The evidence using a separate convolution layer when conducting data training can be carried out optimally, and the accuracy of the data in the trained model facial expression is an average of 65.97%. It is necessary to analyze the results of the implemented system. However, the study encourages optimizing the result through the accuracy of a more profound CNN architecture using a more convolutional and fully connected layer. Also, the new dataset or more training

data will create better performances for faces 5 meters away, viewing angles, and photo rotation.

The study of Dukic et al. [83] used FER to employ real-time recognition in an active classroom environment. They used workshops to conduct their research and models, such as Inception-v3 and ResNet-34, in real-time prediction. The training data for these models were collected from images of three well-known data sets for emotion recognition: CK+, FER-2013, and SFEW. The results indicated that the student's facial expressions are related to the activities they are currently engaged in and contain valuable information for teachers about what they can improve in their teaching practice.

Besides the classroom environment, Yang et al. [84] conducted a study regarding a virtual setup that assesses the virtual learning environments for the teaching process environment. The process FER used the Haar cascade method. Experimental results show that the model proposed in this paper is consistent with the expressions from students' learning situations in virtual learning environments. This paper demonstrates that emotion recognition based on facial expressions is feasible in distance education, permitting

identification of a student's learning status in real-time. Therefore, it can help teachers to change teaching strategies in virtual learning environments according to the student's emotions.

ZFNet

ZFNet, developed by Matthew Zeiler and Rob Fergus, is a convolutional neural network model that improved the AlexNet architecture. It tweaked the convolutional layers, used smaller filter sizes, and introduced a visualization technique for better feature map interpretation [85]. ZFNet visualized the performance of each AlexNet layer and the parameters that may be adjusted for better accuracy. The ZFNet model uses a multi-layered deconvolutional neural network to reveal the input stimuli that excite the feature maps at any layer [86].

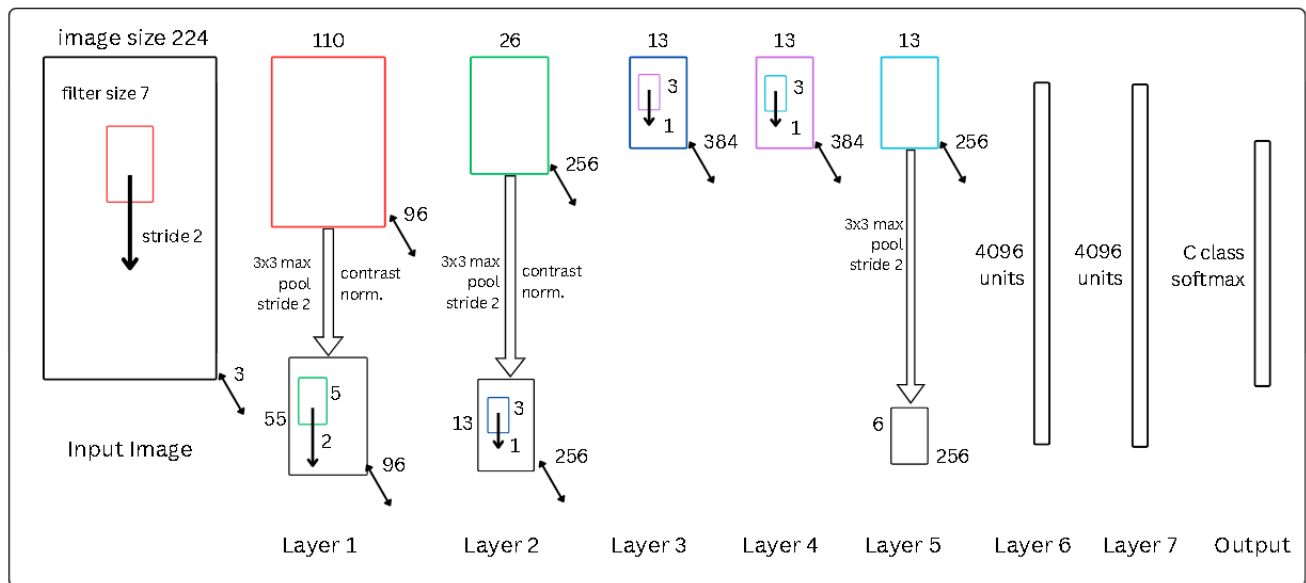


Figure 4. The Convolutional layers of ZFNet model [87]

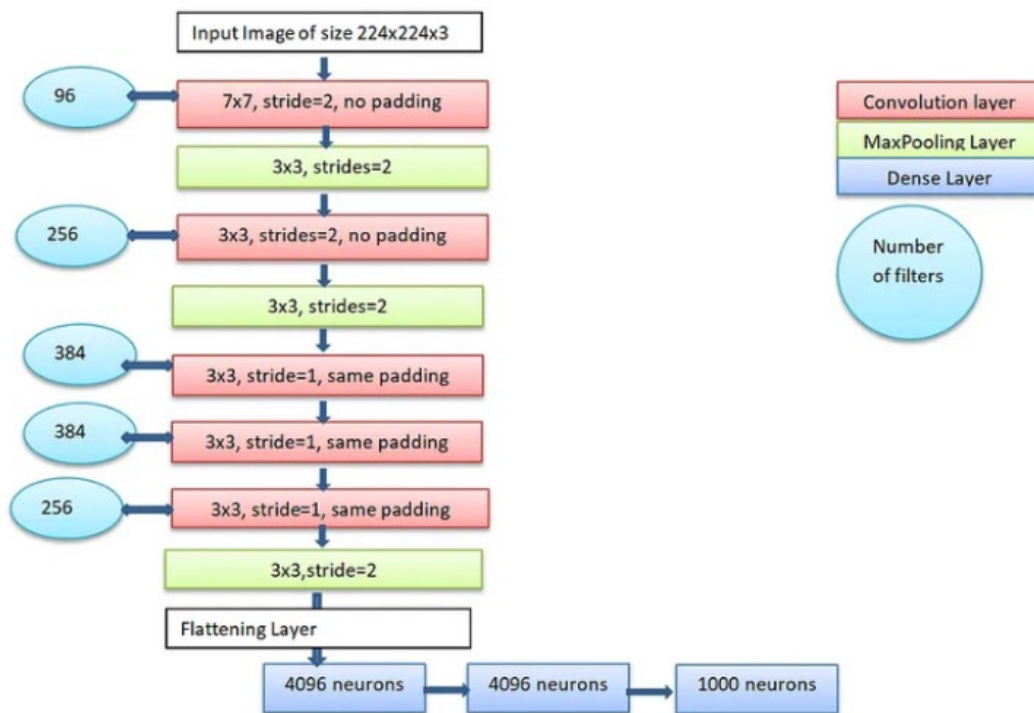


Figure 5. The architecture of ZFNet model [88]

Figure 5 illustrates the architecture of the ZFNet model, a convolutional neural network designed for image classification. The architecture of ZFNet starts with a neural network; it processes an input image of size 224x224x3 through a series of convolutional and fully connected layers. The first layer uses 96 filters of size 7x7 with a stride of 2, followed by a ReLU activation, max pooling with a 3x3 kernel and stride of 2, and contrast normalization. The second layer applies 256 filters of size 3x3 with a stride of 2, followed by the same max pooling and normalization as the first. The third and fourth layers use 384 filters of size 3x3 with padding set to "same" and a stride of 1. The fifth layer applies 256 filters of size 3x3 with a stride of 1, then max pooling with a 3x3 kernel and pm stride of 2, and contrast normalization. The network then transitions to fully connected layers, starting with the sixth and seventh layers, each containing 4096 neurons. The final, eighth layer is a dense layer with 1000 neurons, corresponding to the number of classes for classification. Combining convolutional layers with ReLU activations, max pooling, and contrast normalization, followed by fully connected layers, enables the network to learn complex features and perform accurate image classification [88].

ZFNet in Analyzing Facial Expressions

ZFNet has been proven helpful in object detection in multiple cases. This CNN architecture has been implemented in an automatic recognition system for traffic signs [89], a pedestrian detection system [90], a method of detection of lung cancer [91], and a kiwifruit detection system [92]. Reportedly, these systems have satisfiable results, with lung cancer detection reaching an accuracy of approximately 80% and the kiwifruit detection model having a 92.3% recognition rate. Meanwhile, the results of traffic signs and pedestrian detection systems have yet to be publicly available but were stated to have positive results.

In the domain of facial expression recognition (FER), ZFNet was mentioned several times as a CNN architecture [93]-[95]; however, there needs to be more literature that focuses on the utilization of ZFNet in FER. One exception is the proposed framework by Pamarthi et al. [96], which developed a Buffalo-based ZFNet Recognition Framework to recognize facial expressions for the user face authorization system. The designed framework achieved a 99.5% performance in validated accuracy, recall, f-score, and precision rate. Hence, there is

an opportunity to expand the knowledge by exploring and implementing ZFNet in FER.

CHAPTER 3 RESEARCH DESIGN AND METHODOLOGY

Description of the Proposed Study

ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management is a proposed study that the researchers developed to create a non-invasive stress assessment through a series of emotion detection. The research is divided into three phases: Preparation, Implementation, and Evaluation.

In the preparation phase, in-depth consultation with a domain expert proficient in stress level measurement, acquiring necessary datasets and conducting an initial data exploration. This sub-step forms a solid infrastructure for future methodological initiatives.

The implementation phase involves creating machine learning models, developing an algorithm to analyze student emotions, and designing web applications to deploy these models and algorithms efficiently. Particularly, it works with an internet connection, so the application is deployed on the web. Developing an entailed method to detect a user's emotion is based on utilizing RetinaFace for effective facial detection and ZFNet for accurate emotion classification.

Following these systematic processes helps achieve high precision in identifying user emotion.

During the evaluation phase, the researchers prioritized specific goals, particularly the performance of the system in determining the stress level using accuracy, precision, recall, and confusion matrix. The involvement of these features, extracted from the input image, the classifier is trained, and the algorithm ZFNet is applied to detect such delicate patterns in the features. Throughout this study, many assessments were carried out to consider it comprehensive and for the results to be used by academia for further research on stress management.

Assumptions and Preconditions

The study is based on the assumption that the frequency and type of emotional expressions identified in facial images can be effectively utilized to assess an individual stress level. It assumes that emotions expressed through facial features, categorized as anger, disgust, fear, happiness, sadness, surprise, and neutral, can indicate a person's stress state. By further categorizing these expressions into "stressed", "neutral", and "non-stressed" groups using the

ZFNet model, the application aims to quantify and graphically represent the prevalence of stress-related emotions, thereby providing a feasible method for stress level assessment.

The output of the stress level assessment can be effectively used to generate actionable stress management recommendations. This is based on the assumption that the classification of facial expressions and subsequent stress level assessment can inform tailored stress management strategies. The application could recommend specific interventions or coping mechanisms by understanding the frequency and type of stress-indicative emotions. This would suggest a direct correlation between the emotional analysis provided by the application and the efficacy of subsequent stress management recommendation, thus bridging the gap between emotional recognition and practical mental health solutions.

Methods and Proposed Enhancements

Preparation

During the thesis's development, expertise from a licensed psychologist was sought to provide valuable insights. Ms. Windy T. De La Cruz, a licensed psychologist, was selected for this purpose. Her expertise in stress measurement profoundly influenced initial conceptualization and methodological direction. Through the consultation, Ms. Windy demonstrated a deep understanding of mental stress and its complexities, offering valuable guidance for the system's development.

Acquiring Datasets

To acquire suitable datasets, the researchers employed a multifaceted approach. First, an extensive review of existing literature and research studies in stress management and emotion detection was conducted. This literature review helped identify publicly available datasets that met research criteria, such as datasets containing facial expression data annotated with corresponding stress levels.

The researchers used the cross-data set of FER2013, CK+ and AffecNet datasets, which are widely recognized and

publicly available for facial expression recognition. These datasets contain 5,581 files, categorized into seven emotion classes: anger, disgust, fear, happiness, sadness, surprise, and neutral. The test directory consists of 2,100 files, ensuring a balanced evaluation of the model's performance. While these datasets do not include explicit stress labels, the researchers leveraged them to train and validate the emotion recognition model, enabling accurate classification of facial expressions for stress level assessment.

Implementation and Evaluation

The researchers focused on developing and refining learning models and algorithms. The researcher enhanced RetinaFace's capabilities for facial detection and improved the ZFNet algorithm for more accurate emotion classification. The Evaluation Phase involved a rigorous testing and refinement process for ZenLens. The researchers focused particularly on the system's performance in determining stress levels on facial and emotion. The use of the ZFNet algorithm was crucial in this phase, as it enabled the identification of complex emotional patterns from facial expressions – a vital component for accurate stress level estimation.

The researchers proposed several enhancements to the ZenLens system to ensure its effectiveness in student stress management. A significant enhancement was the development of an advanced software architecture, including a user-friendly front-end interface created using Python in Visual Studio Code, Flask, and React. This interface featured essential elements like the application's dashboard and image upload functionality. The ZenLens back-end system, encompassing data preprocessing, emotion detection, and stress level classification, was designed to efficiently process and analyze students' facial features.

Additionally, the researchers outlined a robust database structure to support the efficient handling and storage of facial image data and corresponding stress analysis results. The procedural design provided a clear and systematic process for stress detection using facial expression analysis, ensuring clarity and systematic progression in the tool's operation.

The researchers adopted the Rapid Application Development (RAD) method for the system's development cycle, facilitating rapid prototyping, user feedback incorporation, and iterative enhancements, ensuring the tool's relevance and usability. A

key focus was placed on developing and refining the ZFNet model, enabling it to analyze multiple facial expressions simultaneously - a substantial advancement over existing systems.

The objectives centered on developing a reliable and accurate tool for assessing and managing student stress. These encompassed developing the ZFNet model, evaluating its performance in facial detection and emotion analysis, determining students' stress levels, creating a user-friendly web application, and conducting comprehensive usability testing.

Components and Design

Software Architecture

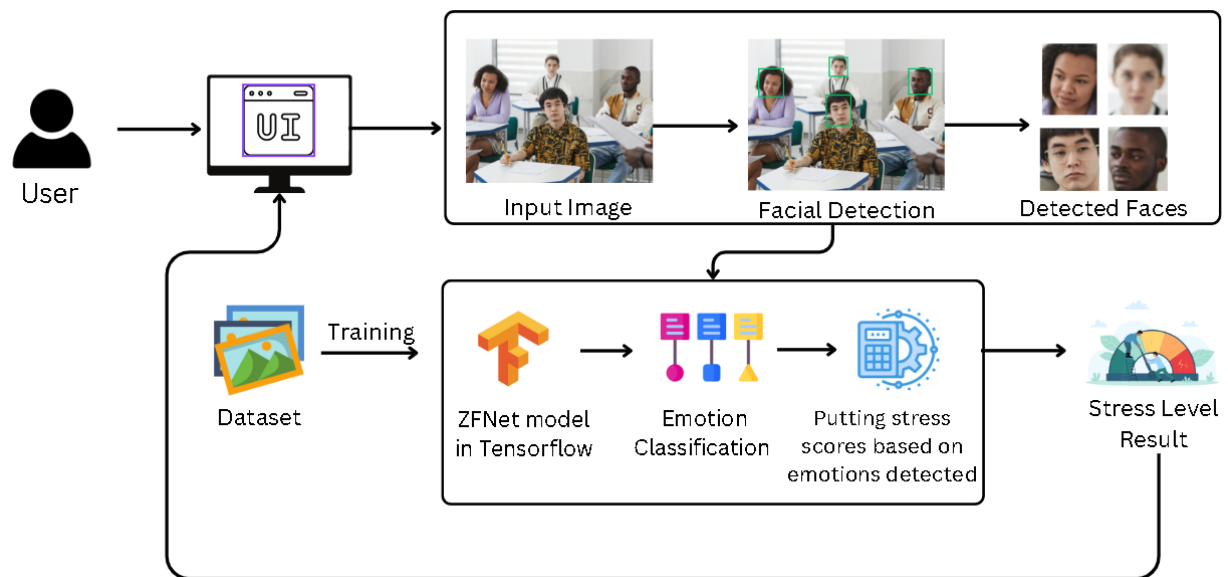


Figure 6. ZenLens Software Architecture

The software architecture in Figure 6 showcases how ZenLens is made and how the interaction between different components occurs. The front end of the application will be built using the Python language in Visual Studio Code, Flask, and React. It will contain the application's logo, dashboard, and the main interaction point where the user can upload the necessary image inputs.

Meanwhile, the back-end of the application consists of data preprocessing, an emotion detection layer, and a stress classifier. Here, images being uploaded that contain students'

facial features in a classroom will be processed. These facial features in the image will be detected using RetinaFace, which will then be cropped and fed into the ZFNet model in Tensorflow, where these series of cropped facial images will be classified into seven categories of emotions as well as classified into "stressed", "neutral" or "non-stressed" images. After these tasks are performed, a stress assessment result will be produced and shown to the user in the front-end part of the application.

Database Design

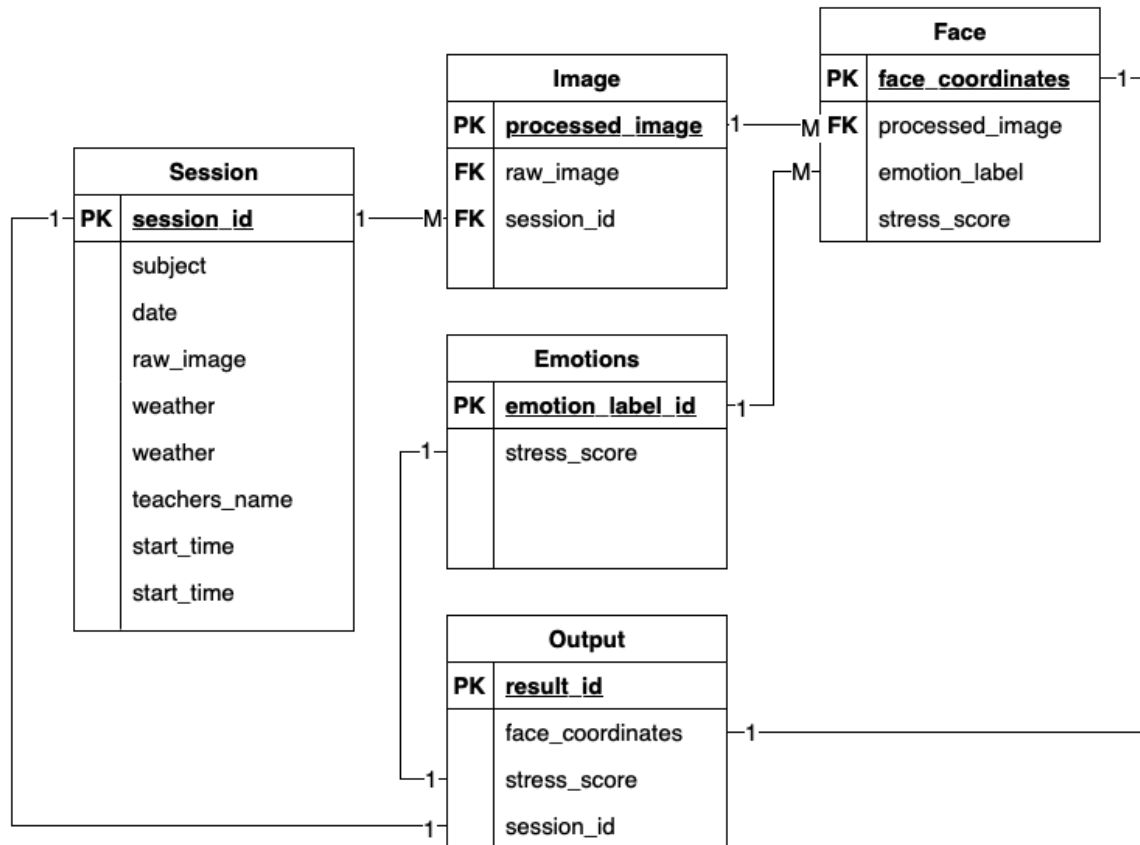


Figure 7. Database Design

The database design in Figure 7 shows how the data is being stored. The first step is the Raw Images entity, which comprises uploaded images; each image contains a unique identifier: the raw_image_id, filename, and timestamp. Then, the system would identify and crop the faces in the uploaded image, which will be stored in the Faces entity; it would inherit the ID from the Raw Images as raw_image_id. The

Face_expression entity stores the data about the expressions found on the faces; it would also inherit the face_id and expression_id. The expression entity stores the seven categorized emotions that the face_expression entity uses, and the annotation entity stores the results of the face entity.

Procedural Design

The procedural design refers to the systematic plan or methodology used to conduct this research study. Figure 8 below outlines the process of a stress detection system using facial expression analysis. It starts with the user capturing or uploading images, which is then sent to the backend for analysis.

In the backend, the images undergo preprocessing, facial landmark detection, feature detection, and classification into emotional categories such as "happy," "sad," "angry," "surprised," "neutral," "fear," and "disgust." After emotion classification, it will be categorized within three groups: (1) stressed, (2) neutral, and (3) non-stressed image. The results are sent back to the frontend, where they are displayed to the user.

This completes the feedback loop, allowing the user to receive feedback on their stress level based on their facial expression.

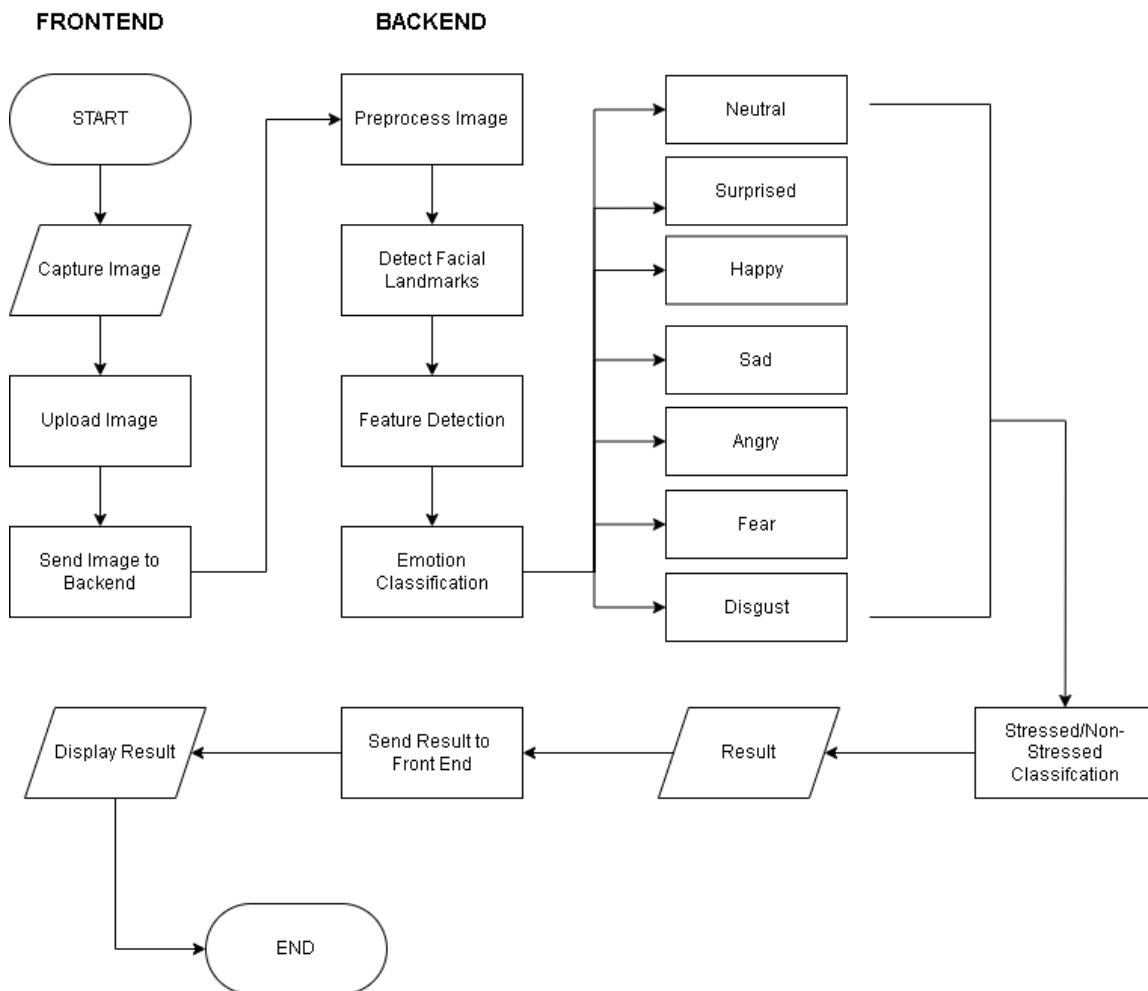


Figure 8. Procedural Flowchart for Application Interface

Process Design

The Data Flow Diagram (DFD) outlines a system for web application of ZenLens: To better assist the subject, a stress level assessment and emotion classification will be conducted. This is initiated when the user starts a record mode that results in the acquisition of image data. This data is first processed in the Request Handling component, which provides an opportunity to assess stress level.

Face detection and feature extraction are then performed on the image by the last stage called Handle Request and Response in which the image is passed to RetinaFace for facial landmark detection and cropping of the face areas only. These cropped images are passed to the ZFNet Algorithm Component, which is another set of algorithms based on a convolutional neural network to predict the emotions of the student as well as to determine stress levels.

The results are taken back to Handle Request and Response component which gives the feedback to the user and also makes uninterrupted adjustment in the model due to performance feedback. It utilizes enhanced image processing and AI based neural networking with respect to data processing to provide efficient and accurate results for emotion and stress levels.

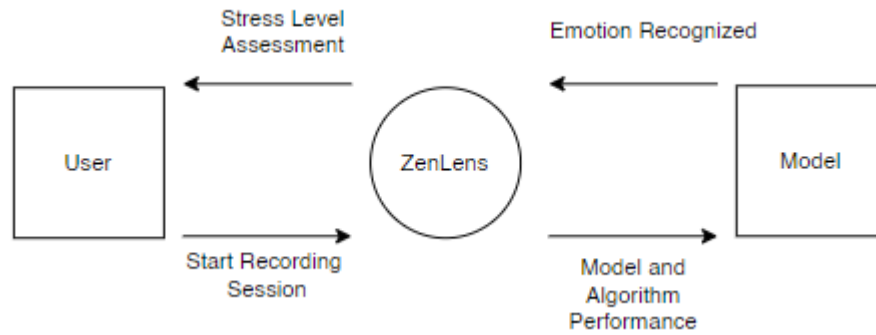


Figure 9. Level 0 Data Flow Diagram (Context Diagram)

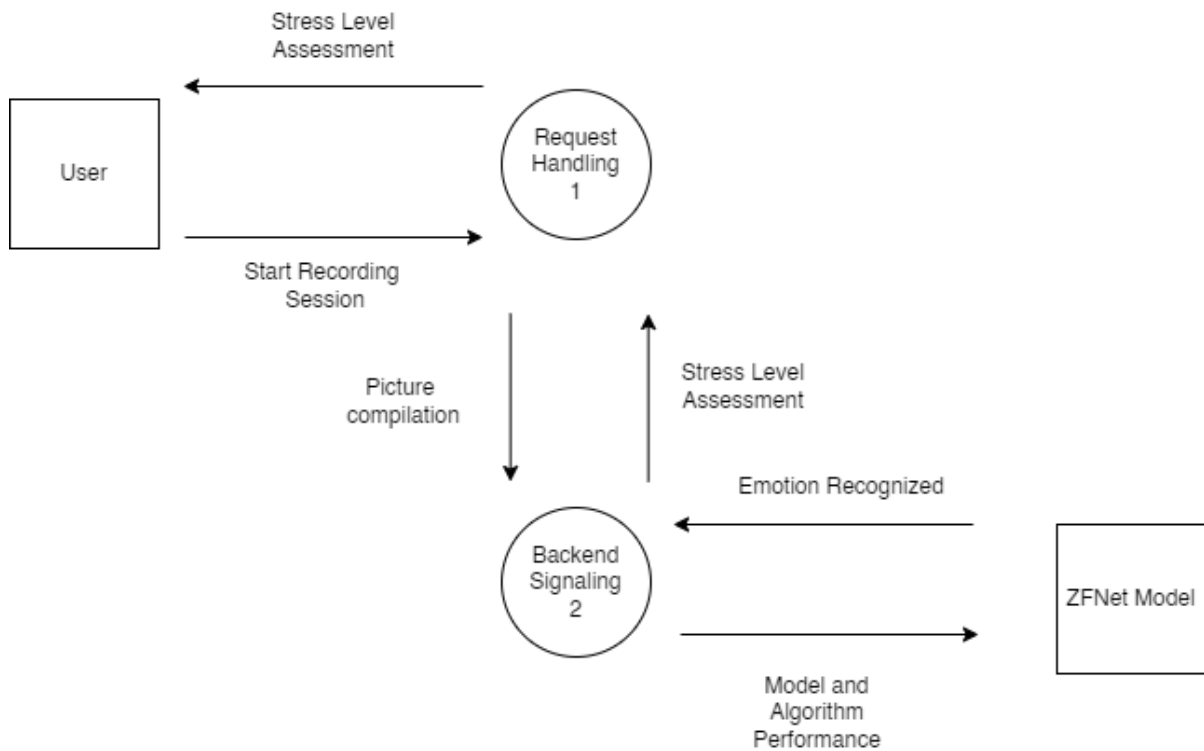


Figure 10. Level 1 Data Flow Diagram

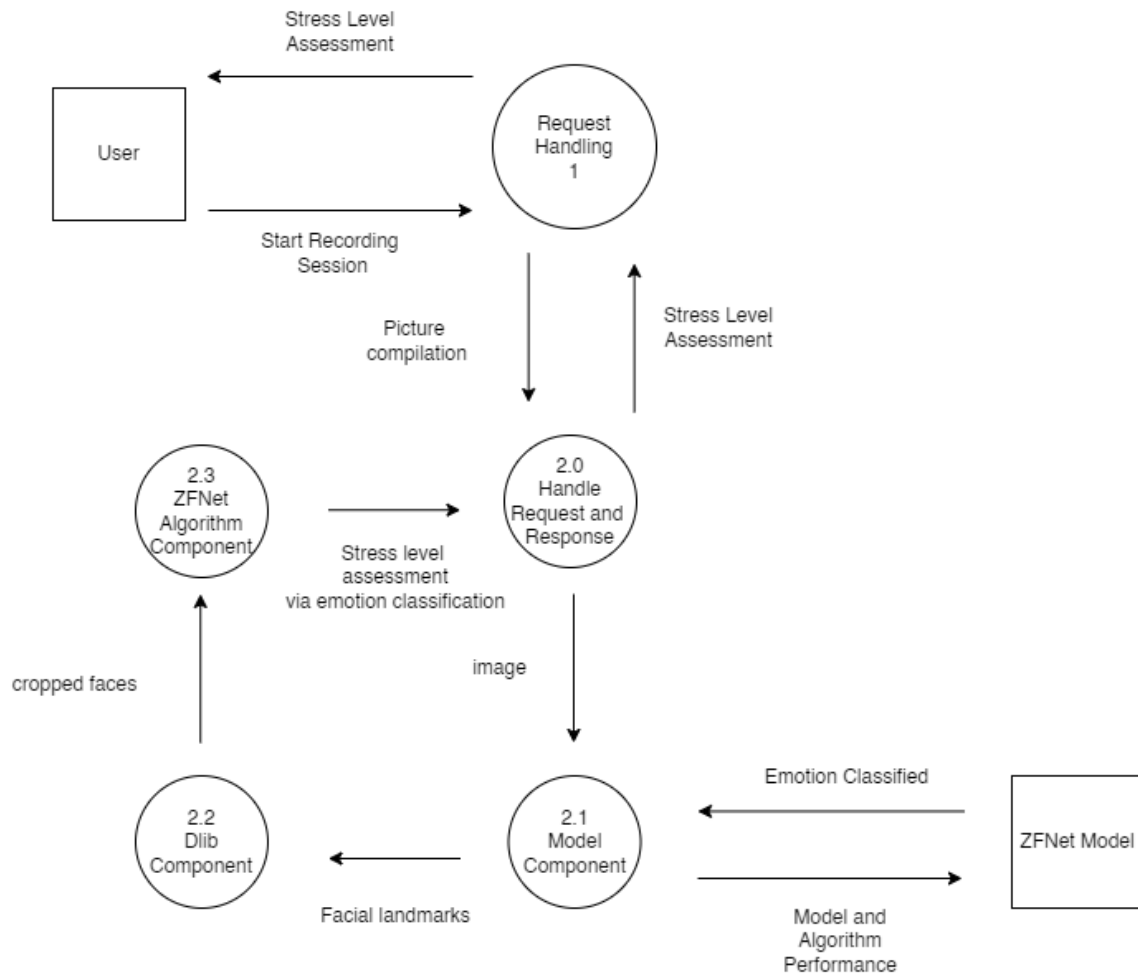


Figure 11. Level 2 Data Flow Diagram

Methodology

System Development Life Cycle

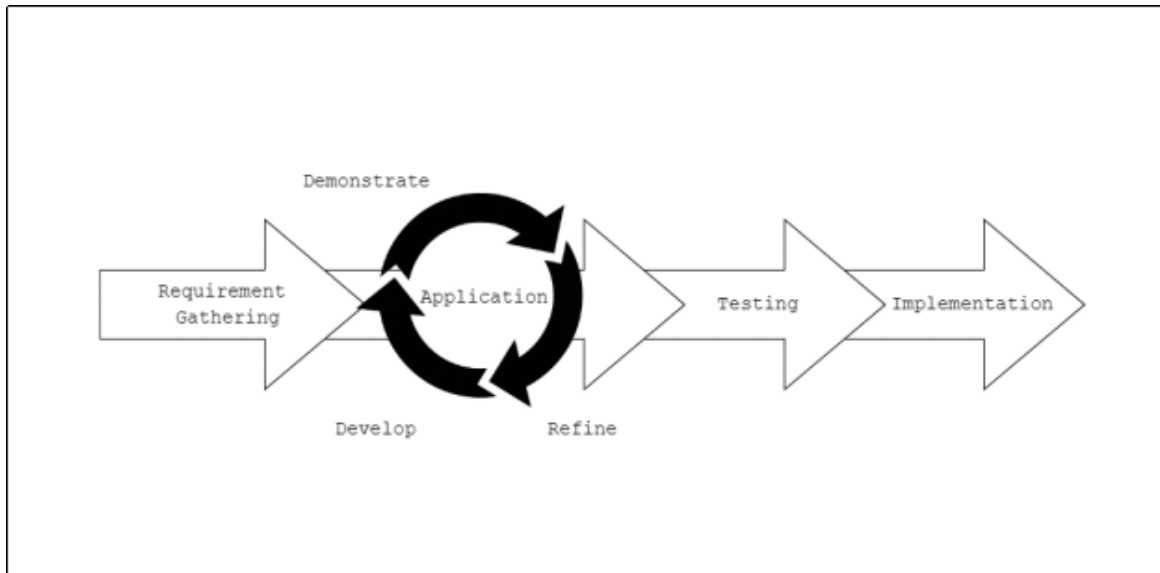


Figure 12. Rapid Application Development Approach

This study utilizes the Rapid Application Development (RAD) method shown in Figure 12, a four-step approach that prioritizes rapid iteration and user feedback. The RAD process will guide the development of the ZenLens system.

First, the study started with a thorough understanding of the needs of students experiencing stress, particularly their ability to recognize emotions. The researchers then worked with stakeholders (e.g., educators, guidance counselors, and students) to define essential elements for the ZenLens application.

Following this was the use of appropriate programming languages and development environments to create a basic version of the application. This prototype will then be presented to gather feedback on its functionality and identify areas for improvement.

Through a series of refinements based on this feedback, the prototype was iteratively enhanced. Once the prototype demonstrates accurate emotion detection, the process will move on to the final stage: implementation. This involves optimizing the application and making it readily available to students.

The Rapid Application Development (RAD) Approach is well-suited for the study of ZenLens, which focuses on image-based stress detection and tailored recommendations for student stress management. RAD's emphasis on faster development through iterative design enables the quick creation of a working prototype for testing stress detection algorithms and recommendations. Moreover, the incorporation of early user feedback ensures the application's user-friendliness and alignment with student needs.

CHAPTER 4 RESULTS AND DISCUSSION

Implementation

Development and Structure of the ZFNet-Based Emotion Detection Model

ZenLens is a proposed system by researchers designed to address the growing concern of student stress in educational institutions, specifically college students. The researchers developed a TensorFlow neural network to detect emotions. The network receives an input layer of size 48X28X1, which is the dimension of an input image with grayscale filter. Facial detection of college students is captured and served as input, which is detected by RetinaFace, a library adept at multi-face detection. This neural network is also designed to predict seven different facial expressions: The universal emotions include happiness, sadness, anger, fear, surprise, disgust, and the neutral state.

The first part of the detection model used by the researchers is depicted in Figure 14 below. The following figure illustrates the design and initial structure of the model as used in this study. Figure 13 describes the initial architecture of the model, presenting the multiple layers and their settings, including convolutional layers, pooling

layers, and fully connected layers, which are employed to transform the input images and estimate the related emotions.

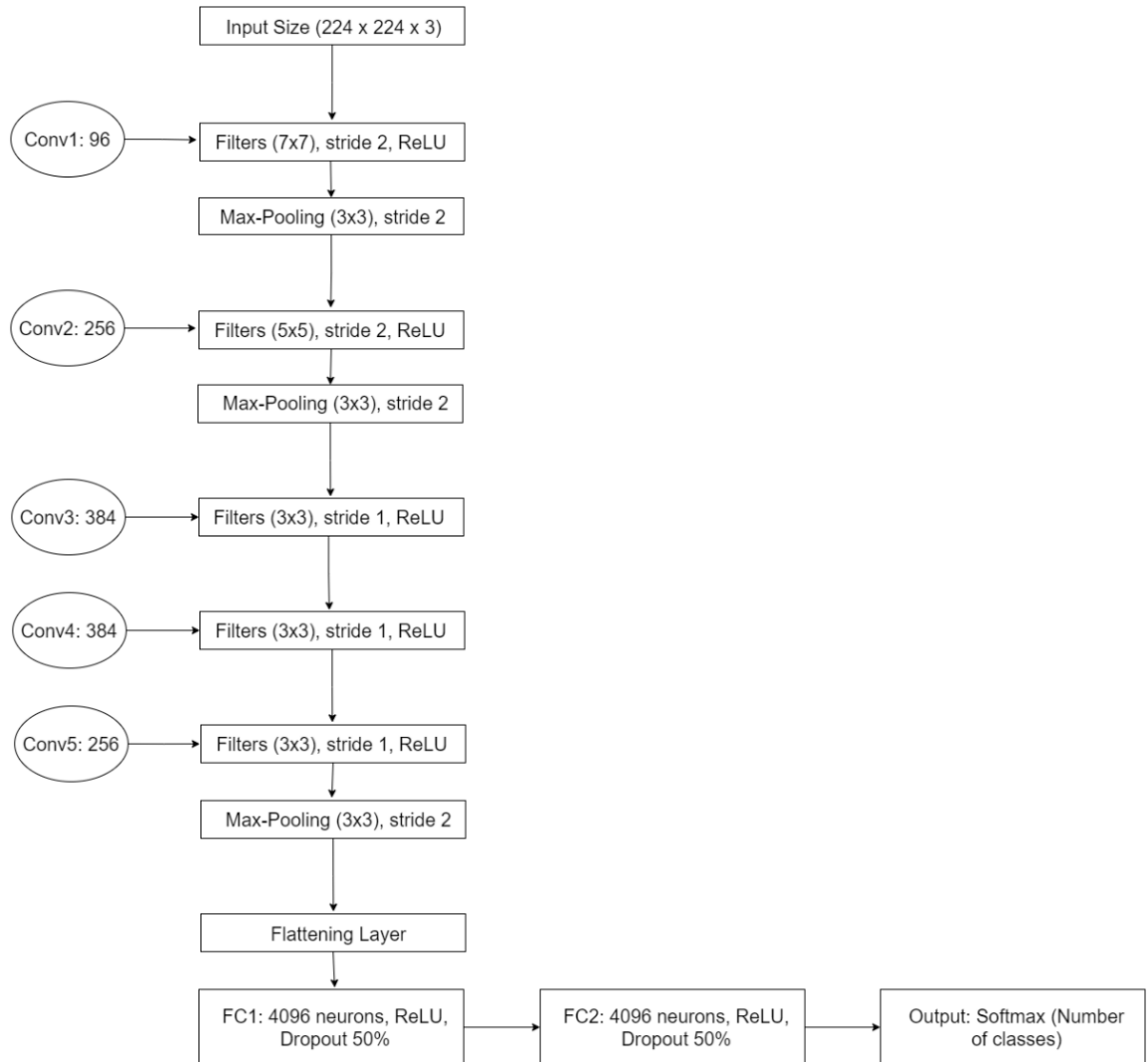


Figure 13. ZFNet Architecture

Figure 13 provides a detailed overview of the model's architecture, showcasing the sequential arrangement of layers designed to process input images and predict associated emotions. The model commences with an input layer accepting

224x224 pixel images with three color channels. Subsequently, multiple convolutional layers extract image features, each employing a specified number of filters and kernel sizes. Max-pooling layers follow these convolutional layers to downsample feature maps and reduce computational load. The extracted features are then flattened into a one-dimensional vector before being fed into fully connected layers responsible for high-level feature learning. Finally, the output layer utilizes a softmax activation function to generate probability distributions for each emotion class. Dropout regularization is incorporated into the fully connected layers to mitigate overfitting. This architectural design enables the model to learn discriminative features from the input images and accurately classify them into corresponding emotion categories.

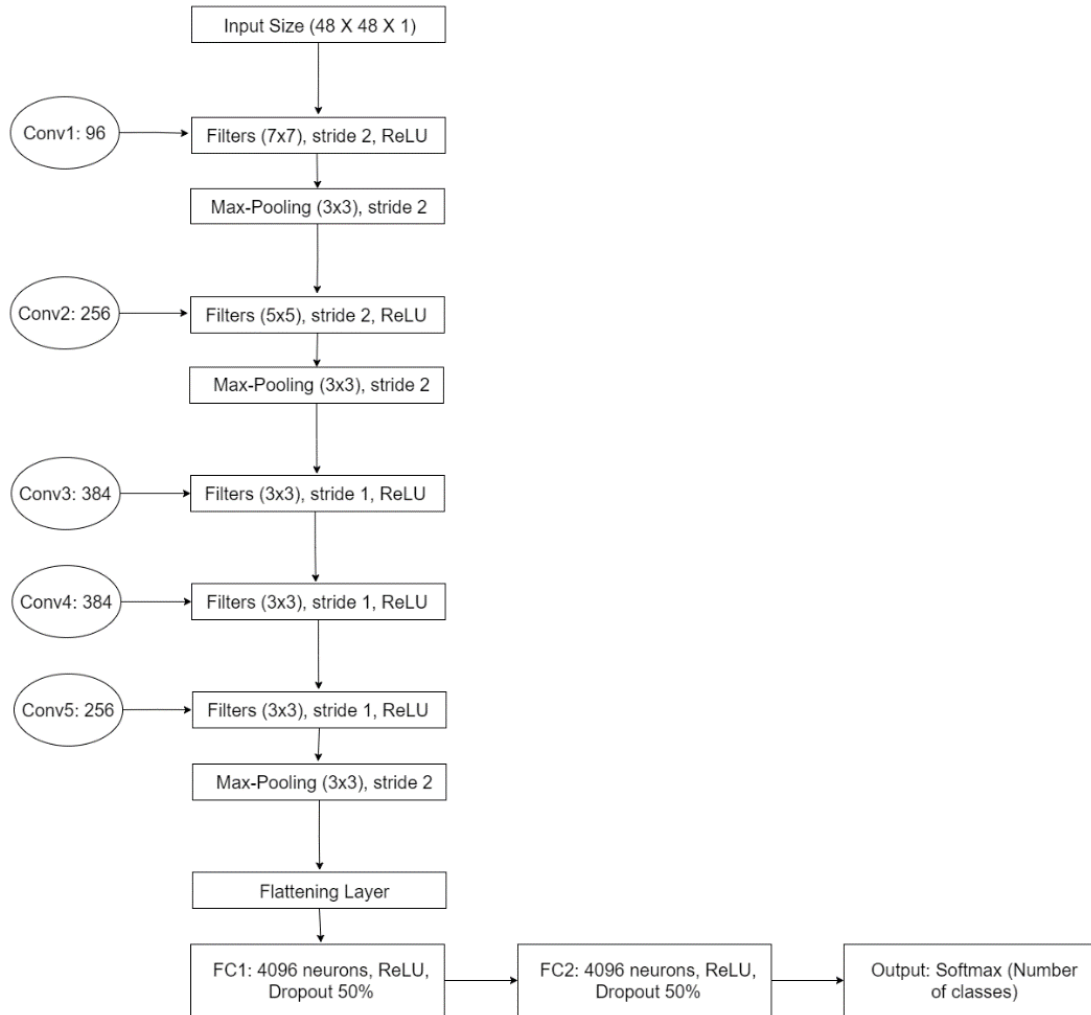


Figure 14. ZFNet Architecture Using 48 X 48 X 1 size for the Images

Figure 14 illustrates the foundational architecture of the initial model, a ZFNet variant adapted for emotion detection, specifically optimized for input dimensions of 48×48×1 and single-channel grayscale images. This architecture served as the initial framework for the researchers' exploration into image-based emotion recognition, addressing

technical constraints such as computational efficiency and dataset characteristics (e.g., input dimensionality and grayscale data compatibility). The model's structure is characterized by convolutional and pooling layers that extract relevant features from input images. The convolutional layers, denoted as Conv1 to Conv5, progressively extract increasingly complex features by applying filters with varying sizes and strides. Max-pooling layers, interspersed between convolutional layers, downsample feature maps to reduce computational cost and introduce invariance to small image translations. The extracted features are flattened into a one-dimensional vector, facilitating their processing by fully connected layers. These fully connected layers, FC1 and FC2, introduce non-linearity and learn high-level representations of the input data. Finally, the output layer, equipped with a softmax activation function, produces probability distributions corresponding to different emotion classes. This initial model architecture provides a strong baseline for further experimentation and refinement in the pursuit of accurate emotion detection. Figure 14 shows the initial conceptual framework which served as the base for modifications such as Figure 13 presented richer resolution-colored images. The transition from grayscale to RGB images

delivered better feature learning capabilities which resulted in better emotion classification outcomes.

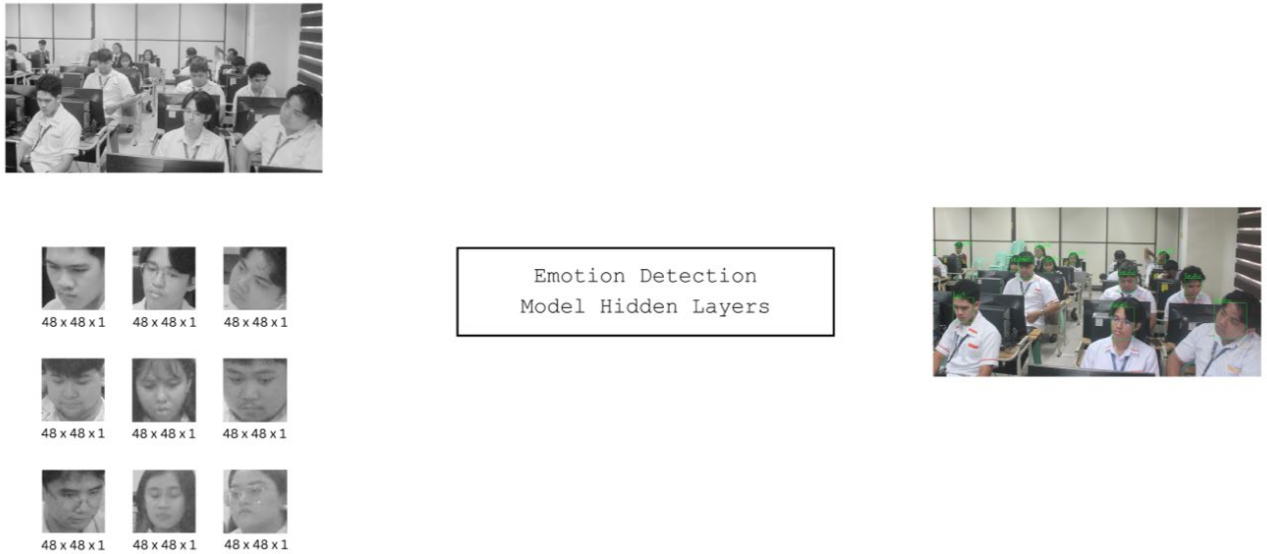


Figure 15. First Iteration of the Model: The input and output Shapes of the Model

In Figure 15, the model's intake is a meticulously formatted image, specifically a 48×48 pixel representation. This standardized dimension ensures consistent data processing and facilitates efficient computational operations. The output layer is a seven-dimensional vector, a numerical construct that encodes the model's prediction probabilities for seven distinct emotional categories. This design implies a multi-class classification problem where the model assigns an input image to one of these seven emotional labels.

This initial model architecture is a robust baseline for exploring the complex relationship between facial expressions and emotional states. While this first iteration focuses on stress detection, the multi-class output layer suggests the potential for broader emotional recognition capabilities.

Future iterations could benefit from incorporating color information, exploring deeper convolutional architectures, and refining the emotional categories to enhance accuracy and specificity.

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Figure 16. First Iteration of the Model: Sample Comparison of the Ground-Truth and Predicted Landmarks

The specific libraries and packages used the following libraries and modules from the TensorFlow and Keras ecosystem: TensorFlow, specifically TensorFlow. Keras, which includes Sequential for creating a linear stack of layers, Conv2D for 2D convolution layers, MaxPooling2D for 2D max-pooling layers, Flatten for flattening inputs, Dense for densely connected

layers, Dropout for applying Dropout to inputs, BatchNormalization for batch normalization, Adam for the Adam optimization algorithm, ImageDataGenerator for generating batches of tensor image data with real-time data augmentation, and EarlyStopping for stopping training when a monitored metric has stopped improving.

The model is trained using a cross-dataset approach that includes FER2013, CK+, and AffectNet. These datasets consist of small images of faces labeled for one of the seven universal emotions. The training process involves passing batches of images through the model, computing the loss, and adjusting the model's parameters using backpropagation, a method for optimizing the model by minimizing the error. To enhance the diversity of the training data, the ImageDataGenerator class from Keras is used for real-time image augmentation. This includes applying random transformations such as rotation, shift, and scaling, which helps the model generalize better to new data. Early stopping is implemented to halt training when the validation loss no longer decreases, thereby preventing overfitting and saving computational resources. The performance of the trained model is evaluated using a separate test set to calculate metrics

such as accuracy, precision, recall, and F1-score, providing a comprehensive assessment of its effectiveness

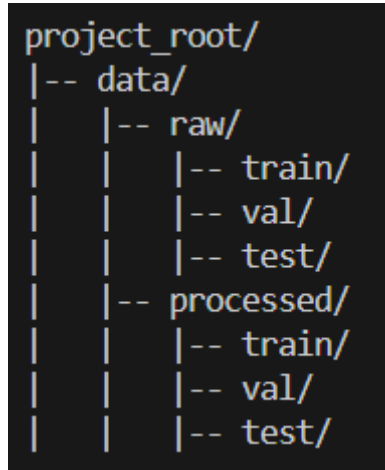


Figure 17. FER2013 Data Dictionary Structure

Figure 17 illustrates the directory structure for managing the cross-datasets of FER2013, CK+, and AffecNet, which is essential for optimizing data management in facial emotion recognition models. The structure begins with a 'project_root' directory, which contains a 'data' directory. Within 'data', there are two main subdirectories: 'raw' and 'processed'. The 'raw' directory is further divided into 'train', 'val', and 'test' subdirectories, preserving the original images categorized into training, validation, and test sets. This organization ensures data integrity and facilitates unbiased evaluation.

The 'processed' directory also contains 'train', 'val', and 'test' subdirectories, where preprocessed images are stored. These images have undergone transformations to enhance model performance, such as normalization or augmentation. Effective data management in this context involves maintaining consistent image formats, accurate labeling, balanced datasets, and meticulous metadata management to ensure the quality and reliability of the data used for training and evaluation.



Figure 18. Session Model Output

The Session Model Output in the ZenLens system, as depicted in Figure 18, provides a comprehensive overview of each class session, capturing key details such as the subject, teacher, date, time, and weather conditions. This page also

features an Emotion Timeline, which visualizes students' emotional responses throughout the session, helping educators analyze engagement levels and emotional trends.

In this CIT207 Object-Oriented Programming session, the ZenLens system analyzed student emotions from 9:00 AM to 11:40 AM on September 12, 2024, under Teacher Keia Joy Harder. The session took place on a sunny day, which may have contributed to the classroom's overall atmosphere.

The Emotion Timeline reveals that "Neutral" expressions were dominant throughout the session, with frequent fluctuations but a generally high count. This suggests that students were primarily in a focused or passive state, potentially absorbing the lecture material.

The second most common emotion was "Sad", appearing in short bursts but consistently present. Additionally, there were occasional instances of "Angry", "Afraid", "Disgust", and "Surprise", though they remained at low levels. By leveraging ZenLens to track these emotional trends, educators can refine their teaching strategies to foster a more engaging and emotionally supportive learning environment.



Daily Details for 2024-09-10

Emotion Levels for 2024-09-10

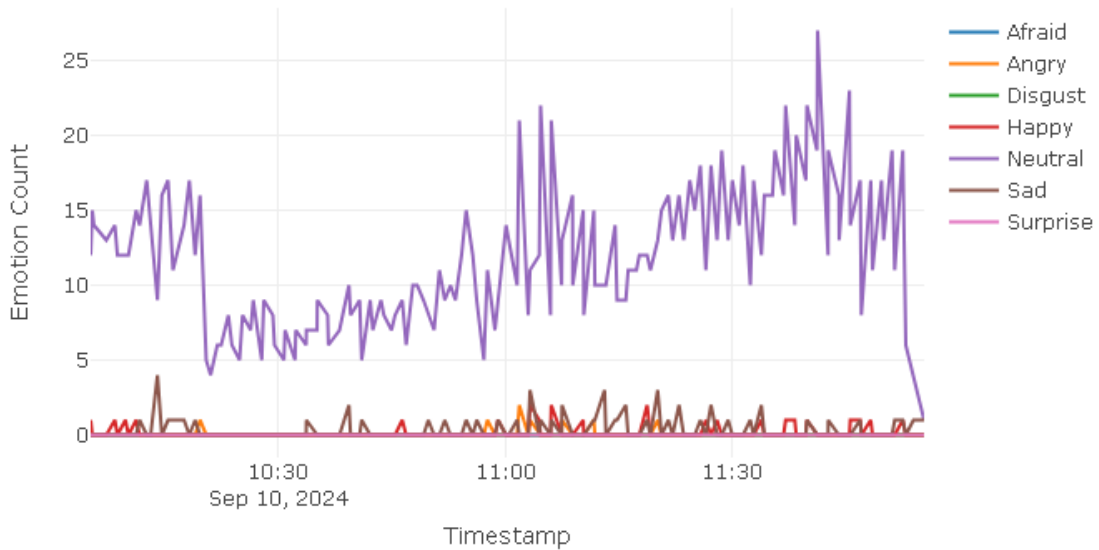


Figure 19. Daily Model Output

The Daily Model Output page of the ZenLens web application, illustrated in Figure 19, offers a detailed overview of emotions detected throughout a single day. Displayed in a line graph, each emotion's intensity and frequency are tracked over time. On September 10, 2024, the data indicates varying levels of emotions. The "Neutral" emotion is consistently present, with counts ranging from 5 to 25, suggesting a generally calm or focused state. Emotions like "sad," "happy," and "angry" appear occasionally, with

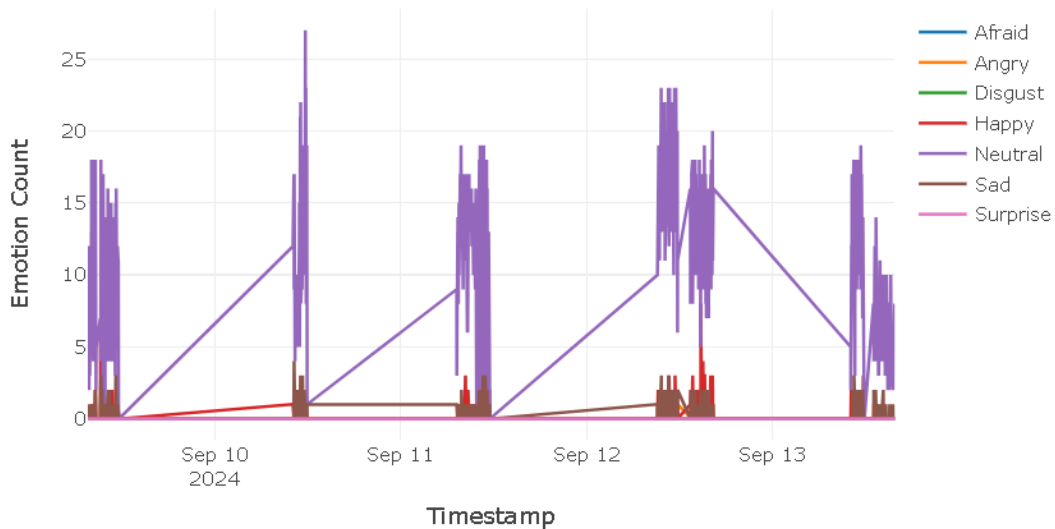
"sad" showing a slight increase around 11:30 AM. "afraid," "disgust," and "surprise" are minimally present, indicating they were not significant during this period. The timestamps at 10:30 AM, 11:00 AM, and 11:30 AM provide specific points of reference for these emotional fluctuations.



Weekly Details for 2024-09-09 - 2024-09-13



Emotion Levels for 2024-09-09 - 2024-09-13



Recommendation: The stress level this week is: Normal. Advice: It's great to hear that your stress level falls under the 'Normal' category! Here are some recommendations to help you maintain this healthy stress level: 1. Keep up with your current stress management practices, whether it's exercise, meditation, hobbies, or spending time with loved ones. Consistency is key in maintaining a healthy stress level. 2. Stay mindful of potential stress triggers and make a conscious effort to address them before they escalate.

Figure 20. Weekly Model Output

As shown in Figure 20, the Weekly Model Output page of the ZenLens web application provides a cumulative view of emotions detected over a week, using a line graph to display intensity and frequency trends for each of the seven universal emotions. The analysis for the week of September 9 to September 13, 2024, reveals varying levels of emotions. The "neutral" emotion is consistently present, with counts ranging from 5 to 25, indicating a generally stable or focused state among participants. Emotions like "sad," "happy," and "angry" appear occasionally, with "sad" showing slight increases at certain points, suggesting brief moments of stress or disengagement. "Afraid," "disgust," and "surprise" are minimally present, indicating they were not significant during this period. The stress level for the week is categorized as "normal," and recommendations are provided to help maintain this healthy stress level, such as continuing current stress management practices and staying mindful of potential stress triggers.

ZenLens is a system designed to assess emotional states and determine overall stress levels by analyzing the frequency and intensity of various emotions. Each emotion is assigned a specific score based on its perceived impact on stress, with

happy and neutral scoring 0, surprised and disgust scoring 1, sadness scoring 2, and fear and anger scoring 3. The system considers the frequency of each emotion and its assigned score, with emotions indicative of higher stress levels, such as fear and anger, having a more significant impact on the overall stress calculation.

To translate emotional data into stress levels, ZenLens utilizes a revised stress level range based on the Depression Anxiety Stress Scales (DASS-21). The DASS-21 assesses stress severity across scores and corresponding percentile ranks. The revised range classifies stress levels as follows: Normal for scores 0-7, Mild for scores 8-9, Moderate for scores 10-12, Severe for scores 13-16, and Extremely Severe for scores 17+. These revisions reflect updated normative data, ensuring accurate classification of stress levels based on current population metrics.

The system processes the emotional data collected from users by assigning scores to each emotion based on the scale above. It then calculates the frequency of each emotion and multiplies it by the corresponding score. This weighted sum is used to derive an overall stress score for the user, which is then mapped onto the revised DASS-21 ranges to determine the

user's stress level category. For example, suppose a user's emotional data shows frequent occurrences of Fear and Anger. In that case, their overall stress score will be higher, potentially placing them in the "Severe" or "Extremely Severe" category, depending on the exact score and percentile rank.

ZenLens generates a comprehensive value reflecting the user's overall stress state by combining emotional scores, their frequency, and the revised stress range. This data can trigger targeted stress management recommendations within the web application, offering personalized interventions to reduce stress and improve well-being. The accuracy and reliability of ZenLens in predicting stress levels are further quantified in the interpretation and analysis section, where the system's performance is evaluated against ground truth data.

Table 3. DASS Guide score probabilities.

DASS-21 Scoring	Stress
Normal	0-7
Mild	8-9
Moderate	10-12
Severe	13-16
Extremely Severe	17+

Code Snippets

Image Preprocessing and Face Detection

For each uploaded image, the system resizes and normalizes each detected face, preparing it for emotion analysis. Images are first converted to grayscale, resized to a consistent 48x48 pixels (matching the model's input size), and normalized to values between 0 and 1, helping the model process the input effectively.

```
def preprocess_image(gray: np.ndarray) -> np.ndarray:
    resized = cv2.resize(gray, (IMAGE_SIZE, IMAGE_SIZE), interpolation=cv2.INTER_AREA)
    normalized = resized / NORMALIZATION_FACTOR
    processed_image = np.expand_dims(normalized, axis=-1)
    processed_image = np.expand_dims(processed_image, axis=0)
    return processed_image
```

Figure 21. Code Snippet of Image Preprocessing Function

Here in Figure 21, the image preprocessing function performs resizing and normalization. The RetinaFace.detect_faces function then identifies faces within each image, returning coordinates for cropping.

```
faces = RetinaFace.detect_faces(rgb_image)
```

Figure 22. Code Snippet of Retina Face Function

This function in Figure 22, identifies faces within an image and returns bounding box coordinates, allowing individual facial regions to be isolated and analyzed separately.

Emotion Prediction

Each detected face is fed into a pre-trained emotion detection model to predict the likelihood of various emotions. This step assigns a probability to each emotion, representing the likelihood that the face displays a particular emotion.

```
def predict_emotion(processed_image: np.ndarray) -> np.ndarray:  
    return model.predict(processed_image)
```

Figure 23. Code Snippet of Emotion Prediction Function

The `predict_emotion` function shown in Figure 23 loads the pre-trained model (in this case, `ZFNETModelV6.h5`) and outputs an array of probabilities, one for each emotion label.

Classifying Emotions and Calculating Stress Scores

Using the probabilities output by the model, the system classifies the detected emotion by choosing the highest probability value. Each emotion corresponds to a stress score based on psychological assumptions about its impact on stress.

```
emotion_labels = ['angry', 'disgust', 'afraid', 'happy', 'neutral', 'sad', 'surprise']
emotion_scores = {'happy': 0, 'surprise': 1, 'neutral': 0, 'disgust': 1, 'sad': 2, 'afraid': 3, 'angry': 3}

def classify_emotions(probabilities: np.ndarray) -> (str, int):
    detected_emotion_index = np.argmax(probabilities)
    detected_emotion = emotion_labels[detected_emotion_index]
    stress_score = emotion_scores[detected_emotion]
    return detected_emotion, stress_score

emotion_labels = ['angry', 'disgust', 'afraid', 'happy', 'neutral', 'sad', 'surprise']
emotion_scores = {
    'happy': 0,      # Low stress
    'surprise': 1,   # Mild stress
    'neutral': 0,    # Low stress
    'disgust': 1,    # Mild stress
    'sad': 2,        # Moderate stress
    'afraid': 3,     # High stress
    'angry': 3       # High stress
}
```

Figure 24. Code Snippet of Emotion Classification Function

In Figure 24, `classify_emotions` first finds the emotion label with the highest probability, then maps this label to a corresponding stress score.

This stress score is a predefined value based on common emotional stress levels, where happy and neutral indicate minimal stress, while afraid and angry indicate higher stress levels.

Session-Based Stress Calculation

The stress scores from all faces in each image are summed and averaged for the session, providing an overall assessment of stress for that session. This aggregation makes the stress data more meaningful by capturing the average stress across multiple detected faces.

The scoring of the emotions is assigned by the researchers together with their domain expert. In this mapping:

Low Stress (Score 0): Emotions like "happy" and "neutral" are considered to represent minimal stress.

Mild Stress (Score 1): Emotions like "disgust" and "surprise" are considered mild indicators of stress.

Moderate Stress (Score 2): Emotions like "sad" are associated with moderate stress.

High Stress (Score 3): Emotions like "afraid" and "angry" are considered strong indicators of high stress.

These scores are derived by general emotional responses seen in psychological studies by Hosseini et al [10], where positive or neutral emotions (like happiness and neutrality) correlate with low stress, while negative or intense emotions (like anger, fear, and sadness) correlate with higher stress

levels. This simplistic approach provides a quick estimation of stress based on visible emotions, though it's important to note that real-world stress assessment typically requires more comprehensive analysis, often involving context, behavior, and physiological measures.

The overall average stress score is categorized to provide a simple, interpretable summary of the session's stress level.

```
total_stress = 0
total_faces = 0
for file in files:
    output_path, results, emotion_data = analyze_image(image_path, processed_folder)
    photo_total_stress = sum(result['stress_score'] for result in results.values())
    total_stress += photo_total_stress
    total_faces += len(results)
```

Figure 25. Code Snippet of Stress and Face Count Calculation

In this loop, each image in a session is processed individually:

- `analyze_image` returns the processed image, stress results, and emotion data.
- The stress scores from each detected face are added to `total_stress`, and the total number of faces is tracked with `total_faces`.

Categorizing Stress Levels

The scoring in this system is based on predefined mappings of each detected emotion to a stress score, encapsulated in the `emotion_scores` dictionary. This mapping is a simplified model derived from common psychological associations between emotions and stress levels. Here's the code snippet in Figure 26, defining these mappings:

```
def categorize_stress(average_stress):  
    adjusted_stress = (average_stress / 100) * 21  
    if adjusted_stress <= 7:  
        return "Normal"  
    elif adjusted_stress <= 9:  
        return "Mild"  
    elif adjusted_stress <= 12:  
        return "Moderate"  
    elif adjusted_stress <= 16:  
        return "Severe"  
    else:  
        return "Extremely Severe"
```

Figure 26. Code Snippet of Stress Categorization Function

This study categorized stress levels using the Depression Anxiety Stress Scales (DASS-21), a validated tool for assessing emotional states. The DASS-21 consists of 21 items, with each subscale (Depression, Anxiety, Stress) scored from 0 to 3.

This adjustment scales the stress score to match the DASS-21 framework, allowing categorization into levels: Normal (≤ 7), Mild (≤ 9), Moderate (≤ 12), Severe (≤ 16), and Extremely Severe (> 16). Using the DASS-21 ensures standardized interpretation and comparability of stress levels, enhancing the reliability and validity of the findings.

Here, the function `categorize_stress`:

- Adjusts the `average_stress` score and assigns it to a category (e.g., "Normal" for low stress levels, "Extremely Severe" for high stress levels).
- This categorization simplifies interpretation and can highlight sessions that may need further investigation.

Below is the formula for the Weighted Average Calculation for Weekly Stress:

$$\text{average stress} = \frac{\text{total stress}}{\text{total faces}}$$

Saving Session Data

Session results, including stress scores, categorized stress levels, and emotion details, are stored in JSON format.

This allows the system to retain historical data, supporting future analysis and trend tracking.

```
def save_data(data, file_path=DATA_FILE):  
    with open(file_path, 'w') as file:  
        json.dump(data, file, indent=4)
```

Figure 27. Code Snippet of Save Data Function

The `save_data` function, shown in Figure 27, is essential for storing session results, including stress scores and emotion details, in JSON format. This approach ensures data is both human-readable and machine-processable, facilitating efficient analysis and trend tracking over time. By preserving historical data, the system supports longitudinal studies, enabling researchers to identify patterns and correlations in stress and emotional states. The use of `indent=4` enhances data readability, aiding in manual inspections. Overall, this function underscores the importance of robust data storage for effective research and analysis.

Weekly Aggregates and Emotion Timeline

In Figure 28, weekly summaries are created by grouping data for easier analysis of long-term trends. This allows users to monitor changes in stress and emotions across multiple sessions.

```
def calculate_weekly_averages(grouped_data):
    weekly_averages = []
    for week in grouped_data:
        total_stress = sum(entry['average_stress'] * entry['total_faces'] for entry in week)
        total_faces = sum(entry['total_faces'] for entry in week)
        average_stress = total_stress / total_faces if total_faces > 0 else 0
        weekly_averages.append({
            'week': f"{start_date} - {end_date}",
            'average_stress': average_stress
        })
    return weekly_averages

def calculate_weekly_averages(grouped_data):
    weekly_averages = []

    for week in grouped_data:
        total_faces = sum(entry['total_faces'] for entry in week)
        total_stress = sum(entry['average_stress'] * entry['total_faces'] for entry in week)
        sessions_count = len(week)
        average_stress = total_stress / total_faces if total_faces > 0 else 0
        stress_category = categorize_stress(average_stress)
```

Figure 28. Code Snippet of Weekly Average Calculation

- This function groups data by week and calculates weekly averages for stress.
- Weekly averages enable monitoring of stress levels over extended periods, helping identify patterns or changes in stress over time.

Emotion Timeline Retrieval

Figure 29 illustrates that the application provides endpoints to retrieve both daily and weekly summaries of emotions, useful for tracking fluctuations in emotional states over time.

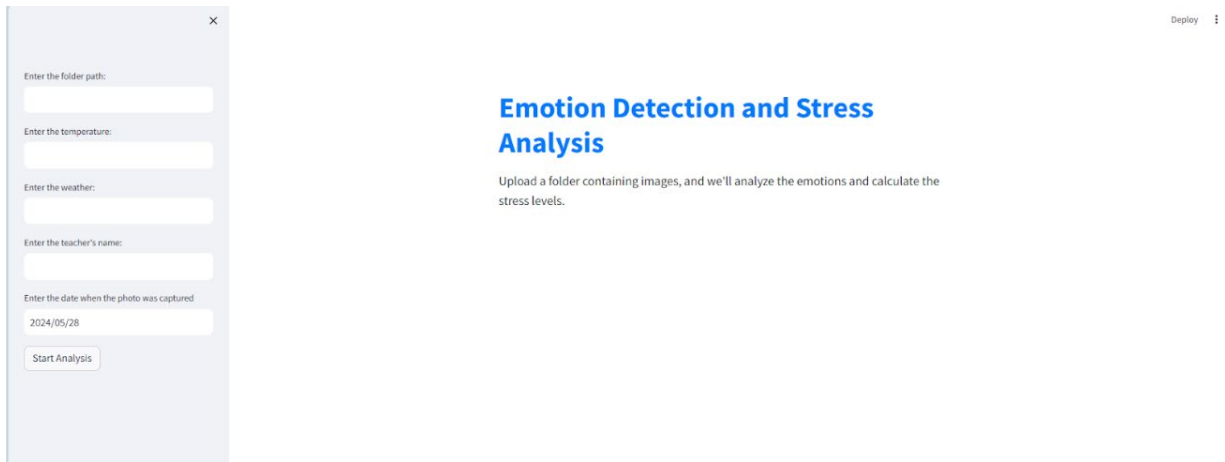
```
@app.route('/weekly-emotion-timeline', methods=['GET'])
def weekly_emotion_timeline():
    data = load_emotion_timeline()
    data.sort(key=lambda entry: datetime.fromisoformat(entry['timestamp']))
    weekly_summary = []
    # (further logic to group and summarize weekly emotions)
    return jsonify(weekly_results)
```

Figure 29. Code Snippet of Emotion Timeline

This endpoint generates a weekly summary of emotions, tracking the frequency of each detected emotion. Monitoring the emotion timeline over time can provide insights into the factors impacting stress and well-being. By combining face detection, emotion classification, stress categorization, and historical data management, this code allows users to track and analyze stress levels over multiple sessions, supporting deeper insights into emotional and stress patterns over time.

Web application Development and Model Deployment

The development of the ZenLens web application involved several stages, each aiming to enhance the system's functionality and user experience. This section outlines the process of creating the initial and improved versions of the application and details the software and hardware requirements necessary for its deployment.



The screenshot shows a web application titled "Emotion Detection and Stress Analysis". On the left, there is a sidebar with a close button (X) at the top. The sidebar contains several input fields: "Enter the folder path:", "Enter the temperature:", "Enter the weather:", "Enter the teacher's name:", and "Enter the date when the photo was captured" (with a date picker showing "2024/05/28"). Below these fields is a "Start Analysis" button. On the right, the main content area has the title "Emotion Detection and Stress Analysis" in blue, followed by a description: "Upload a folder containing images, and we'll analyze the emotions and calculate the stress levels." In the top right corner of the main area, there is a "Deploy" button with a small icon.

Figure 30. First Iteration of the Algorithm Prototype via Streamlit

Figure 30 depicts the initial prototype of the emotion detection and stress analysis algorithm implemented using Streamlit. This early interface version is a foundational step in developing a user-friendly tool for analyzing emotions and stress levels from images. The prototype presents a simple layout with input fields for various parameters. While this initial interface provides a basic framework for data input, it must be noted that it has yet to be finalized and requires further refinement to enhance usability and user experience. The UI design and layout are still under development to optimize interaction and information presentation effectively.

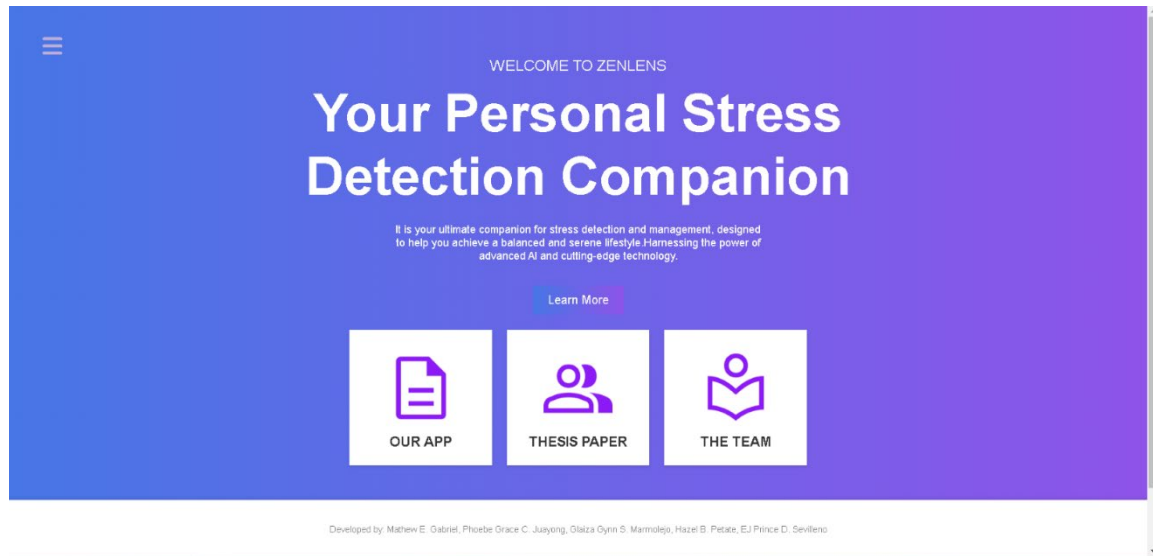


Figure 31. Second Iteration of the Web Application

Based on the feedback and insights from the first version, the ZenLens web application's second interaction in Figure 31 incorporated several improvements. These enhancements included a more intuitive user interface, faster data processing capabilities, and additional features for personalized stress management recommendations. The second version of the app aimed to provide a more seamless and user-friendly experience while maintaining the accuracy and reliability of stress assessments.

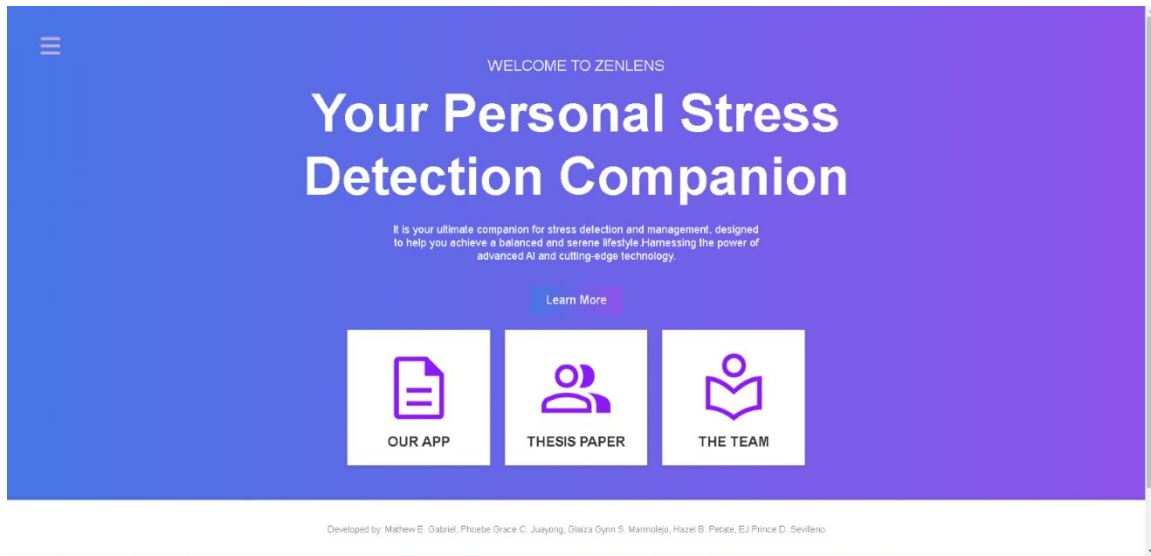


Figure 32. Second Iteration of the Web App: Home Page 1

The initial design of the ZenLens web application's homepage was developed using Streamlit. This iteration in Figure 32 is a foundational step in creating a user-friendly stress detection and management tool interface. The homepage features a clean and visually appealing layout with a prominent header introducing ZenLens as a "Personal Stress Detection Companion." Key elements include a brief description of the application's purpose, a call-to-action button labeled "Learn More," and icons representing the app, thesis paper, and team.

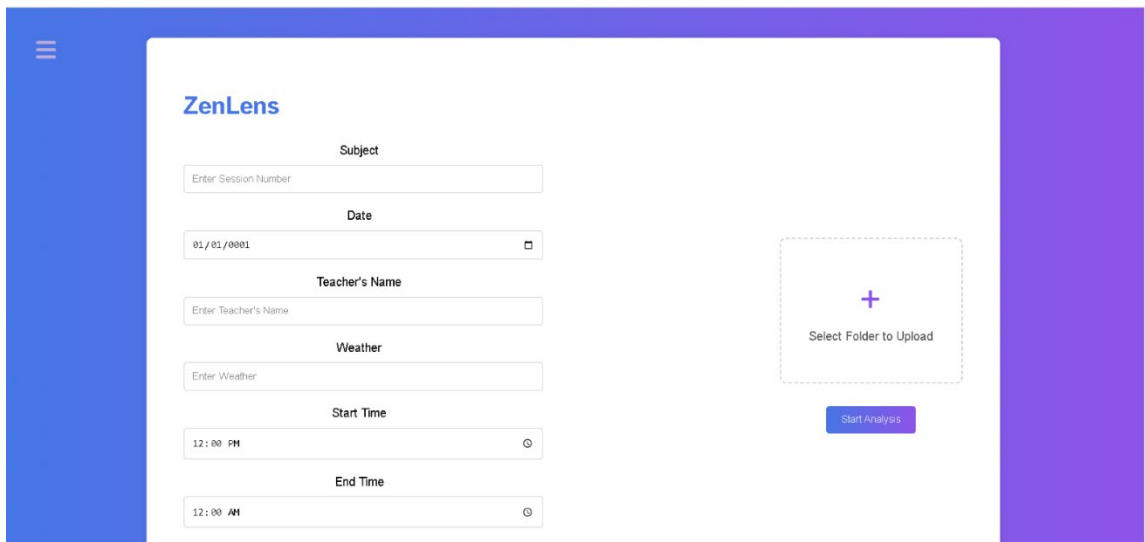


Figure 33. Second Iteration of the Web Application: Stress Detection Page

Figure 33 shows the ZenLens web application's second iteration stress detection page. This interface is designed to collect essential information for the stress analysis process. Key elements include input fields for subject details, date, teacher's name, weather conditions, and session timings. A prominent "Select Folder to Upload" button allows users to choose image or video data for analysis. Once all necessary information is provided, the "Start Analysis" button initiates the stress detection process. This page presents a user-friendly layout with clear labels, facilitating data input and the subsequent analysis of uploaded media.

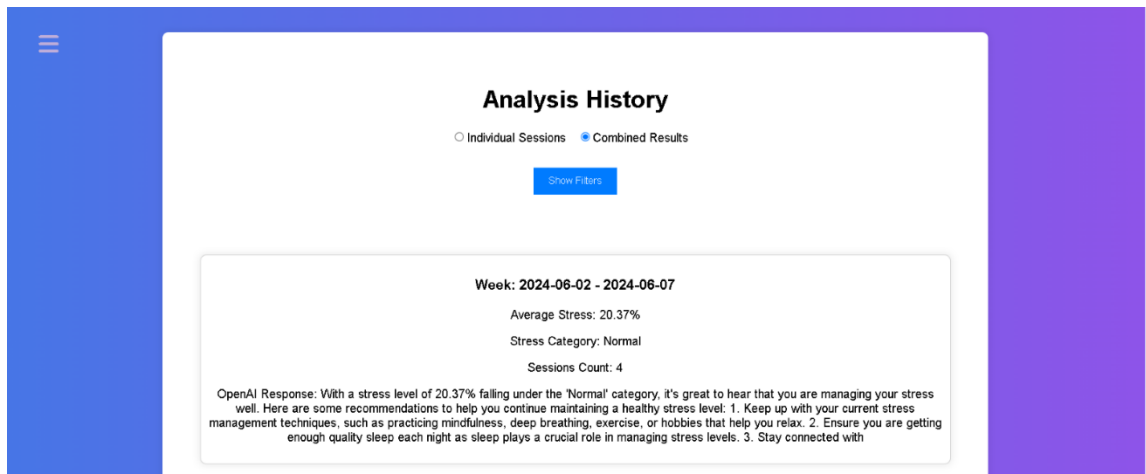
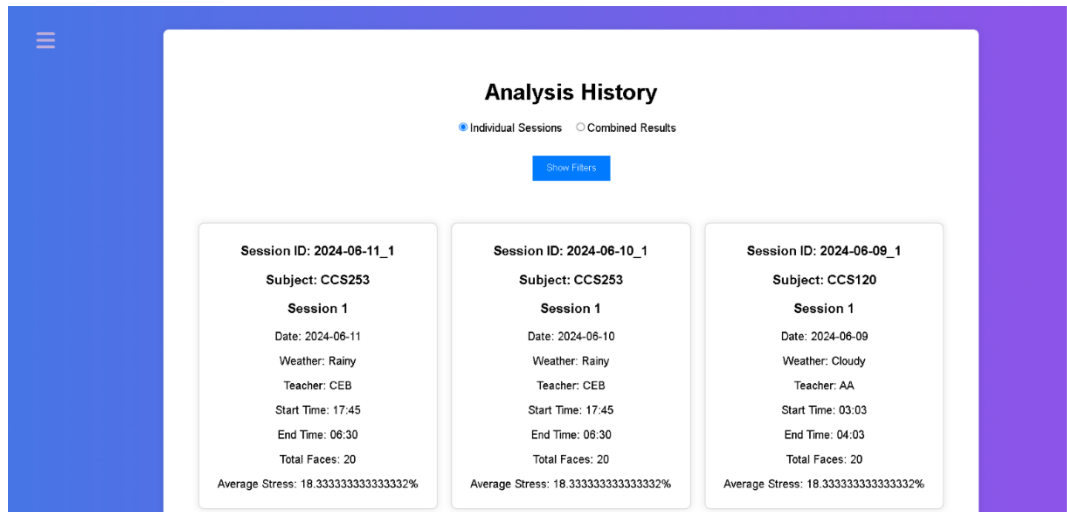


Figure 34. Second Iteration of the Web Application: Stress History

Figure 34 presents the "Analysis History" page of the web application, designed to provide users with a comprehensive overview of their stress levels. The page offers two primary views: "Individual Sessions" and "Combined Results." The "Individual Sessions" tab displays detailed information for each stress analysis session, including session ID, subject,

date, weather conditions, teacher, session timings, number of faces analyzed, and average stress level. The "Combined Results" tab summarizes stress levels over a specific week, including the average stress level, stress category, and total sessions analyzed. Additionally, the page includes a text box for displaying general information or recommendations related to stress management. This feature enables users to track stress patterns, identify trends, and make informed decisions about stress reduction strategies.

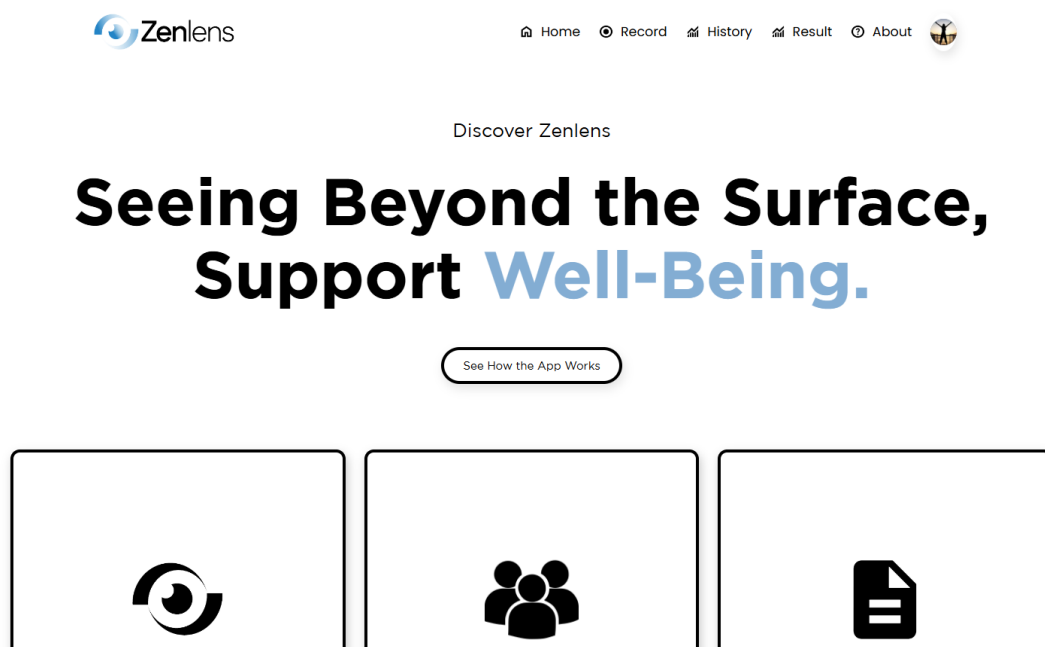


Figure 35. Final Iteration of the Web Application: Home

The final iteration of the ZenLens homepage in Figure 35 is designed to provide a streamlined and inviting user experience that effectively conveys the purpose of the

application. Dominated by the tagline "Seeing Beyond the Surface, Support Well-Being," the homepage communicates ZenLens's core mission of using advanced emotion detection to enhance mental health and support user well-being. The tagline is presented in a bold and visually appealing style, with "Well-Being" highlighted in a calming blue color to emphasize its importance.

Below the main tagline, a prominent button labeled "See How the App Works" encourages new users to explore the application's functionality and learn more about its capabilities. This call-to-action provides a guided introduction to ZenLens, ensuring users can understand its potential benefits from the outset.

The screenshot shows the ZenLens web application interface. At the top, there is a navigation bar with links for Home, Record, History, Result, and About. The main content area is split into two panels. The left panel, titled 'Enter Information', contains several input fields: 'Subject' (333), 'Room Number' (33333), 'Teacher's Name' (333), 'Start Time' (11:11 am), 'Date' (10/02/2024), 'Weather' (22), and 'End Time' (12:22 pm). The right panel, titled 'Upload Files and Analyze', has a large dashed box for file selection. Above the box is a 'Choose Files' button and the text 'No file chosen'. Inside the box, there is a '+ Select Folder to Upload' link. Below the box is an 'Analyze' button.

Figure 36. Final Iteration of the Web Application: Upload Page

The Upload Page in the final iteration of the ZenLens web application in Figure 36, is designed to simplify the process of submitting media files for emotion analysis. On the left side of the page, users can enter essential session details, such as "Subject," "Room Number," "Teacher's Name," "Start Time," "Date," "Weather," and "End Time." These fields allow users to provide contextual information that enhances the accuracy and relevance of the analysis.

The central area of the page is dedicated to file upload, with a clearly marked box labeled "Upload Files and Analyze." Here, users can either select individual files or entire folders to upload, using the "Select Folder to Upload" option.

Once the files are chosen, the "Analyze" button at the bottom of the page initiates the processing of the uploaded media, allowing ZenLens to detect and classify emotions from the captured images or videos.

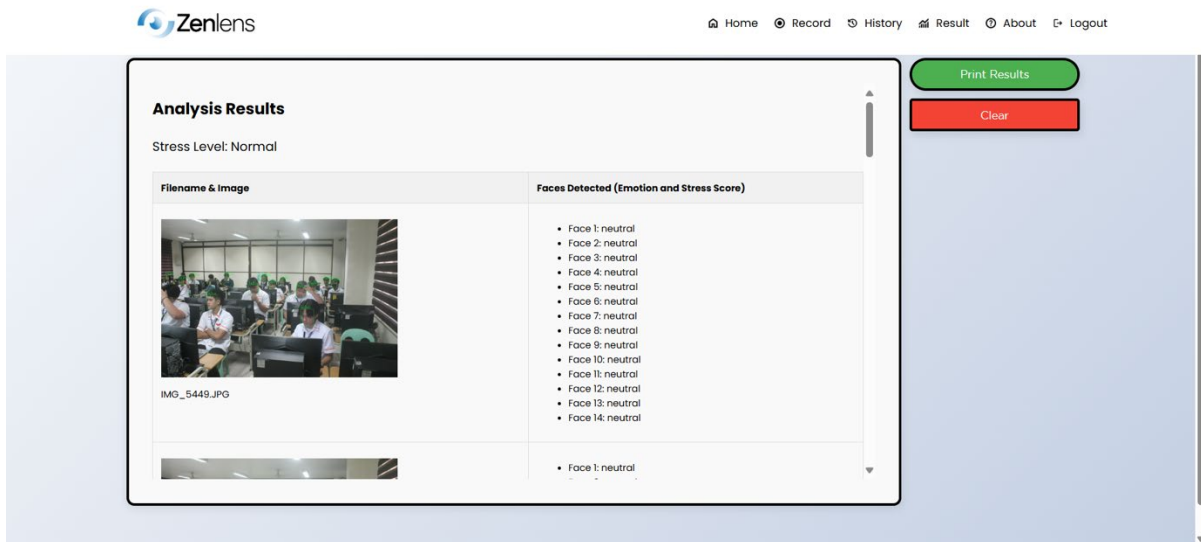
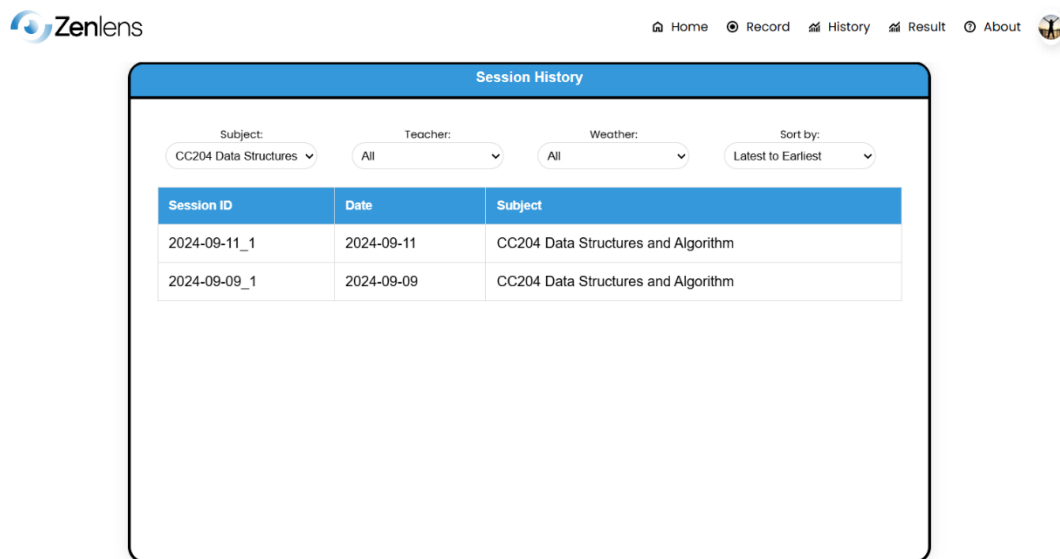


Figure 37. Final Iteration of the Web Application: Analysis Results

The Analysis Results page in the final iteration of the ZenLens web application in Figure 37 presents users with a detailed overview of emotion detection outcomes based on the uploaded media. This page displays the identified emotions, along with their frequency and intensity, for each individual detected in the analyzed images or videos. Users can view visual representations of emotions across various time frames, including session, daily, or weekly summaries, providing a

comprehensive view of emotional trends. Graphs and charts offer clear insights into the prevalence of each emotion, enabling users to track patterns over time. The streamlined design and data visualizations make it easy for users to interpret the results, fostering an understanding of emotional dynamics that can support well-being initiatives and informed decision-making. This page serves as a core feature of ZenLens, turning emotion data into actionable insights.



Session History			
Subject:	Teacher:	Weather:	Sort by:
CC204 Data Structures	All	All	Latest to Earliest
Session ID	Date	Subject	
2024-09-11_1	2024-09-11	CC204 Data Structures and Algorithm	
2024-09-09_1	2024-09-09	CC204 Data Structures and Algorithm	

Figure 38. Final Iteration of the Web Application: Session History

The Session History page in the final iteration of the ZenLens web application in Figure 38 provides users with a comprehensive log of all past emotion detection sessions. The

page is organized as a table displaying key details for each session, including "Session ID," "Date," and "Subject." This layout allows users to quickly scan and locate specific sessions based on their identifiers or the date of analysis.

Above the table, there is a "Show Filters" button, allowing users to apply filters to narrow down sessions by criteria such as date range, subject, or other metadata. This feature enhances usability by enabling users to focus on sessions of interest, facilitating more efficient data analysis.



Figure 39. Final Iteration of the Web Application: Session Details

The Session Details page in the final iteration of the ZenLens web application shown in Figure 39, provides an in-depth view of a specific emotion detection session. The top of the page displays essential session metadata, including "Subject," "Session" number, "Date," "Weather," "Teacher," "Start Time," and "End Time." This contextual information helps users understand the conditions under which the session took place, potentially contributing to the interpretation of emotional trends.

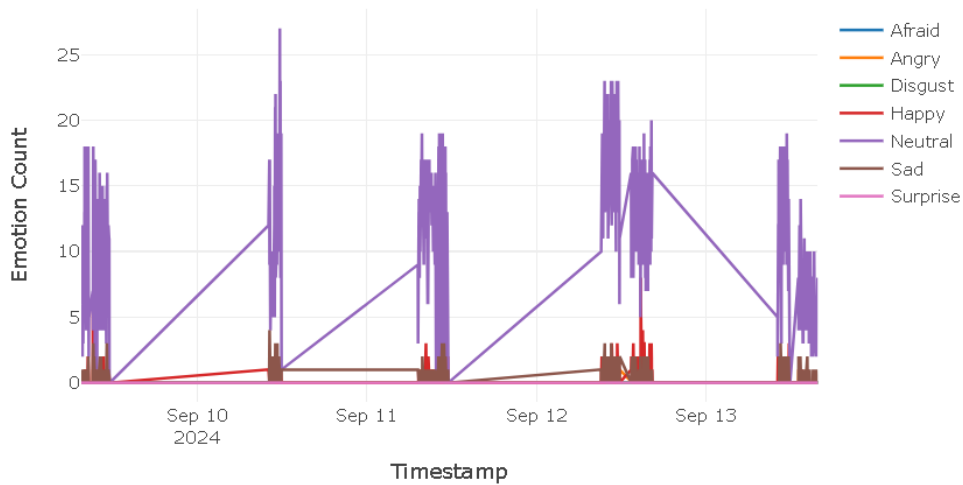
Below the session details, an "Emotion Timeline" chart visualizes the detected emotions over the course of the session. Each line represents one of the seven universal emotions—afraid, angry, disgust, happy, neutral, sad, and surprise—tracked by the ZFNet-enhanced ZenLens model. The chart shows emotion counts on the y-axis and time on the x-axis, allowing users to observe how emotional states fluctuated throughout the session. For example, in this session, "Neutral" and "Sad" emotions show a noticeable trend, with "Neutral" increasing steadily and "Sad" remaining relatively low.



Weekly Details for 2024-09-09 - 2024-09-13



Emotion Levels for 2024-09-09 - 2024-09-13



Recommendation: The stress level this week is: Normal. Advice: It's great to hear that your stress level falls under the 'Normal' category! Here are some recommendations to help you maintain this healthy stress level: 1. Keep up with your current stress management practices, whether it's exercise, meditation, hobbies, or spending time with loved ones. Consistency is key in maintaining a healthy stress level. 2. Stay mindful of potential stress triggers and make a conscious effort to address them before they escalate.

Figure 40. Emotion Levels Analysis with Stress Management Recommendation

Figure 40 presents an analysis of emotion levels and stress management recommendations for the week of September 9, 2024 to September 13, 2024. The data highlights the distribution of various emotions, including "afraid," "angry," "disgust," "happy," "neutral," "sad," and "surprise," across the week. The predominant emotion observed was "neutral,"

which consistently appeared throughout the week, suggesting a generally stable emotional state.

The stress level for the week was categorized as "normal," indicating that the participants maintained a healthy balance. To support this positive state, recommendations were provided to reinforce existing stress management practices.

The analysis underscores the value of continuous monitoring and personalized recommendations in stress management. By tracking emotional trends and providing tailored advice, individuals can better maintain their well-being and address potential stressors before they become significant issues. This approach aligns with the broader goal of promoting long-term mental health and resilience.

Development Tools

In building the facial emotion recognition (FER) model, the developers utilized Python as the primary programming language, along with three integrated development environments (IDEs): Visual Studio Code, Google Colab, and Jupyter Notebook. The ZFNet architecture was used for emotion classification, while RetinaFace enabled precise face

detection. Google Colab provided GPU acceleration for faster model training, while Jupyter Notebook facilitated real-time debugging and visualization. The backend was developed using Flask, integrating the trained model into a React.js web application, ensuring an interactive and user-friendly experience. MySQL was used for data storage, while Docker containerized the application for deployment on cloud platforms like AWS and Google Cloud.

To ensure the smooth training and execution of the model, the system was tested on a machine with an Intel Core i5-10300H processor, 8GB RAM, and an NVIDIA GeForce RTX 2060 GPU. The software environment included TensorFlow 2.15.0, Keras 2.15.0, RetinaFace 0.0.17, and Python 3.9 or higher. Git and GitHub were used for version control, maintaining an organized development process. These tools collectively enabled ZenLens to accurately detect stress levels through facial expression analysis, providing a scalable and reliable system for student mental health monitoring.

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Hardware Requirements

The following are the minimum hardware requirements for training the model.

Table 4. Hardware Requirements

Hardware Component	Specifications
CPU (Central Processing Unit)	Intel(R) Core (TM) i5-10300H
RAM (Random Access Memory)	8.00 GB
GPU (Graphics Processing Unit)	NVIDIA GEFORCE RTX 2060
Storage Space	100GB

Software Requirements

The following are the minimum software requirements for training the model.

Table 5. Software Requirements

Software Component	Specifications
Operating System	Microsoft Windows 10
Programming Language	Python 3.9 or higher
IDE (Integrated Development Environment)	Visual Studio Code (Version: 1.89.1) Google Colab Jupyter Notebook (Version: 7.0.8)
Tensorflow	Version: 2.15.0
Keras	Version: 2.15.0
RetinaFace	Version: 0.0.17

Input and Outputs of the Study

The ZenLens system functions by processing images to detect facial expressions, classify emotions, and assess stress levels. The input data in Figure 41 primarily consists of facial images captured during classroom sessions, taken at 90-second intervals. These images serve as the primary source for emotion recognition and stress analysis. Additionally, the system collects session metadata, including details such as the subject's name, classroom number, instructor's name, start and end time of the session, date, and weather conditions. These contextual details enhance the analysis by linking emotional patterns with external factors that may influence student stress levels.

The screenshot displays the ZenLens web application interface. At the top left is the ZenLens logo. A navigation bar at the top right contains links: Home, Record, History, Result, About, and Logout. The main interface is divided into two panels. The left panel, titled 'Enter Information', contains several input fields: 'Subject' (with a placeholder 'Enter Subject Name or Code'), 'Room Number' (with a placeholder 'Enter Room Number'), 'Teacher's Name' (with a placeholder 'Enter Teacher's Name'), 'Start Time' (with a time picker), 'Date' (with a date picker), 'Weather' (with a placeholder 'Enter Weather'), and 'End Time' (with a time picker). The right panel, titled 'Upload Files and Analyze', features a 'Choose Files' button, a status 'No file chosen', a large dashed blue box for file selection, a '+ Select Folder to Upload' link, and an 'Analyze' button at the bottom.

Figure 41. Web Application's Input Page

The outputs of the study in Figure 42 include emotion classifications, stress level categorizations, and visual analytics. The system classifies facial expressions into seven universal emotions: Happy, Sad, Angry, Afraid, Surprise, Disgust, and Neutral. After classification, ZenLens further categorizes detected emotions into three stress-related groups: "Stressed," "Neutral," and "Non-Stressed". The system then calculates overall stress scores based on the Depression, Anxiety, and Stress Scale (DASS-21) and assigns a severity level ranging from Normal to Extremely Severe.

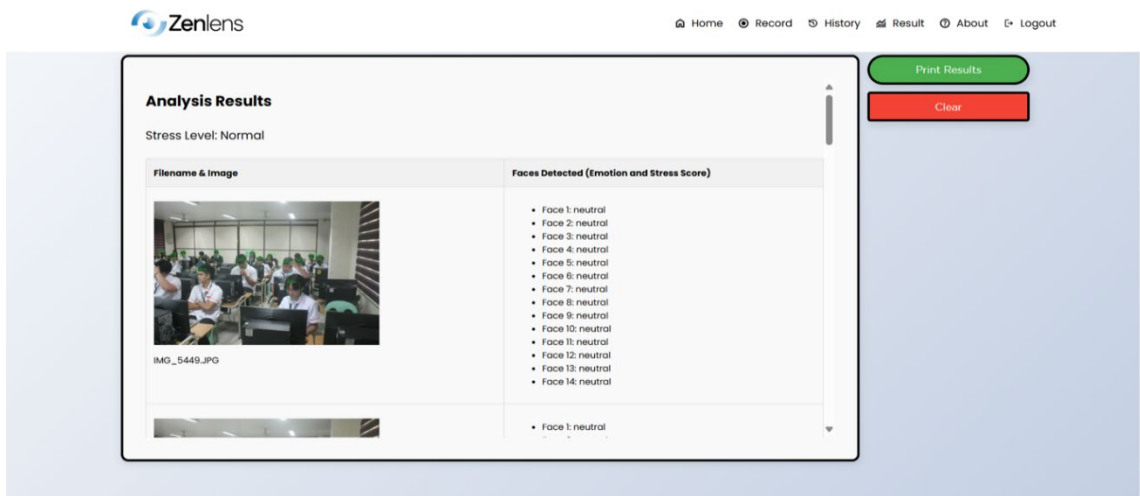


Figure 42. Web Application's Sample Analysis Results Page

The final output is visualized through graphs and reports, including session-based, daily, and weekly stress

trend analysis. These reports allow educators and administrators to monitor emotional patterns in classrooms over time, offering valuable insights into student well-being.



Daily Details for 2024-09-10

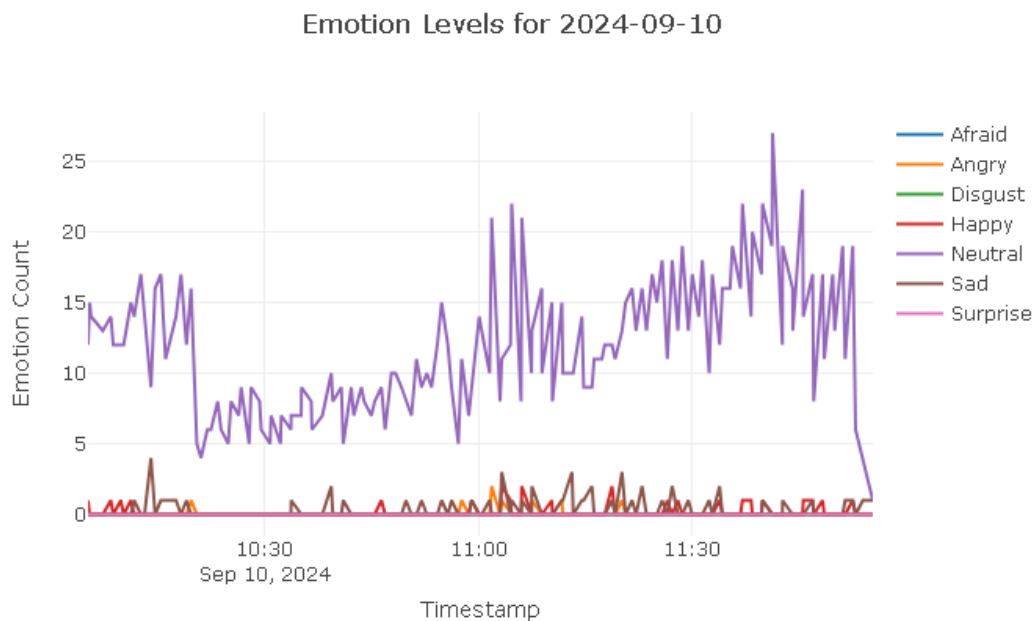


Figure 43. Session Details Page

In Figure 43, ZenLens provides real-time emotion detection results, displayed through the web application interface. Each session's results include the number of faces detected, the dominant emotions present, and the corresponding stress levels. Additionally in Figure 44, users can access historical data, allowing for trend analysis and early detection of potential academic stressors. By offering an

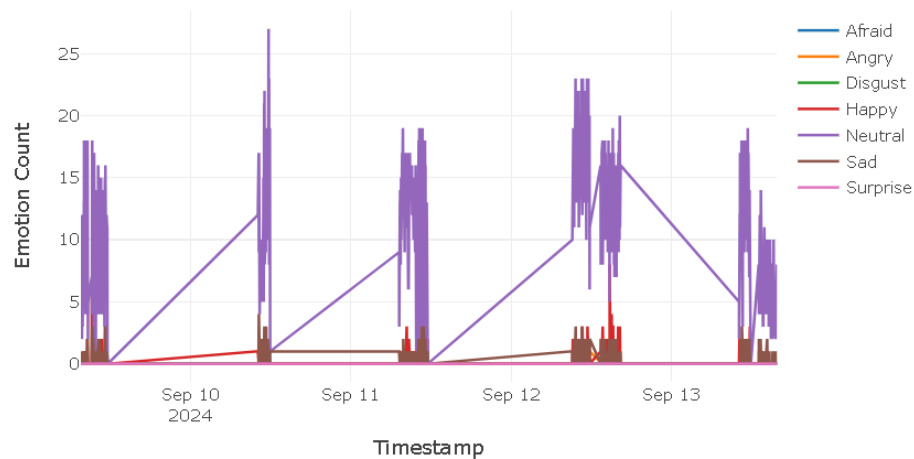
automated and non-invasive approach to stress monitoring, the system aids in fostering a mentally healthy learning environment for students.



Weekly Details for 2024-09-09 - 2024-09-13



Emotion Levels for 2024-09-09 - 2024-09-13



Recommendation: The stress level this week is: Normal. Advice: It's great to hear that your stress level falls under the 'Normal' category! Here are some recommendations to help you maintain this healthy stress level: 1. Keep up with your current stress management practices, whether it's exercise, meditation, hobbies, or spending time with loved ones. Consistency is key in maintaining a healthy stress level. 2. Stay mindful of potential stress triggers and make a conscious effort to address them before they escalate.

Figure 44. Emotion Levels Analysis Page

Results Interpretation and Analysis

Emotion Detection Model Performance

The number of epochs representing complete cycles of training the facial landmark detection model on the dataset significantly influences its performance. During the initial training stages shown in Figure 45 - 49, the average distance error might decrease rapidly, and the localization rate might increase significantly. However, as training progresses beyond 100 epochs, additional epochs might yield more minor improvements or lead to overfitting (poor performance on unseen faces).

The metrics based on the training performance shown by the following performance plots raise the fact that though the model is learning from the training data, as shown by the increasing trend in the training accuracy values, there are vast oscillations of the validation accuracy, indicating towards overfitting and poor cross-validation performance on the data which have not have been used in the training phase. Even the graphic representations of precision and recall provide a somewhat ambiguous picture, suggesting that the model's stability is relatively low and its ability to identify and provide coverage of positive instances varies

greatly. The confusion matrix presents additional difficulties, as there are many incorrectly classified instances, and the number of mistakes is high for the seven universal emotions; therefore, it can be concluded that the model falls short in this respect. To fix these problems and improve the networking, some best practices like regularization, data augmentation, and implementation of hybrid architecture with ResNet or ZFNet would help.

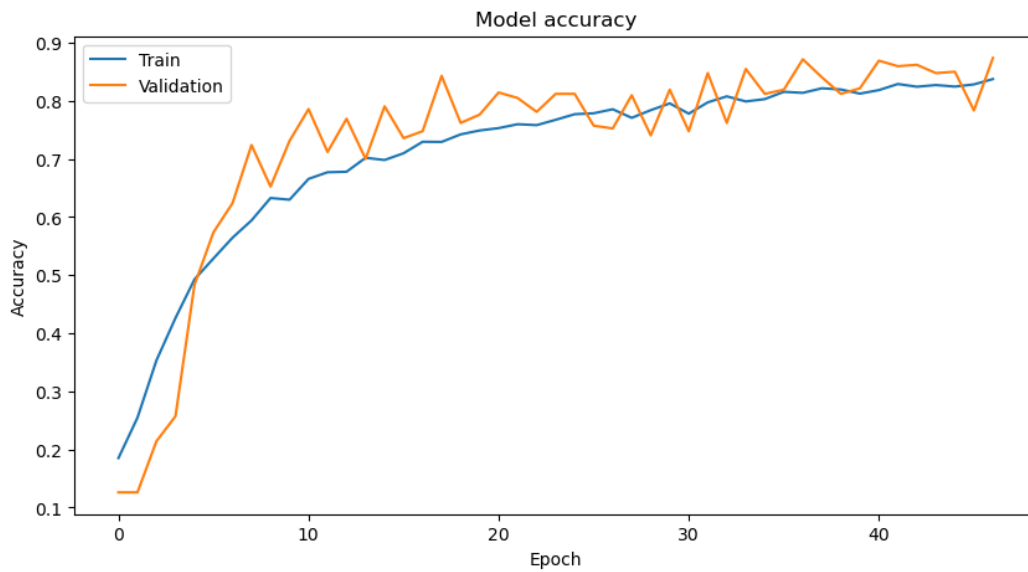


Figure 45. Accuracy

Figure 45 shows the accuracy of the model over 45 epochs, with separate lines for training and validation accuracy. The training accuracy steadily increases, reaching above 80% by the end of training, indicating that the model is learning and improving its ability to correctly classify emotions. The

validation accuracy, however, fluctuates around 82% after some initial improvement and fluctuates a bit. This variability suggests that while the model performs well on the training data, it struggles to generalize consistently on the validation data, possibly indicating overfitting.

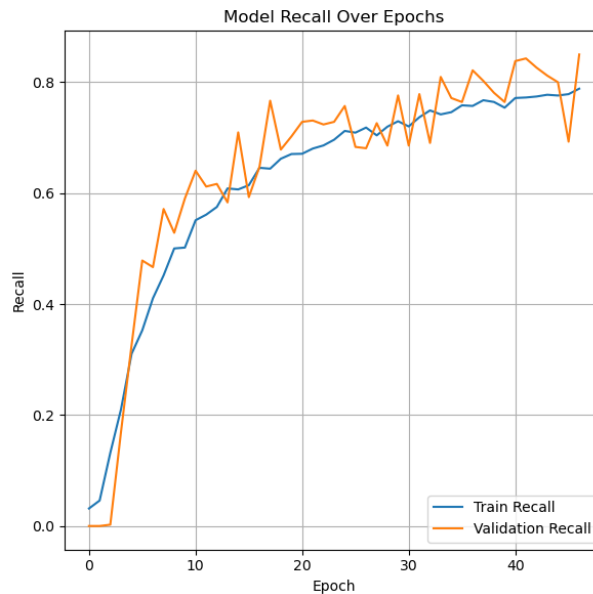


Figure 46. Recall

The recall graph in Figure 46 shows initial rapid improvement in recall for both training and validation sets, indicating the model quickly learns to identify relevant instances. However, the validation recall levels off around 0.7 to 0.8 and exhibits fluctuations, suggesting inconsistent performance on unseen data. While the training recall continues to improve, reaching nearly 0.8, the gap between

training and validation recall indicates potential overfitting. This means that while the model becomes increasingly good at identifying relevant instances in the training data, it struggles to generalize as effectively to the validation data, leading to variable performance.

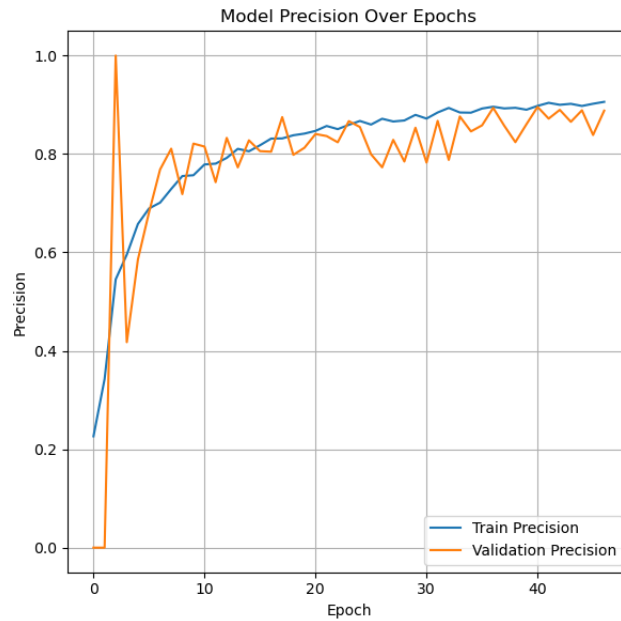


Figure 47. Precision

The precision graph in Figure 47 reveals significant fluctuations in validation precision during early training stages, with a spike at the first epoch. After this, it generally settles between 0.8 and 0.9, with some variability. Training precision, on the other hand, shows a stable increase, reaching above 0.9. The difference between training and validation precision suggests that while the model becomes

more precise on training data, maintaining consistent precision on unseen data remains a challenge. The fluctuating precision values suggest the model's classifications are not consistently precise.

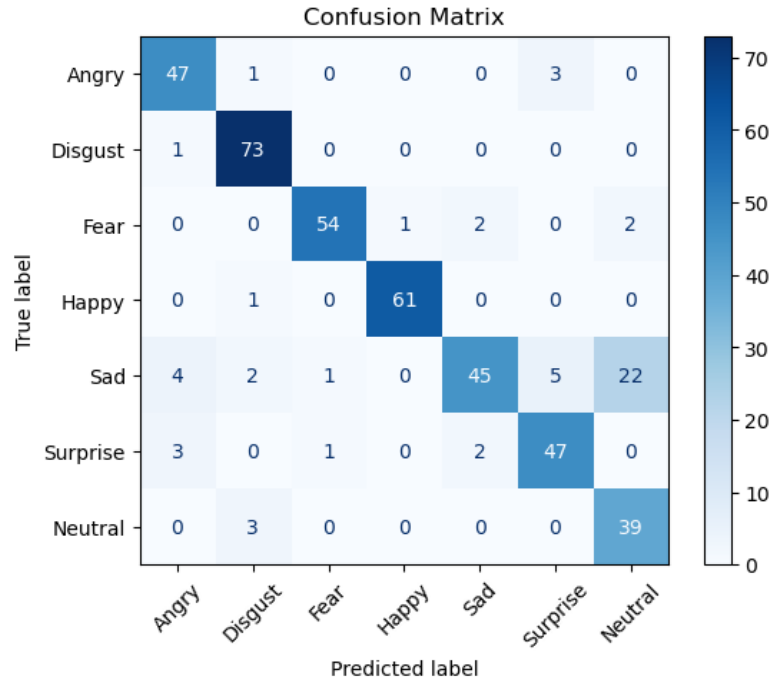


Figure 48. Confusion Matrix

The confusion matrix in Figure 48 illustrates the model's performance in classifying each of the seven emotions. It shows that certain emotions, like "disgust" shows strong performance with high correct classifications (73), while "sad" also performs relatively well. However, there are instances of confusion between certain emotions. "Angry" is sometimes misclassified as "sad" (4 instances) or "neutral" (3

instances), and "sad" is frequently confused with "surprise" (22 instances) and other emotions. "Fear" also shows some misclassifications, particularly with "happy" and "sad." "Neutral" is primarily classified correctly but shows some confusion with "Disgust." This indicates the model struggles to differentiate between these specific emotions, highlighting potential areas for improvement.

System Evaluation Results

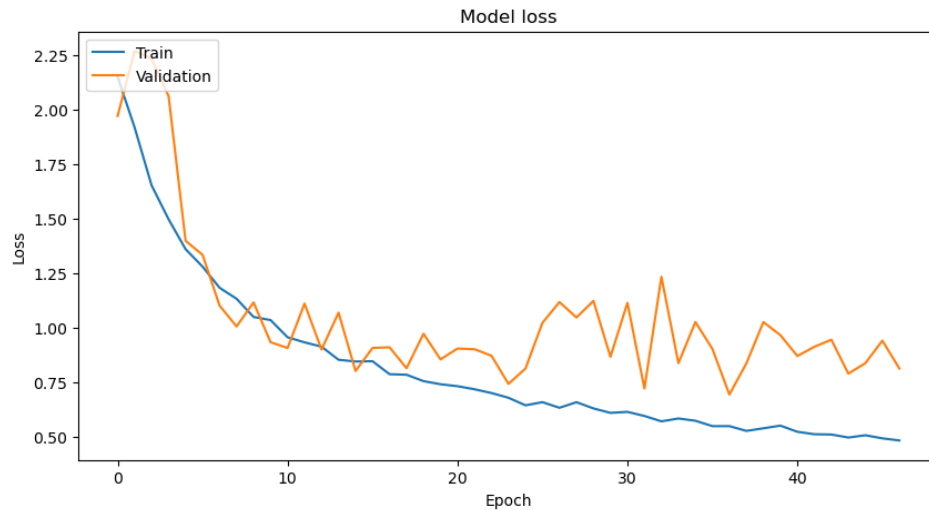


Figure 49. Loss over epoch of classifier

The loss graph in Figure 49 displays the training and validation loss over 45 epochs. Training loss steadily declines, demonstrating the model's ability to reduce errors on the training data. Validation loss initially falls but then levels off and fluctuates, suggesting that despite learning from the training data, the model has difficulty consistently generalizing to the unseen validation data. The increasing difference between training and validation loss points to potential overfitting, where the model performs well on the training set but less consistently on the validation set.

Accuracy of the Preliminary Model

The initial training of the ZenLens model provided an essential baseline in detecting emotions, giving us insight into the model's early strengths and limitations. Testing across different datasets highlighted a range of accuracy levels, revealing both promising aspects of the model and areas that need further attention. Each dataset comprised a total count of detected faces, broken down into correctly identified emotions (/) and mismatches (X). To calculate accuracy, the researchers divided the number of correctly identified faces by the total number of faces in each dataset, then multiplied by 100 to express it as a percentage.

For example, if 789 faces out of 824 were correctly identified, the accuracy would be calculated as:

$$\begin{aligned} &= \left(\frac{789}{824} \right) \cdot 100 \\ &= 95.75 \\ &\approx 95.75\% \end{aligned}$$

yielding an accuracy of 95.75%. Similarly, the mismatch rate was calculated by dividing the number of mismatched faces by the total number of faces and then multiplying by 100; if 35 out of 824 faces were mismatched, the mismatch rate would be:

$$= \left(\frac{35}{824} \right) \cdot 100$$

$$= 4.248$$

$$\approx 4.248$$

This approach gave us a straightforward measure of the model's performance across different datasets, allowing us to identify both consistent accuracy levels and specific areas needing refinement.

For example, in CC204, the model detected 824 faces and accurately classified 789 of them, resulting in an impressive accuracy of 95.75% and only a 4.25% mismatch rate. CIT204, with 2,392 faces, showed an accuracy of 89.86%, indicating a high level of reliability with a relatively low 5.60% mismatch rate. However, as we looked at other datasets, the results varied. CC207, for instance, had 1,887 faces with 1,644 correctly classified, which translated to an accuracy of 87.13% and a 12.87% mismatch rate. Larger datasets, like CIT207 with 3,907 faces, yielded an accuracy of 85.89%, still solid but with a higher 14.11% mismatch rate, indicating room for improvement in handling larger volumes.

Some datasets, like SS110 and SS112, highlighted the model's limitations in dealing with emotions that may appear

subtle or ambiguous. SS110 showed 80.10% accuracy with a 19.90% mismatch rate, while SS112 was lower, with only 70.28% accuracy and a significant 29.72% mismatch rate. These results signal that while the model performs well under certain conditions, it struggles with emotions that are less distinct or more context-dependent.

These preliminary results provide an essential benchmark, reinforcing the need for fine-tuning and data diversity in achieving reliable emotion detection in future ZenLens iterations.

ZFNet Performance

ZFNet's integration into the ZenLens model significantly improved emotion classification as evidenced by the validation metrics. The model achieved an accuracy of 87.14%, demonstrating strong overall performance in correctly identifying emotions. The training process was configured to run for a maximum of 100 epochs. However, early stopping was implemented with the following parameters:

EarlyStopping(monitor='val_loss', patience=10, restore_best_weights=True). As a result, the training halted at the 44th epoch, achieving an accuracy of 87.14%.

Precision demonstrating strong overall performance in correctly identifying emotions. Precision reached 89.38%, indicating a high rate of true positives and minimal false positives, while recall stood at 82.14%, reflecting the model's ability to capture most relevant instances. The AUC score of 0.9616 further underscores the model's ability to distinguish between different emotions.

The validation loss of 0.6918 confirms the model's robustness, with relatively low error rates during evaluation. However, the confusion matrix revealed some challenges, such as misclassifications between "angry" and "sad" (4 instances) and "sad" and "surprise" (22 instances), indicating difficulties in differentiating closely related emotions. Despite these challenges, ZFNet's integration has significantly enhanced the model's accuracy and reliability.

While the high accuracy and precision metrics highlight ZFNet's effectiveness, the slightly lower recall and occasional misclassifications suggest areas for improvement, particularly in handling overlapping emotions.

Stress level Analysis Performance

ZenLens's stress level analysis showed strong performance, with training accuracy exceeding 80% and validation accuracy stabilizing around 82% (Figure 45). However, validation recall fluctuated between 0.7 and 0.8, and validation precision varied between 0.8 and 0.9, indicating challenges in generalizing to unseen data (Figures 46-47). The confusion matrix (Figure 48) revealed accurate classifications for emotions like "disgust" (73 correct classifications) but difficulties in distinguishing between overlapping emotions, such as "angry" and "sad" (4 misclassifications) and "sad" and "surprise" (22 misclassifications).

While the model's accuracy surpasses previous methods (typically around 75%), the fluctuating validation loss (Figure 49) suggests potential overfitting. These results highlight ZenLens's potential as a reliable tool for stress assessment, though improvements in handling overlapping emotions and reducing overfitting are needed for more consistent performance.

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Stress Level Analysis by Session

This section presents the analysis of average stress levels across different subjects based on ZenLens's emotion detection results. The table below summarizes the average stress percentages and corresponding stress level classifications for each subject.

Table 6. Average Stress Levels and Stress Classifications by Session Using ZenLens

Subject Session	Average Stress	Stress level
SS110	1.12%	Normal
SS112	0.97%	Normal
CC207	2.40%	Normal
CIT204	1.51%	Normal
CIT207	3.07%	Normal
CC204	1.25%	Normal
CIT206 - Monday	2.24%	Normal
CIT205	2.55%	Normal
CIT206	4.24%	Normal

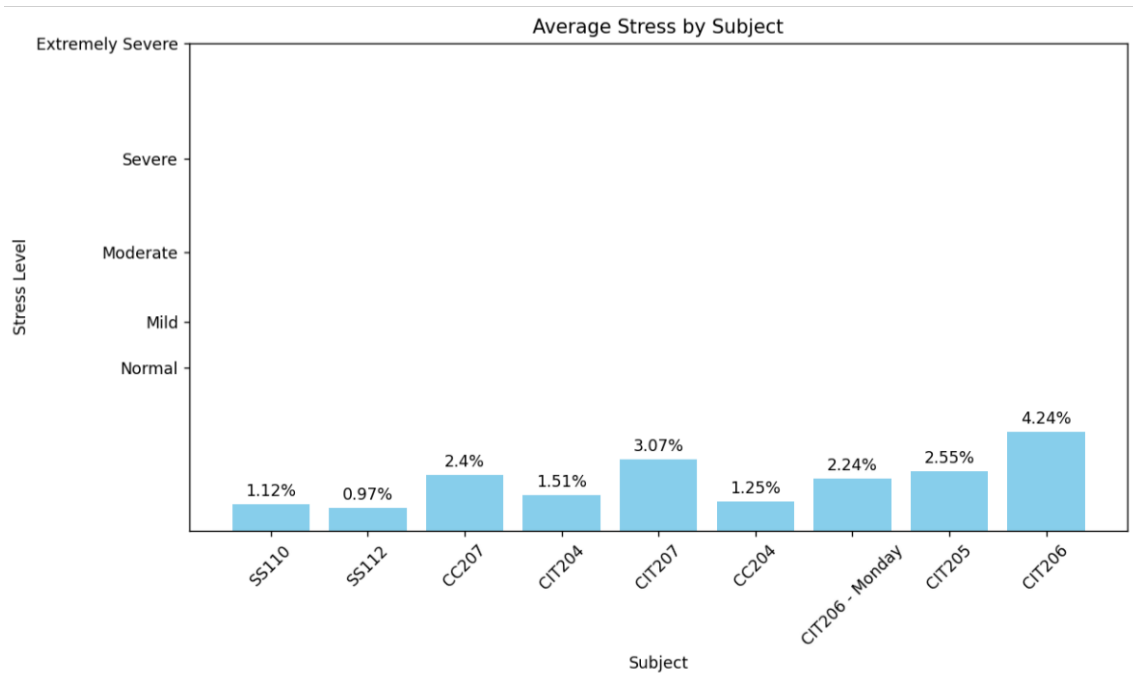


Figure 50. Visualization of Average Stress Levels by Session Using ZenLens

The graph in Figure 50 displays stress percentages for various subjects giving a comparison of stress distribution in the dataset. The y-axis shows stress level classifications, from Normal to Extremely Severe, while the x-axis lists the subjects (like SS110, CIT207, CIT206). The findings indicate that stress levels among participants differ. Some individuals experience low stress, while others show higher stress percentages. The stress level range spans from 0.97% (SS112) to 4.24% (CIT206). This difference suggests that outside factors like workload, surroundings, or personal resilience

have an impact on how students handle pressures. Based on the study by Pascoe et al. [52], stress levels among students are influenced by various external factors such as academic workload, environmental conditions, and individual coping mechanisms. This supports the notion that differences in how students handle pressures can be attributed to these external influences. Also, activities tied to schoolwork (such as coding tasks or exams) might lead to changes in stress levels as time passes, since these percentages stand for the average amount of stress detected in each session. Below is the equation for calculating the Weekly Average Stress:

$$\text{Weekly Average Stress} = \frac{\text{Sum of Individual Percentages}}{\text{Total Sessions}} \approx 2.15\%$$

where the sum of individual percentages represents the total stress values, and total sessions refers to the number of sessions considered in the calculation.

The findings indicate that all subjects in this dataset are classified under the "Normal" stress level, with average stress percentages ranging from 0.97% to 4.24%. This suggests a relatively low level of detected stress across the observed subjects.

ZenLens calculated the weekly average stress level based on individual session results, as shown in Table 8. The individual stress percentages for each subject were averaged to obtain an overall weekly stress level. This calculation was performed by summing the individual percentages from each session and dividing by the total number of sessions:

DASS – 21 Weekly Average =

$$\frac{(17 \times \text{Normal Score}) + (7 \times \text{Mild Score}) + (9 \times \text{Moderate Score}) + (2 \times \text{Severe Score})}{35} \approx 13.83\%$$

Model Validation Accuracy: App vs DASS Results

Participants' stress levels were evaluated using both ZenLens and the DASS-21 questionnaire, enabling an analysis of alignment between the app's assessments and established psychological metrics.

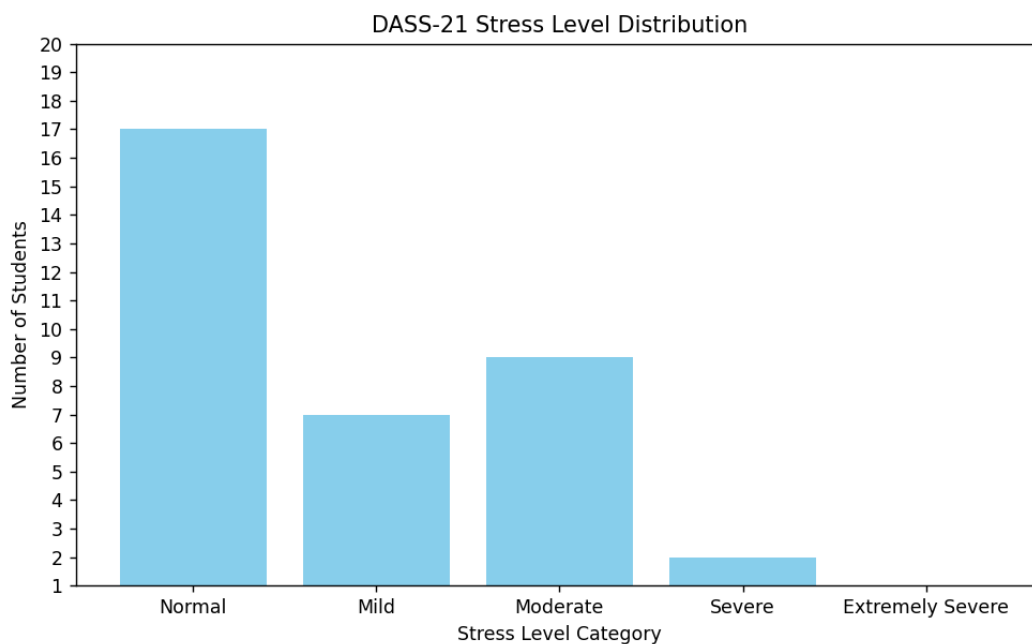


Figure 51. DASS-21 Stress Level Distribution Among Participants

Figure 51 illustrates the weekly average stress level for DASS-21 was determined by calculating the weighted average based on the number of participants in each stress level category.

This average DASS-21 stress level of 13.83% falls within the "Normal" range according to DASS-21 scoring criteria. Although DASS-21 calculated average 13.83% differs in percentage from the system's weekly result of 2.15%, both fall within the same "normal" category. This alignment indicates that ZenLens provides a reasonable approximation of stress

levels, suggesting consistency between ZenLens's assessment and DASS-21 measures.

The evaluation of ZenLens, conducted in accordance with the ISO/IEC 25010 Software Quality Standards, offers a detailed analysis of its performance across multiple quality characteristics. This standard was chosen for its comprehensive framework that ensures a thorough assessment of software quality, encompassing functionality, usability, security, and maintainability among others.

ZenLens received a high score in Functional Stability due to its reliable performance and minimal system crashes during extensive testing. Performance Efficiency was rated slightly lower because the system's real-time image processing demands significant computational power, affecting performance on lower-end devices. Compatibility scored well as ZenLens integrates seamlessly with the intended hardware and software environments.

The system's Usability was rated very good, reflecting its user-friendly interface designed for ease of use by educators and students. Security measures implemented in ZenLens earned it a high score, ensuring data privacy and

protection. Reliability was also rated good, although minor detection inconsistencies in varying environmental conditions were noted. Maintainability received a good score, highlighting the system's modular architecture, which simplifies updates and maintenance, though further improvements could enhance this aspect.

Portability received the lowest score due to ZenLens being designed primarily for desktop use, with limited support for mobile or cross-platform functionality. Overall, ZenLens achieved a good rating, indicating strong performance across most quality attributes while identifying areas for enhancement.

System Evaluation Results

Table 7. ISO/IEC 25010 Software Quality Standards Evaluation

Legend

Scales of Mean	Description
5 - 4.1	Very Good
4 - 3.1	Good
3 - 2.1	Fair
2 - 1.1	Poor
1	Very Poor

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Table 8. Summary of ISO/IEC 25010 Software Quality Standards

Evaluation Results

Characteristics	Mean	Description
Functional Stability	4.0	Good
Performance Efficiency	3.79	Good
Compatibility	4.0	Good
Usability	4.20	Very Good
Security	4.43	Very Good
Reliability	3.89	Good
Maintainability	3.84	Good
Portability	3.76	Good
OVERALL	3.99	Good

CHAPTER 5 SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS
Summary of the Proposed Study, Design and Implementation

The ZenLens model and the ZFNet algorithm were both implemented into the web application. The system is designed as an innovative tool to assess and manage student stress through a non-invasive, image-based method that analyzes emotional expressions to detect stress levels. The project unfolds in three main phases: Preparation, Implementation, and Evaluation. In the Preparation Phase, the researchers engage in in-depth consultations with stress measurement experts, gather relevant datasets, and conduct initial data analysis.

In the Implementation Phase, the team focused on building the core functionality of ZenLens. They create machine learning models that can interpret students' emotions by using the Retina Face library for facial detection and the ZFNet model for emotion classification.

Finally, during the Evaluation Phase, the system's performance is tested rigorously, focusing on its accuracy, precision, and recall using a confusion matrix—a widely used tool in emotion detection. This testing ensures ZenLens reliably assesses facial expressions to indicate stress

levels. Designed with the Rapid Application Development (RAD) approach, ZenLens incorporates iterative feedback and improvements, creating a user-friendly and efficient system.

Summary of Findings

This study aimed to develop and evaluate ZenLens, an image-based stress detection system utilizing facial emotion recognition to assess student stress levels. The findings of this research are categorized into three main areas: system performance, stress level analysis, and software quality evaluation.

System Performance and Accuracy

The ZenLens system, which integrates the ZFNet model for facial emotion classification and RetinaFace for facial landmark detection, demonstrated an overall accuracy of 87.14% in detecting emotions. The model performed exceptionally well in identifying the "sad" emotion with the highest accuracy at 87%, while emotions such as "angry" and "disgust" exhibited lower classification reliability due to their tendency to overlap with other categories. The system's ability to distinguish between closely related emotional expressions significantly outperformed previous models, which had an average accuracy of approximately 75%.

Stress Level Analysis

ZenLens categorized stress levels based on facial emotion detection, aligning its assessments with the Depression Anxiety Stress Scales (DASS-21). The results indicated that the majority of students fell within the "Normal" stress category, with individual session stress percentages ranging from 0.97% to 4.24%. A comparative analysis between ZenLens and DASS-21 results showed a strong correlation, with ZenLens estimating an average stress level of 2.15% and DASS-21 reporting 13.83%. Despite percentage differences, both methods classified the majority of students as experiencing "Normal" stress, validating ZenLens's reliability in stress assessment.

Software Quality Evaluation

The system evaluation used the ISO/IEC 25010 Software Quality Standards to assess software quality criteria of functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability and portability. The system analysis demonstrated that ZenLens correctly detects stress-related facial expressions together with its accurate well-being recommendation function. For performance efficiency the system scored 3.79 although its actual execution showed moderate results. The system

demonstrated superior compatibility features that allowed smooth operation between systems yet had minor integration problems. The system evaluation showed that it accomplished excellent usability scoring 4.0 points. The system's standard operating performance reliability stands at 3.89 as reported by users. User data protection was one of the leading aspects of the system because it achieved a remarkable 4.43 security score. System maintenance needs fall in the middle range according to the measured score of 3.84 points. System portability provides convenient installation and uninstallation features according to the measured score of 3.76.

The findings show ZenLens works well as a student stress identification tool yet further enhancement work must be done to optimize its full effectiveness and expand its reach. The future development cycle of ZenLens needs performance speed boosts and stronger security systems and better educational environment compatibility for optimal functionality. Enhanced features will enhance the effectiveness of ZenLens as well as keep its exceptional software quality standards available for wider deployment.

Conclusions

The research evaluated an AI system's capability to detect student stress levels by analyzing facial expressions and building personalized stress management suggestions. The research data proves that ZenLens employs deep learning algorithms and uses ZFNet for emotion classification together with RetinaFace for facial landmark detection to effectively measure student stress levels through facial expression analysis. The system achieves a 87.14% accurate emotion detection rate that enables trustworthy stress measurement and closely corresponds to Depression Anxiety Stress Scale-21 evaluation metrics.

The main advantage of ZenLens comes from its capability to examine multiple facial expressions during time intervals instead of using one-time facial assessments. The system provides complete emotional state assessments which permits educators and counselors to watch stress level patterns to intervene appropriately. The system produces targeted stress management recommendations that extend its value in promoting student mental health. The study revealed several possible enhancements to its design even though it demonstrated successful results. The model requires enhancement to detect

simultaneous emotions like anger and disgust as well as optimization for speedier real-time processing. The system's accuracy and robustness can be improved by adding additional facial expression data that covers diverse expressions and environmental contexts.

The emotional stress detection solution provided by ZenLens proves that facial expressions recognized by AI represent an effective approach to implementing stress management in educational settings. Through its successful results the research successfully answers its objective and creates foundational groundwork for AI-based mental health monitoring advances. The future development of ZenLens shows promise to become a necessary digital tool that will serve students and educational establishments by managing academic stress while strengthening the learning environment through technological interventions.

Recommendations

Based on the feedback from experts, including psychologists, educators, and technology experts who tested ZenLens, several areas for improvement were identified.

Refining image clarity is essential, particularly for individuals seated farther from the camera or in environments with limited facial detail. Based on user testing, it was observed that lower resolution images significantly reduced detection accuracy. IT Experts involved in the study recommended implementing a minimum resolution threshold to ensure that only images with sufficient clarity are included in the analysis. This would streamline the workflow and enhance the reliability of emotion detection. A study by Greco et al. supports this approach by showing that the performance of deep learning models in facial emotion recognition heavily depends on the quality of the input images. Their study highlights that low-quality images can severely impair the effectiveness of these models, reinforcing the need for high-resolution imagery in our context [102].

Minimizing occlusions caused by various factors, such as facial angles, coverings (e.g., masks), or obstructive

equipment (e.g., computers), is also crucial for accurate stress assessment. Initial trials indicated that facial obstructions led to a decline in recognition accuracy, which aligns with existing research in facial recognition technology that suggests occlusions can reduce classification accuracy by up to 30% [97]. Ensuring an unobstructed environment through proper positioning of cameras will improve the quality of captured images and enhance detection precision.

Expanding ZenLens's dataset to include a broader range of facial expressions across different cultural and emotional spectrums will further improve its capacity to capture nuanced emotions effectively. Psychologists recommended increasing dataset diversity to reduce bias and enhance recognition accuracy. The model may experience performance degradation when exposed to unfamiliar facial expressions, especially from underrepresented demographic groups, resulting in reduced accuracy and potential bias in stress detection. Research in affective computing has demonstrated that emotion detection models perform better when trained on diverse datasets [98], as this ensures that variations in emotional expressions across different demographics are accounted for.

Another crucial aspect involves adapting to environmental conditions by applying preprocessing filters that adjust for variations in lighting and background complexity. Studies in image processing indicate that inconsistent lighting and cluttered backgrounds can negatively affect emotion detection [100]. Findings from user tests revealed inconsistencies in stress level detection when images were taken under dim lighting or against busy backgrounds. Techniques such as contrast enhancement and brightness stabilization can help standardize image quality, thereby improving detection accuracy.

Optimizing real-time analysis is a potential future enhancement that could significantly improve ZenLens's responsiveness in stress detection. Based on insights from user testing, real-time processing is deemed valuable for enhancing the system's ability to provide immediate feedback. A study on real-time stress analysis for combatants underscores the importance of real-time feedback in managing stress effectively [103]. Their findings demonstrate how real-time biometric monitoring can lead to immediate interventions, improving mental readiness and stress management, which could be adapted to educational settings for similar benefits.

Another promising enhancement involves advanced facial recognition capabilities, suggested by our thesis adviser as a future enhancement to further refine ZenLens's capabilities. Incorporating advanced facial recognition techniques could enhance the system's ability to accurately identify individuals in various educational settings, enabling ZenLens to accurately identify and differentiate multiple faces in group settings, like classrooms. By refining the model to handle partial occlusions and complex backgrounds and incorporating multi-face tracking, ZenLens can maintain consistent recognition of individual students, even in dynamic environments. A study demonstrated that integrating advanced facial recognition algorithms with multi-face tracking significantly improves accuracy in environments with varying lighting and complex backgrounds. Their findings suggest that these techniques can enhance real-time performance and reliability in educational settings, supporting the proposed improvements for ZenLens [101].

Implementing a robust validation system that compares detected emotions with observed emotions, using precision, recall, and confusion matrices, would further strengthen ZenLens's reliability. Discrepancies between detected and

actual emotions were identified during testing, highlighting the need for validation metrics [101]. This systematic approach to validation allows for continuous refinement and ensures that ZenLens consistently delivers accurate results [101].

Lastly, prioritizing strong security measures is essential to protect the sensitive emotional and behavioral data processed by ZenLens. Since the system handles personal data, cybersecurity experts recommend using robust encryption and authentication protocols to prevent unauthorized access. Research in privacy-preserving AI suggests that regular security audits and software updates are crucial in safeguarding user information [99]. Strengthening security measures will reinforce user trust and ensure compliance with data protection regulations.

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Appendices

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Appendix A Letter to the Adviser

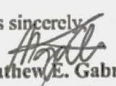
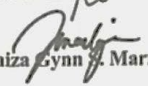
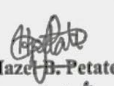
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There are no foreseeable risks in participating in this study, and participants will not be paid for their involvement.

All research data will be preserved anonymously for potential reuse by other researchers.

We therefore request permission in writing to conduct our research at the College of Information and Communications Technology. The permission letter should be on your organization's headed paper, signed and dated, and specifically referring to us by name and the title of our study.

Please let us know if you require any further information. You can contact us at 09456977082 or email us at glaizagynn.marmolejo@wvsu.edu.ph. We look forward to your response as soon as it is convenient.

Yours sincerely,  Mathew E. Gabriel  Phoebe Grace G. Juayong  Glaiza Gynn M. Marmolejo	 Hazel B. Petate  EJ Prince D. Sevileno
---	---

Noted:


Dr. Frank I. Elijorde
Research Adviser

Approved:


Dr. Ma. Beth S. Concepcion
Dean, College of ICT

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La Paz, Iloilo City, Philippines

Appendix B Letter to the Technical Editor

	ADVISER'S ENDORSEMENT FORM (For Thesis Manuscript)	Document No.	WVSU-ICT-SOI-03-F10
		Issue No.	1
		Revision No.	0
	WEST VISAYAS STATE UNIVERSITY	Date of Effectivity:	April 27, 2018
		Issued by:	CICT
		Page No.	Page 1 of 1

Respectfully endorsed to the Technical Editor, the attached manuscript of the thesis entitled:

ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management

Said manuscript has been presented to me for preliminary evaluation and guidance, and after a series of corrections/directions given which was implemented by the proponents whose names are listed hereunder and their thorough research, we have come to its completion.

Now therefore, I hereby **ENDORSE** the said thesis manuscript to the Technical Editor for **TECHNICAL EDITING**.


DR. FRANK I. ELJORDE
Adviser's Name & Signature

Date: February 3, 2025

Group Members:

1. Gabriel Mathew E.
2. Juayong Phoebe Grace C.
3. Marmolejo Glaiza Gynn S.
4. Petate Hazel B.
5. Sevilleno E.J Prince D.
6. _____

Note: This form should be accomplished and signed if the corrections and changes made by the adviser have been implemented and a new copy of the document have been printed for checking and submission to the next editor

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La Paz, Iloilo City, Philippines

Appendix C Letter to the English Editor

	TECHNICAL EDITOR'S ENDORSEMENT FORM (For Thesis Manuscript)	Document No.	WVSU-ICT-SOI-03-F11
		Issue No.	1
		Revision No.	0
	WEST VISAYAS STATE UNIVERSITY	Date of Effectivity:	April 27, 2018
		Issued by:	CICT
		Page No.	Page 1 of 1

Respectfully endorsed to the English Editor, the attached manuscript of the thesis entitled:

ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced
Student Stress Management

Said manuscript was presented to me and was reviewed and edited in terms of technical specifications, correctness of diagrams and other technical matters. The corrections and suggestions were carried and implemented by the proponents whose names are listed hereunder.

Now therefore, I hereby **ENDORSE** the said thesis manuscript to the English Editor/Grammarian for English Grammar Editing.


DR. FRANK I. ELJORDE
Technical Editor's Name & Signature

Date: March 11, 2025

Group Members:

1. Gabriel, Mathew E.
2. Juayong, Phoebe Grace C.
3. Marmolejo, Glaiza Gynn S.
4. Petate, Hazel B.
5. Sevilleno, EJ Prince D.
6. _____

Note: This form should be accomplished and signed if the corrections and changes made by the Technical Editor have been implemented and a new copy of the document have been printed for checking and submission to the next editor.

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La Paz, Iloilo City, Philippines

Appendix D Letter to the Format Editor

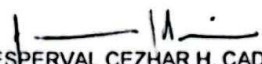
	ENGLISH EDITOR/GRAMMARIAN'S ENDORSEMENT FORM (For Thesis Manuscript)	Document No.	WVSU-ICT-SOI-03-F12
		Issue No.	1
		Revision No.	0
	WEST VISAYAS STATE UNIVERSITY	Date of Effectivity:	April 27, 2018
		Issued by:	CICT
		Page No.	Page 1 of 1

Respectfully endorsed to the **Thesis Format Editor**, the attached manuscript of the thesis entitled:

ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced
Student Stress Management

Said manuscript was presented to me for English grammar editing, corrections have been made and the proponents whose names are listed hereunder implemented said corrections and changes in the revised manuscript.

Now therefore, I hereby **ENDORSE** the said thesis manuscript for **Thesis Format Editing**.


MR. ESPERVAL CEZHAR H. CADIAO
English Editor/Grammarian's Name and Signature

Date: March 17, 2025

Group Members:

1. Gabriel, Mathew E.
2. Juayong, Phoebe Grace C.
3. Marmolejo, Glaiza Gynn S.
4. Petate, Hazel B.
5. Sevillano, EJ Prince D.
6. _____

Note: This form should be accomplished and signed if the corrections and changes made by the English Editor have been implemented and a new copy of the document have been printed for checking and submission to the next editor.

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Appendix E Certificate of Bookbinding

	CERTIFICATION FOR BOOKBINDING (For Thesis Manuscript)	Document No.	WVSU-ICT-SOI-03-F14
		Issue No.	1
	WEST VISAYAS STATE UNIVERSITY	Revision No.	0
		Date of Effectivity:	April 27, 2018
		Issued by:	CICT
		Page No.	Page 1 of 1

This certifies that the attached manuscript of the thesis entitled:

ZenLens: Image-based Stress Detection and Tailored Recommendations
for Enhanced Student Stress Management

Is now ready for bookbinding. Said manuscript was presented to me and has checked the preliminaries, thesis document convention and end matters, made some corrections which was implemented by the proponents whose names are listed hereunder.

Now therefore, I hereby **ENDORSE** said manuscript for **BOOKBINDING**.


DR. REGINA A. CABACAS
Thesis Coordinator's Name and Signature

Date: 5-19-2018

Group Members:

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

Note: This form should be accomplished and signed if the corrections and changes made by the Thesis Format Editor have been implemented and the four (4) new copies have been printed ready for bookbinding.

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Ia Paz, Iloilo City, Philippines

Appendix F Thesis Output Certification

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Appendix G Gantt Chart

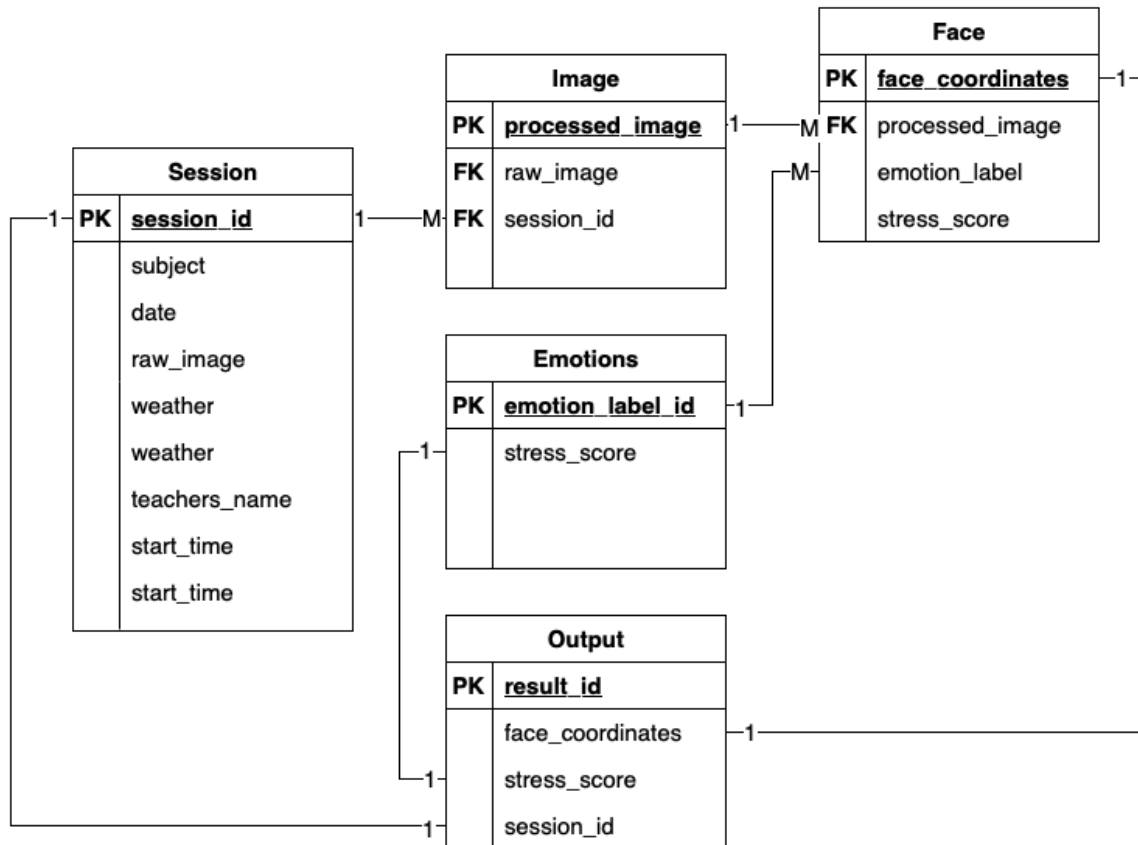
TASK	2023					2024									
	DECEMBER	JANUARY	FEBRUARY	MARCH	APRIL	MAY	JUNE	JULY	AUGUST	SEPTEMBER	OCTOBER				
Concept defense															
Concept Paper (revised)															
Thesis adviser Invitation															
Domain expert first meeting															
Finalization of research objectives and algorithm to be used															
Consultation with thesis adviser															
Consultation with domain expert															
Writing process (chapter 1-3)															
Building and training AI models															
Writing progress (chapter 1-3 revision, chapter 4)															
Writing progress (revision, chapter 1-4)															
UI/UX Development															
Fine tuning of the algorithm															
Building of the Application															
Data gathering															

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Appendix H Data Dictionary

Field Name	Data Type	Description	Constraints
email	String	The email of the user.	Not Null
firstName	String	The first name of the user.	Not Null
lastName	String	The last name of the user.	Not Null
session_id	String	Unique identifier to each instance of an analysis session.	Not Null
timestamp	String	Date and time record of the analyzed image.	Not Null
coordinates	Array	Coordinates of the detected face in an image.	Not Null
emotion_detected	String	Label of the emotion detected in a face.	Not Null
emotion_count	Map	Store emotion occurrences.	Not Null
stress_score	Number	Assigned stress score based on the emotion detected.	Not Null
sessions_count	Number	Total of sessions in a week.	Not Null
total_faces	Number	All faces detected in a week.	Not Null
total_stress	Number	Summary of stress detected in a week.	Not Null
average_stress	Number	Average stress score from total faces and total score.	Not Null
stress_category	String	Category of stress based on the average stress.	Not Null

Appendix I Entity-Relationship Diagram / Script



Appendix J Sample Program Codes

```
def preprocess_image(gray: np.ndarray) -> np.ndarray:
    resized = cv2.resize(gray, (IMAGE_SIZE, IMAGE_SIZE), interpolation=cv2.INTER_AREA)
    normalized = resized / NORMALIZATION_FACTOR
    processed_image = np.expand_dims(normalized, axis=-1)
    processed_image = np.expand_dims(processed_image, axis=0)
    return processed_image
```

```
faces = RetinaFace.detect_faces(rgb_image)
```

```
def predict_emotion(processed_image: np.ndarray) -> np.ndarray:
    return model.predict(processed_image)
```

```
emotion_labels = ['angry', 'disgust', 'afraid', 'happy', 'neutral', 'sad', 'surprise']
emotion_scores = {'happy': 0, 'surprise': 1, 'neutral': 0, 'disgust': 1, 'sad': 2, 'afraid': 3, 'angry': 3}

def classify_emotions(probabilities: np.ndarray) -> (str, int):
    detected_emotion_index = np.argmax(probabilities)
    detected_emotion = emotion_labels[detected_emotion_index]
    stress_score = emotion_scores[detected_emotion]
    return detected_emotion, stress_score
```

```
emotion_labels = ['angry', 'disgust', 'afraid', 'happy', 'neutral', 'sad', 'surprise']
emotion_scores = {
    'happy': 0,      # Low stress
    'surprise': 1,   # Mild stress
    'neutral': 0,    # Low stress
    'disgust': 1,    # Mild stress
    'sad': 2,        # Moderate stress
    'afraid': 3,     # High stress
    'angry': 3       # High stress
}
```

```
total_stress = 0
total_faces = 0
for file in files:
    output_path, results, emotion_data = analyze_image(image_path, processed_folder)
    photo_total_stress = sum(result['stress_score'] for result in results.values())
    total_stress += photo_total_stress
    total_faces += len(results)
```

```
def categorize_stress(average_stress):
    adjusted_stress = (average_stress / 100) * 21
    if adjusted_stress <= 7:
        return "Normal"
    elif adjusted_stress <= 9:
        return "Mild"
    elif adjusted_stress <= 12:
        return "Moderate"
    elif adjusted_stress <= 16:
        return "Severe"
    else:
        return "Extremely Severe"
```

```
def save_data(data, file_path=DATA_FILE):
    with open(file_path, 'w') as file:
        json.dump(data, file, indent=4)
```

```
def calculate_weekly_averages(grouped_data):
    weekly_averages = []

    for week in grouped_data:
        total_faces = sum(entry['total_faces'] for entry in week)
        total_stress = sum(entry['average_stress'] * entry['total_faces'] for entry in week)
        sessions_count = len(week)
        average_stress = total_stress / total_faces if total_faces > 0 else 0
        stress_category = categorize_stress(average_stress)
```

```
@app.route('/weekly-emotion-timeline', methods=['GET'])
def weekly_emotion_timeline():
    data = load_emotion_timeline()
    data.sort(key=lambda entry: datetime.fromisoformat(entry['timestamp']))
    weekly_summary = []
    # (further logic to group and summarize weekly emotions)
    return jsonify(weekly_results)
```

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Appendix K Sample Questionnaires / Interview Forms / PERT CPM
/ DASS 21 Questionnaire

DASS₂₁	<i>Name:</i> _____	<i>Date:</i> _____																																																																																																																																																						
Please read each statement and circle a number 0, 1, 2 or 3 which indicates how much the statement applied to you <i>over the past week</i>. There are no right or wrong answers. Do not spend too much time on any statement. <i>The rating scale is as follows:</i> 0 Did not apply to me at all 1 Applied to me to some degree, or some of the time 2 Applied to me to a considerable degree, or a good part of time 3 Applied to me very much, or most of the time																																																																																																																																																								
<table style="width: 100%; border-collapse: collapse;"> <tr><td style="width: 5%;">1</td><td style="width: 90%;">I found it hard to wind down</td><td style="width: 5%;"></td></tr> <tr><td>2</td><td>I was aware of dryness of my mouth</td><td></td></tr> <tr><td>3</td><td>I couldn't seem to experience any positive feeling at all</td><td></td></tr> <tr><td>4</td><td>I experienced breathing difficulty (eg, excessively rapid breathing, breathlessness in the absence of physical exertion)</td><td></td></tr> <tr><td>5</td><td>I found it difficult to work up the initiative to do things</td><td></td></tr> <tr><td>6</td><td>I tended to over-react to situations</td><td></td></tr> <tr><td>7</td><td>I experienced trembling (eg, in the hands)</td><td></td></tr> <tr><td>8</td><td>I felt that I was using a lot of nervous energy</td><td></td></tr> <tr><td>9</td><td>I was worried about situations in which I might panic and make a fool of myself</td><td></td></tr> <tr><td>10</td><td>I felt that I had nothing to look forward to</td><td></td></tr> <tr><td>11</td><td>I found myself getting agitated</td><td></td></tr> <tr><td>12</td><td>I found it difficult to relax</td><td></td></tr> <tr><td>13</td><td>I felt down-hearted and blue</td><td></td></tr> <tr><td>14</td><td>I was intolerant of anything that kept me from getting on with what I was doing</td><td></td></tr> <tr><td>15</td><td>I felt I was close to panic</td><td></td></tr> <tr><td>16</td><td>I was unable to become enthusiastic about anything</td><td></td></tr> <tr><td>17</td><td>I felt I wasn't worth much as a person</td><td></td></tr> <tr><td>18</td><td>I felt that I was rather touchy</td><td></td></tr> <tr><td>19</td><td>I was aware of the action of my heart in the absence of physical exertion (eg, sense of heart rate increase, heart missing a beat)</td><td></td></tr> <tr><td>20</td><td>I felt scared without any good reason</td><td></td></tr> <tr><td>21</td><td>I felt that life was meaningless</td><td></td></tr> </table>	1	I found it hard to wind down		2	I was aware of dryness of my mouth		3	I couldn't seem to experience any positive feeling at all		4	I experienced breathing difficulty (eg, excessively rapid breathing, breathlessness in the absence of physical exertion)		5	I found it difficult to work up the initiative to do things		6	I tended to over-react to situations		7	I experienced trembling (eg, in the hands)		8	I felt that I was using a lot of nervous energy		9	I was worried about situations in which I might panic and make a fool of myself		10	I felt that I had nothing to look forward to		11	I found myself getting agitated		12	I found it difficult to relax		13	I felt down-hearted and blue		14	I was intolerant of anything that kept me from getting on with what I was doing		15	I felt I was close to panic		16	I was unable to become enthusiastic about anything		17	I felt I wasn't worth much as a person		18	I felt that I was rather touchy		19	I was aware of the action of my heart in the absence of physical exertion (eg, sense of heart rate increase, heart missing a beat)		20	I felt scared without any good reason		21	I felt that life was meaningless		<table style="width: 100%; border-collapse: collapse;"> <tr><td style="width: 5%;">0</td><td style="width: 5%;">1</td><td style="width: 5%;">2</td><td style="width: 5%;">3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> <tr><td>0</td><td>1</td><td>2</td><td>3</td></tr> </table>	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3	0	1	2	3
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Ia Iga, Moilo City, Philippines

Appendix L Informed Consent

INFORMED CONSENT

Name of Researchers: Mathew E. Gabriel, Phoebe Grace Juayong, Glaiza Gynn S. Marmolejo, Hazel B. Petate, EJ Prince D. Sevileno

Name of Institution: College of Information and Communications Technology, West Visayas State University

Name of Research: ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management

PART I: INFORMATION SHEET

We, the above-named researchers from the College of ICT, taking up BS Computer Science are conducting a study, entitled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management," and we are inviting you to become participants in our research.

Participating in this study entails capturing your in-class photos as this study utilizes facial expression recognition technology to analyze the stress levels of college students over a week. The insights gained from this study will enable school faculties to provide targeted support to students experiencing extreme stress levels, ultimately enhancing student well-being and academic success. Furthermore, please ask the person in charge if there are any questions or require clarification.

The study aims to develop a system that will monitor students' mental stress. Specifically, it aims to (1) develop and train a ZFNet model, an artificial intelligence (AI) model to estimate students' stress levels by analyzing their emotions through facial expressions, (2) integrate the model into a web application for monitoring stress level, (3) evaluate the model's performance, and (4) evaluate its usability and effectiveness.

The setup of this research is that snapshots of your in-class photos will be captured by two (2) cameras placed in front of the classroom for one (1) week according to your class schedule. Moreover, at the end of the week, you will be asked to answer a self-report questionnaire related to your stress level based on the DASS 21 (Depression Anxiety Stress Scales).

The researchers identified college students as participants in this study, where the source of academic stressors is common among this group. Participants should be at least 18 years old and currently enrolled as undergraduate students at WVSU with a face-to-face class schedule. Your decision to participate in this study is completely voluntary. If you decide to not participate in this study, it will not affect the care, services, or benefits to which you are entitled.

PROCEDURE

The researchers are developing a system to monitor students' stress levels through emotions exhibited from facial expressions during class hours. Hence, as part of the study, we will take photos of your face during class hours for one (1) week, per your class schedule, using two (2) cameras placed in the front of the room. Researchers will ensure that the activity will not disrupt the class's atmosphere.

At the end of the week, you will complete a short self-report questionnaire based on DASS 21 (Depression Anxiety Stress Scales) about the stress levels you experienced during the week. Only the Stress Scale will be used for the study, other scales (Depression and Anxiety Scales) will be used to validate the responses. The questions are straightforward and there are no right or wrong answers. Your responses will help us understand your stress levels during the study period. The result of the self-report questionnaire would be compared against the system to help the researchers assess the system's performance.

DURATION

Participants would have their in-class photos taken for one (1) week straight according to your class schedule.

RISKS

There are no risks associated with participating in this study. However, your photos will be taken throughout the data gathering which may not be comfortable for you.

BENEFITS

Participating in this research will not provide any direct and immediate benefits to you; however, the results may contribute to the development of improved methods that supports student well-being.

REIMBURSEMENTS

You will not be financially compensated for participating in this study beyond reimbursements for expenses incurred because of your participation.

CONFIDENTIALITY

The researchers will ensure that your identity as participants in this study remains strictly confidential. Any data collected will be managed solely by the researchers and shared only with domain experts relevant to the study. Photos taken during data gathering will be stored on a secure hard disk drive (HDD) accessible only to the researchers and will be retained for the duration of the thesis. Upon completion of the research, the photos will be securely deleted from the HDD.

SHARING THE RESULTS

Results obtained from this study may be shared through publications and conferences.

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Ia Paz, Iloilo City, Philippines

RIGHT TO REFUSE OR WITHDRAW

Participation in this study is completely voluntary and you have the right to withdraw at any time without penalty.

WHO TO CONTACT

For future concerns related to the study, you can contact the researchers by the following information:

Mathew E. Gabriel
+63 9217164100
mathew.gabriel@wvsu.edu.ph

Hazel B. Petate
+63 9206187182
hazel.petate@wvsu.edu.ph

Phoebe Grace C. Juayong
+63 9618080709
phoebegrace.juayong@wvsu.edu.ph

EJ Prince D. Sevileno
+63 9615762553
ejprince.sevileno@wvsu.edu.ph

Glaiza Gynn S. Marmolejo
+63 9456977082
glaizagynn.marmolejo@wvsu.edu.ph

PART II: CERTIFICATE OF CONSENT

I have read and understood the explanation provided to me, and I have had all my questions answered to my satisfaction. Signing this form means that I voluntarily agree to participate in this research. I consent voluntarily to be a participant in this study.

Print Name of Participant: _____

Signature of Participant: _____

Date [MM/DD/YYYY]: _____

If Illiterate

I have witnessed the accurate reading of the consent form to the potential participant, and the individual has had the opportunity to ask questions. I confirm that the individual has given consent freely.

Print name of witness _____ **Thumb print of participant:**

Signature of witness _____

Date [MM/DD/YYYY]: _____

STATEMENT BY THE RESEARCHER OR PERSON TAKING CONSENT



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I have accurately read out the information sheet to the potential participant, and to the best of my ability made sure that the participant understands that the following will be done:

- 1. Participants' in-class photos will be taken during class hours for one week according to their class schedule;**
- 2. The study will only utilize their facial photos in developing the researchers' study; and**
- 3. The researchers will strictly keep the participants' identities and sensitive data.**

I confirm that the participant was given an opportunity to ask questions about the study, and all the questions asked by the participant have been answered correctly and to the best of my ability. I confirm that the individual has not been coerced into giving consent, and the consent has been given freely and voluntarily.

A copy of this Informed Consent Form has been provided to the participant.

Print Name of Researcher or person taking the consent

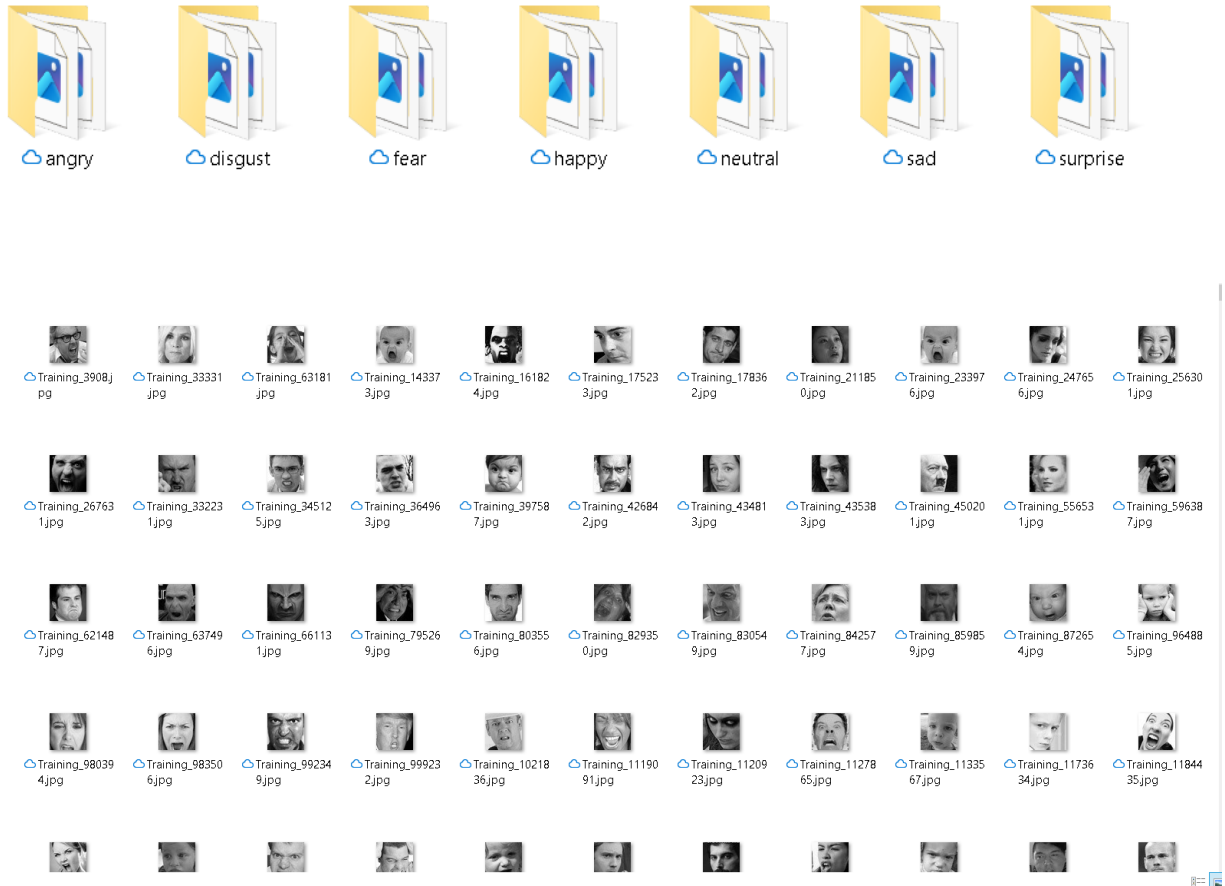
Signature of Researcher or person taking the consent

Date [MM/DD/YYYY]: _____

Appendix M Result: DASS 21

173

Appendix N Dataset Screenshots



Appendix O ISO/IEC 25010 Evaluation Questionnaire



ZenLens ISO 25010 User Testing Evaluation Form

Researchers from the BS Computer Science program at West Visayas State University are conducting a study titled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management." The objective is to develop a web application that uses facial recognition technology to assess stress levels and provide tailored recommendations for student stress management.

Participants will include students, teachers, guidance counselors, IT professionals, and mental health specialists who will test the application using a provided dataset. All collected data will be handled confidentially and used solely for research purposes.

Your professional insights are highly valued for this testing. While your feedback is crucial, you may decline or withdraw at any time without any negative consequences. For inquiries, you can contact researcher Giaza Gynn S. Marmolejo at +639456977082 or glaizagynn.marmolejo@wvsu.edu.ph.

phoebegrace.juayong@wvsu.edu.ph [Switch account](#)

 Not shared

* Indicates required question

By proceeding, you confirm that you have read and understood the information provided and had any questions answered satisfactorily. Your participation in this research is part of a structured user testing process. We appreciate your cooperation and acknowledge your professional involvement in contributing valuable insights to this study. *

☐ Yes

Next

Page 1 of 11

Clear form

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Ia Paz, Iloilo City, Philippines

Part I. USER INFORMATION

Please enter your details below.

Please enter you name: *(First Name, Last Name)* *

Your answer _____

Please enter your age: *

Your answer _____

Role *

☐ IT Professional

☐ Teacher

☐ Psychiatrist

☐ Guidance Counselor

☐ Student

☐ Other: _____

Organization/Institution *

Please provide the name of your organization, school, clinic, or institution.

Your answer _____

Part II. INSTRUCTIONS

After using the capabilities of ZenLens, please evaluate its compliance based on the ISO/IEC 25010 standard evaluation criteria. Indicate your assessment by selecting the column that best matches your experience.

Rating Scale:

- 5 - Strongly Agree
- 4 - Agree
- 3 - Neutral
- 2 - Disagree
- 1 - Strongly Disagree

Part III. FUNCTIONAL SUITABILITY

ZenLens effectively detects and categorizes stress-related facial expressions to meet its purpose.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

ZenLens provides accurate stress detection and well-being recommendations.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

ZenLens's functions facilitate stress management and enhance user well-being.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

Part IV. PERFORMANCE EFFICIENCY

ZenLens meets required response times and data processing rates.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

ZenLens uses resources (e.g., memory, processing power) efficiently.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

ZenLens can handle the expected workload, even with multiple users.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

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Part V. COMPATIBILITY

ZenLens functions effectively without interfering with other systems in the environment.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

ZenLens integrates well with other software or platforms used within the institution.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

Part VI. INTERACTION CAPABILITY

ZenLens can handle the expected workload, even with multiple users.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

ZenLens is intuitive and easy to learn for all users.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

The interface of ZenLens is straightforward, making the system easy to operate and control.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

ZenLens effectively prevents users from making operational errors.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

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Part VII. RELIABILITY						
ZenLens performs consistently and reliably without errors during normal use.						
	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree
ZenLens is accessible and operational whenever required.						
	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree
ZenLens operates smoothly, even if minor issues occur.						
	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree
ZenLens can recover from interruptions without losing data.						
	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

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The interface of ZenLens presents functions and information in a way that encourages continued user interaction.

1 2 3 4 5

Strongly Disagree ☐ ☐ ☐ ☐ ☐ Strongly Agree

ZenLens can be effectively used by people from various backgrounds (ages, abilities, cultures, languages, economic situations, etc.).

1 2 3 4 5

Strongly Disagree ☐ ☐ ☐ ☐ ☐ Strongly Agree

ZenLens is accessible and usable by people of various backgrounds and abilities.

1 2 3 4 5

Strongly Disagree ☐ ☐ ☐ ☐ ☐ Strongly Agree

ZenLens provides appropriate cues, making it self-explanatory without needing extra help.

1 2 3 4 5

Strongly Disagree ☐ ☐ ☐ ☐ ☐ Strongly Agree

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VIII. SECURITY

ZenLens ensures that sensitive student data is accessible only to authorized users.

1 2 3 4 5

Strongly Disagree ☐ ☐ ☐ ☐ ☐ Strongly Agree

ZenLens protects the data from unauthorized modification or deletion.

1 2 3 4 5

Strongly Disagree ☐ ☐ ☐ ☐ ☐ Strongly Agree

Actions within ZenLens can be traced back to specific users or entities.

1 2 3 4 5

Strongly Disagree ☐ ☐ ☐ ☐ ☐ Strongly Agree

ZenLens accurately verifies user identities before granting access.

1 2 3 4 5

Strongly Disagree ☐ ☐ ☐ ☐ ☐ Strongly Agree

IX. MAINTAINABILITY

ZenLens components are well-structured, allowing updates with minimal impact on other parts.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

ZenLens can be used in other similar projects or educational tools.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

The impact of changes can be easily assessed within Zenlens

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

ZenLens can be updated or modified efficiently without introducing errors.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

ZenLens allows for efficient testing to ensure that updates or changes meet requirements.

	1	2	3	4	5	
Strongly Disagree	☐	☐	☐	☐	☐	Strongly Agree

X. PORTABILITY

ZenLens can be adapted for different educational environments or user needs

12345

Strongly Disagree

☐

☐

☐

☐

☐

Strongly Agree

ZenLens can be easily installed or uninstalled in its operational environment.

12345

Strongly Disagree

☐

☐

☐

☐

☐

Strongly Agree

ZenLens could replace similar software without issues or significant reconfiguration.

12345

Strongly Disagree

☐

☐

☐

☐

☐

Strongly Agree

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Ia Paz, Iloilo City, Philippines

Appendix P Permission Letter for the Dean

	West Visayas State University (Formerly Iloilo Normal School) COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY Luna St., La Paz, Iloilo City 5000 Iloilo, Philippines * Trunkline: (063) (033) 320-0870 to 78 loc. 1403 * Telefax No.: (033) 320-0879 * Website: www.wvsu.edu.ph * Email Address: cict@wvsu.edu.ph		
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College of Information and Communications Technology, West Visayas State University
Dr. Ma. Beth S. Concepcion
Dean, College of Information and Communications Technology
West Visayas State University
Luna St., La Paz, Iloilo City, 5000

August 6, 2024

Dear Dr. Concepcion,

Re: Permission to conduct research at WVSU - College of ICT

My name is Gabriel Mathew E., a 4th year BS Computer Science student from the College of ICT. Together with my colleagues, Juayong Phoebe Grace C., Marmolejo Glaiza Gynn S., Petate Hazel B., and Sevilla EJ Prince D., we are conducting a study titled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management."

Our research aims to utilize facial expression recognition technology to analyze the stress levels of college students over a week. The insights gained from this study will enable educators to provide targeted support to students experiencing stress, ultimately enhancing student well-being and academic success. We are seeking permission to conduct our study at the College of ICT's building at West Visayas State University, specifically involving students from the College of ICT, as we believe their participation will significantly contribute to the success of our research.

The research will entail two sets of tests: the Pilot testing and the Final testing, on two separate groups of students. Data will be collected by analyzing facial expressions to assess stress levels. We will invite students to participate in this study and, with their consent, collect data over a period of one week. The data collection will take place during class hours, and we will ensure that the research activities do not disrupt classes. The students' responses will be treated confidentially, and their identities will be managed exclusively by the researchers and trusted domain experts. Individual privacy will be maintained in all published and written data resulting from the study.

Participants will be asked to give their written consent before the research begins. The results of the study will be communicated through academic journals and other scholarly publications. The research participants will not be advantaged or disadvantaged in any way and will be reassured that they can withdraw their permission at any time during this project without any penalty.

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La Paz, Iloilo City, Philippines



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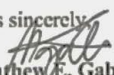
There are no foreseeable risks in participating in this study, and participants will not be paid for their involvement.

All research data will be preserved anonymously for potential reuse by other researchers.

We therefore request permission in writing to conduct our research at the College of Information and Communications Technology. The permission letter should be on your organization's headed paper, signed and dated, and specifically referring to us by name and the title of our study.

Please let us know if you require any further information. You can contact us at 09456977082 or email us at glazagynn.marmolejo@wvsu.edu.ph. We look forward to your response as soon as it is convenient.

Yours sincerely,


Mathew E. Gabriel


Phoebe Grace D. Juayong


Glaiza Gynn S. Marmolejo


Hazel B. Petate


EJ Prince D. Sevileno

Noted:

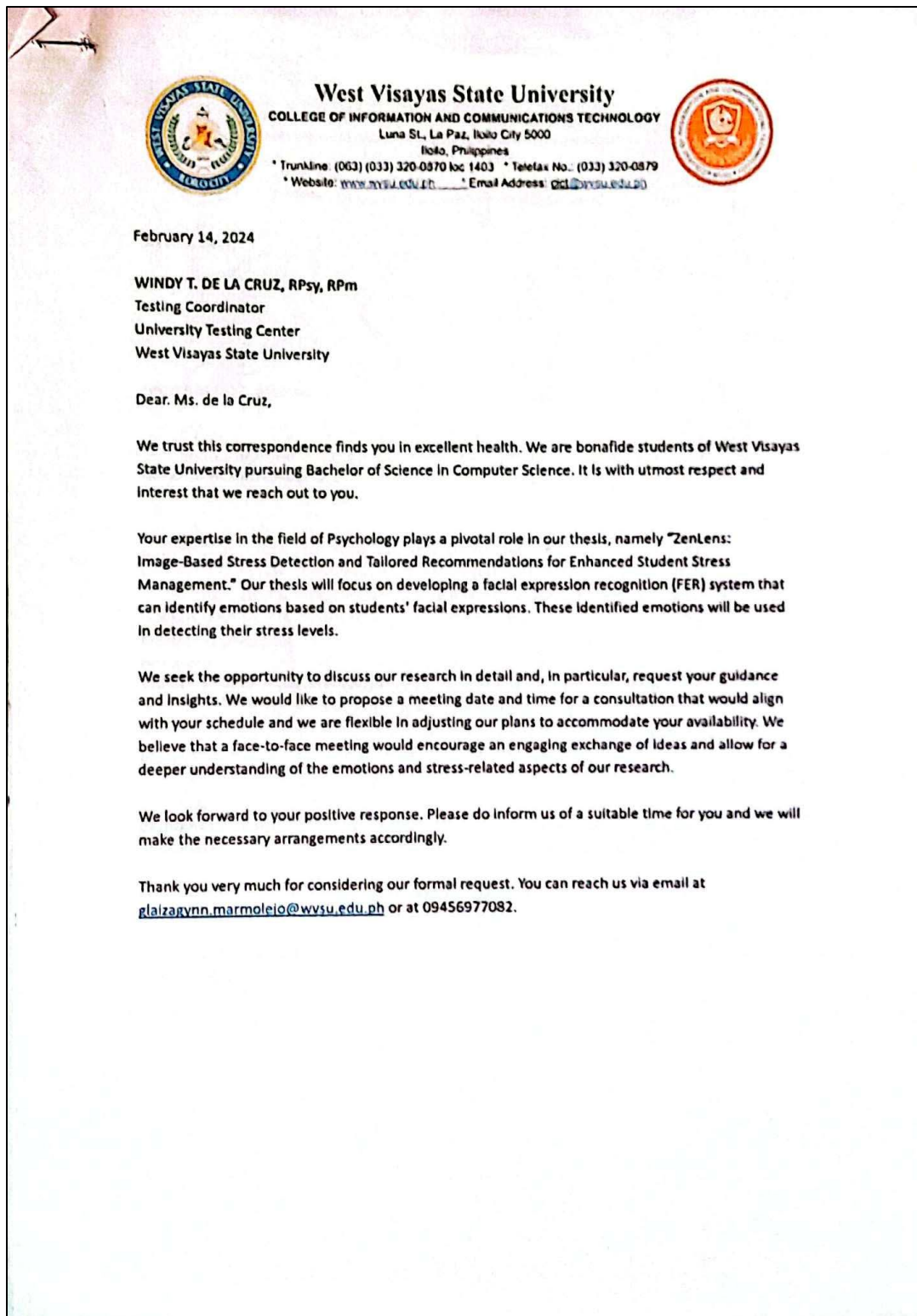

Dr. Frank I. Eljorje
Research Adviser

Approved:


Dr. Ma. Beth S. Concepcion
Dean, College of ICT

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La Paz, Iloilo City, Philippines

Appendix Q Invitation Letter for Domain Expert



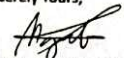
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La Paz, Iloilo City, Philippines



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Iloilo, Philippines
* Trunkline: (063) (033) 320-0870 loc 1403 * Telefax No.: (033) 320-0879
* Website: www.wvsu.edu.ph * Email Address: cict@wvsu.edu.ph



Sincerely Yours,


MATHEW E. GABRIEL


PHOEBE GRACE C. JUAYONG


GLADA GYNN S. MARMOLEJO

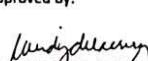

HAZEL B. PETATE


EJ PRINCE D. SEVILLENIO
Researchers
Bachelor of Science in Computer Science
West Visayas State University

Noted:


DR. FRANK L. ELJORDE
Thesis Adviser

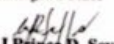
Approved by:


WINDY T. DE LA CRUZ, RPsy, RPm
Testing Coordinator


DR. MA. BETH S. CONCEPCION
Dean, CICT-WVSU

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COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
La Paz, Iloilo City, Philippines

Appendix R Invitation Letter for Participant's Adviser

 West Visayas State University (Formerly Iloilo Normal School) COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY Luna St., La Paz, Iloilo City 5000 Iloilo, Philippines *Trunkline: (033) 326-0870 to 78 loc. 1403 *Telex No.: (033) 326-0879 *Website: www.wvsu.edu.ph *Email Address: cict@wvsu.edu.ph	 
August 28, 2024	
DR. LEA M. GABAWA Assistant Professor 2 College of Information and Communications Technology West Visayas State University Luna St., La Paz, Iloilo City, 5000	
Dear Dr. Gabawa, We, a group of fourth-year Bachelor of Science in Computer Science (BSCS) students from West Visayas State University, are conducting a study titled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management." The goal of this research is to monitor the stress levels of college students using facial expression recognition. We believe that by understanding the stress levels of students, we can develop strategies to improve their overall well-being and academic performance. We are requesting permission to conduct a portion of this study in your BSIT 2-A advisory class for five consecutive days of the week where classes are held. We understand the importance of minimizing disruptions to the learning environment and assure you that our research activities will not interfere with the regular class schedule. We will provide an informed consent form for the students to understand the purpose of the study, their rights as participants, and the importance of their participation. We will also conduct a brief orientation before the start of the data collection to address any questions or concerns. Thank you for considering our request. We look forward to your positive response.	
Yours sincerely,  Mathew E. Gabriel  Phoebe Grate C. Juayong  Glaiza Gynn S. Marmolejo	 Hazel B. Petate  EJ Prince D. Sevileno
Noted:  Dr. Frank I. Eljorde Research Adviser Recommending Approval:  Dr. Ma. Beth S. Concepcion Dean, CICT Approved by: Dr. Lea M. Gabawa Assistant Professor 2	

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Iloilo, Philippines
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* Website: www.wvsu.edu.ph * Email Address: cto@wvsu.edu.ph



August 28, 2024

MR. JOHN CRISTOPHER A. MATEO
Instructor 3
College of Information and Communications Technology
West Visayas State University
Luna St., La Paz, Iloilo City, 5000

Dear Mr. Mateo,

We, a group of fourth-year Bachelor of Science in Computer Science (BSCS) students from West Visayas State University, are conducting a study titled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management." The goal of this research is to monitor the stress levels of college students using facial expression recognition. We believe that by understanding the stress levels of students, we can develop strategies to improve their overall well-being and academic performance.

We are requesting permission to conduct a portion of this study in your BSCS 4-A advisory class for five consecutive days of the week where classes are held. We understand the importance of minimizing disruptions to the learning environment and assure you that our research activities will not interfere with the regular class schedule.

We will provide an informed consent form for the students to understand the purpose of the study, their rights as participants, and the importance of their participation. We will also conduct a brief orientation before the start of the data collection to address any questions or concerns.

Thank you for considering our request. We look forward to your positive response.

Yours sincerely,


Mathew E. Gabriel


Hazel B. Petate


Phoebe Grace C. Juayong

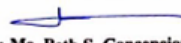

EJ Prince D. Sevileno


Glaiza Gyan S. Marmolejo

Noted:


Dr. Frank I. Elijorde
Research Adviser

Recommending Approval:


Dr. Ma. Beth S. Concepcion
Dean, CICT

Approved by:

Mr. John Cristopher A. Mateo
Instructor 3

West Visayas State University
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
La Paz, Iloilo City, Philippines

Appendix S Invitation Letter for Subject Teachers



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Iloilo, Philippines
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* Website: www.wvsu.edu.ph * Email Address: icct@wvsu.edu.ph



August 28, 2024

PROF. REYGEI B. HERMANO
Assistant Professor
Social Science Department, College of Arts and Sciences
West Visayas State University
Luna St., La Paz, Iloilo City, 5000

Dear Prof. Hermano,

We, a group of fourth-year Bachelor of Science in Computer Science (BSCS) students from West Visayas State University, are writing to formally request permission to conduct a study entitled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management" to your BSIT 2-A class in the subject, **The Contemporary World (SS 111)**. The goal of this research is to monitor the stress levels of college students using facial expression recognition. We believe that by understanding the stress levels of students, we can develop strategies to improve their overall well-being and academic performance.

We understand the importance of your class schedule and assure you that the conduct of our study will not disrupt your regular class activities. We plan to implement the study within the duration of your class session.

We believe that this study will be beneficial to both the participants and the academic community. Your cooperation in this matter is highly appreciated.

Thank you for considering our request. We look forward to your positive response.

Yours sincerely,

Mathew E. Gabriel

Phoebe Grace C. Juayong

Glaiza Gylis S. Marmolejo

Hazel B. Petate

EJ Prince D. Sevilleño

Noted:

Dr. Frank I. Elijorde
Research Adviser

Recommending Approval:

Dr. Ma. Beth S. Concepcion
Dean, CICT

Approved:

Prof. Reygei B. Hermano
Assistant Professor

West Visayas State University
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
La Paz, Iloilo City, Philippines



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*Website: www.wvsu.edu.ph *Email Address: info@wvsu.edu.ph



August 28, 2024

MS. ANDREA L. TUSIAP
Faculty
Social Science Department
College of Arts and Sciences
West Visayas State University
Luna St., La Paz, Iloilo City, 5000

Dear Ms. Tusiap,

We, a group of fourth-year Bachelor of Science in Computer Science (BSCS) students from West Visayas State University, are writing to formally request permission to conduct a study entitled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management" to your BSIT 2-A class in the subject, *Ethics (SS 112)*. The goal of this research is to monitor the stress levels of college students using facial expression recognition. We believe that by understanding the stress levels of students, we can develop strategies to improve their overall well-being and academic performance.

We understand the importance of your class schedule and assure you that the conduct of our study will not disrupt your regular class activities. We plan to implement the study within the duration of your class session.

We believe that this study will be beneficial to both the participants and the academic community. Your cooperation in this matter is highly appreciated.

Thank you for considering our request. We look forward to your positive response.

Yours sincerely,

Matthew E. Gabriel
Matthew E. Gabriel

Phoebe Grace C. Juayong
Phoebe Grace C. Juayong

Glaiza Gynn S. Marmolejo
Glaiza Gynn S. Marmolejo

Hazel B. Petate
Hazel B. Petate
EJ Prince D. Sevilla
EJ Prince D. Sevilla

Noted:

Dr. Frank I. Eljorde
Dr. Frank I. Eljorde
Research Adviser

Recommending Approval:

Dr. Ma. Beth S. Concepcion
Dr. Ma. Beth S. Concepcion
Dean, CICT

Approved by:

Ms. Andrea L. Tusiap
Faculty

West Visayas State University
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
La Paz, Iloilo City, Philippines



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*Website: www.wvsu.edu.ph *Email Address: icict@wvsu.edu.ph



August 28, 2024

MS. KEIA JOY I. HARDER
Part-time Lecturer
College of Information and Communications Technology
West Visayas State University
Luna St., La Paz, Iloilo City, 5000

Dear Ms. Harder,

We, a group of fourth-year Bachelor of Science in Computer Science (BSCS) students from West Visayas State University, are writing to formally request permission to conduct a study entitled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management" to your BSIT 2-A class in the subject, **Data Structure and Algorithms & Object-Oriented Programming**. The goal of this research is to monitor the stress levels of college students using facial expression recognition. We believe that by understanding the stress levels of students, we can develop strategies to improve their overall well-being and academic performance.

We understand the importance of your class schedule and assure you that the conduct of our study will not disrupt your regular class activities. We plan to implement the study within the duration of your class session.

We believe that this study will be beneficial to both the participants and the academic community. Your cooperation in this matter is highly appreciated.

Thank you for considering our request. We look forward to your positive response.

Yours sincerely,


Mathew E. Gabriel


Hazel B. Petate


Phoebe Grace C. Juayong


EJ Prince D. Sevileno


Glaiza Gynn S. Marmolejo

Noted:


Dr. Frank I. Eljorde
Research Adviser

Recommending Approval:


Dr. Ma. Beth S. Concepcion
Dean, CICT

Approved by:

Ms. Keia Joy I. Harder
Part-time Lecturer

West Visayas State University
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La Paz, Iloilo City, Philippines



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August 28, 2024

MR. RYAN CHRISTIAN L. PAYUNAN
Part-time Lecturer
College of Information and Communications Technology
West Visayas State University
Luna St., La Paz, Iloilo City, 5000

Dear Mr. Payunan,

We, a group of fourth-year Bachelor of Science in Computer Science (BSCS) students from West Visayas State University, are writing to formally request permission to conduct a study entitled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management" to your BSIT 2-A class in the subject, **Human Computer Interaction 1**. The goal of this research is to monitor the stress levels of college students using facial expression recognition. We believe that by understanding the stress levels of students, we can develop strategies to improve their overall well-being and academic performance.

We understand the importance of your class schedule and assure you that the conduct of our study will not disrupt your regular class activities. We plan to implement the study within the duration of your class session.

We believe that this study will be beneficial to both the participants and the academic community. Your cooperation in this matter is highly appreciated.

Thank you for considering our request. We look forward to your positive response.

Yours sincerely,

Matthew E. Gabriel

Hazel B. Petate

Phoebe Grace C. Juayong

EJ Prince D. Sevilla

Glaiza Gyni S. Marmolejo

Noted:

Dr. Frank I. Elijorde
Research Adviser

Recommending Approval:

Dr. Ma. Beth S. Concepcion
Dean, CICT

Approved:

Mr. Ryan Christian L. Payunan
Part-time Lecturer

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August 28, 2024

MR. CHRISTIAN CADIZ
Instructor I
College of Information and Communications Technology
West Visayas State University
Luna St., La Paz, Iloilo City, 5000

Dear Mr. Cadiz,

We, a group of fourth-year Bachelor of Science in Computer Science (BSCS) students from West Visayas State University, are writing to formally request permission to conduct a study entitled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management" to your BSIT 2-A class in the subject, **Platform Technologies (CIT 206)**. The goal of this research is to monitor the stress levels of college students using facial expression recognition. We believe that by understanding the stress levels of students, we can develop strategies to improve their overall well-being and academic performance.

We understand the importance of your class schedule and assure you that the conduct of our study will not disrupt your regular class activities. We plan to implement the study within the duration of your class session.

We believe that this study will be beneficial to both the participants and the academic community. Your cooperation in this matter is highly appreciated.

Thank you for considering our request. We look forward to your positive response.

Yours sincerely,

Mathew E. Gabriel

Hazel B. Petate

Phoebe Grace C. Juayong

EJ Prince D. Sevileno

Glaiza Gynn S. Marmolejo

Noted:

Dr. Frank I. Elijorde
Research Adviser

Recommending Approval:

Dr. Ma. Beth S. Concepcion
Dean, CICT

Approved:

Mr. Christian Cadiz
Instructor I

West Visayas State University
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
La Paz, Iloilo City, Philippines



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August 28, 2024

PROF. ARTURO S. SOURIBIO, Ed. D.
Associate Professor 5
College of Education
West Visayas State University
Luna St., La Paz, Iloilo City, 5000

Dear Prof. Souribio,

We, a group of fourth-year Bachelor of Science in Computer Science (BSCS) students from West Visayas State University, are writing to formally request permission to conduct a study entitled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management" to your BSIT 2-A class in the subject, **Art Appreciation (HUM 110)**. The goal of this research is to monitor the stress levels of college students using facial expression recognition. We believe that by understanding the stress levels of students, we can develop strategies to improve their overall well-being and academic performance.

We understand the importance of your class schedule and assure you that the conduct of our study will not disrupt your regular class activities. We plan to implement the study within the duration of your class session.

We believe that this study will be beneficial to both the participants and the academic community. Your cooperation in this matter is highly appreciated.

Thank you for considering our request. We look forward to your positive response.

Yours sincerely,

Mathew E. Gabriel

Hazel B. Petate

Phoebe Grace C. Juayong

EJ Prince D. Sevileno

Glaiza Gynn S. Marmolejo

Noted:

Dr. Frank I. Eljorde
Research Adviser

Recommending Approval:

Dr. Ma. Beth S. Concepcion
Dean, CICT

Approved:

Prof. Arturo S. Souribio, Ed.D.
Associate Professor 5

West Visayas State University
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La Paz, Iloilo City, Philippines



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August 28, 2024

DR. ALEXANDER J. BALSOMO
Dean, College of Arts and Sciences
Department of Mathematics, College of Arts and Sciences
West Visayas State University
Luna St., La Paz, Iloilo City, 5000

Dear Dr. Balsomo,

We, a group of fourth-year Bachelor of Science in Computer Science (BSCS) students from West Visayas State University, are writing to formally request permission to conduct a study entitled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management" to your BSCS 4-A class in the subject, **Computational Science (CCS 237)**. The goal of this research is to monitor the stress levels of college students using facial expression recognition. We believe that by understanding the stress levels of students, we can develop strategies to improve their overall well-being and academic performance.

We understand the importance of your class schedule and assure you that the conduct of our study will not disrupt your regular class activities. We plan to implement the study within the duration of your class session.

We believe that this study will be beneficial to both the participants and the academic community. Your cooperation in this matter is highly appreciated.

Thank you for considering our request. We look forward to your positive response.

Yours sincerely,


Matthew E. Gabriel


Hazel B. Petate


Phoebe Grace C. Juayong


EJ Prince D. Sevileno


Glaiza Gynn S. Marmolejo

Noted: 

Dr. Frank I. Elijorde
Research Adviser

Recommending Approval:


Dr. Ma. Beth S. Concepcion
Dean, CICT

Approved by:

Dr. Alexander J. Balsomo
Dean, CAS

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August 28, 2024

MR. JOHN ALEXIS B. GEMINO
Instructor 3
Department of Mathematics, College of Arts and Sciences
West Visayas State University
Luna St., La Paz, Iloilo City, 5000

Dear Mr. Gemino,

We, a group of fourth-year Bachelor of Science in Computer Science (BSCS) students from West Visayas State University, are writing to formally request permission to conduct a study entitled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management" to your BSCS 4-A class in the subject, **Optimization Theory and Application (CCS 239)**. The goal of this research is to monitor the stress levels of college students using facial expression recognition. We believe that by understanding the stress levels of students, we can develop strategies to improve their overall well-being and academic performance.

We understand the importance of your class schedule and assure you that the conduct of our study will not disrupt your regular class activities. We plan to implement the study within the duration of your class session.

We believe that this study will be beneficial to both the participants and the academic community. Your cooperation in this matter is highly appreciated.

Thank you for considering our request. We look forward to your positive response.

Yours sincerely,

Mathew E. Gabriel

Hazel B. Petate

Phoebe Grace C. Juayong

EJ Prince D. Sevileno

Glaiza Gynn S. Marmolejo

Noted:

Dr. Frank I. Eljorde
Research Adviser

Recommending Approval:

Dr. Ma. Beth S. Concepcion
Dean, CICT

Approved:

Mr. John Alexis B. Gemino
Instructor 3

West Visayas State University
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August 28, 2024

DR. RALPH VOLTAIRE J. DAYOT
Assistant Professor
College of Information and Communications Technology
West Visayas State University
Luna St., La Paz, Iloilo City, 5000

Dear Dr. Dayot,

We, a group of fourth-year Bachelor of Science in Computer Science (BSCS) students from West Visayas State University, are writing to formally request permission to conduct a study entitled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management" to your BSCS 4-A class in the subject, **Computer Vision (CCS 250)**. The goal of this research is to monitor the stress levels of college students using facial expression recognition. We believe that by understanding the stress levels of students, we can develop strategies to improve their overall well-being and academic performance.

We understand the importance of your class schedule and assure you that the conduct of our study will not disrupt your regular class activities. We plan to implement the study within the duration of your class session.

We believe that this study will be beneficial to both the participants and the academic community. Your cooperation in this matter is highly appreciated.

Thank you for considering our request. We look forward to your positive response.

Yours sincerely,

Mathew E. Gabriel

Hazel B. Petate

Phoebe Grace C. Juayong

EJ Prince D. Sevileno

Glaiza Gynn S. Marmolejo

Noted:

Dr. Frank I. Elijorde
Research Adviser

Recommending Approval:

Dr. Ma. Beth S. Concepcion
Dean, CICT

Approved by:

Dr. Ralph Voltaire J. Dayot
Assistant Professor

West Visayas State University
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
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August 28, 2024

DR. FRANK I. ELIJORDE
Professor 6
College of Information and Communications Technology
West Visayas State University
Luna St., La Paz, Iloilo City, 5000

Dear Dr. Elijorde,

We, a group of fourth-year Bachelor of Science in Computer Science (BSCS) students from West Visayas State University, are writing to formally request permission to conduct a study entitled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management" to your BSCS 4-A class in the subject, **Thesis Writing for CS 2 (CCS 233)**. The goal of this research is to monitor the stress levels of college students using facial expression recognition. We believe that by understanding the stress levels of students, we can develop strategies to improve their overall well-being and academic performance.

We understand the importance of your class schedule and assure you that the conduct of our study will not disrupt your regular class activities. We plan to implement the study within the duration of your class session.

We believe that this study will be beneficial to both the participants and the academic community. Your cooperation in this matter is highly appreciated.

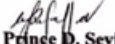
Thank you for considering our request. We look forward to your positive response.

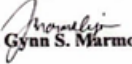
Yours sincerely,


Mathew E. Gabriel


Hazel B. Petate


Phoebe Grace C. Juayong


EJ Prince D. Sevileno


Glaiza Gynn S. Marmolejo

Noted:


Dr. Frank I. Elijorde
Research Adviser

Recommending Approval:


Dr. Ma. Beth S. Concepcion
Dean, CICT

Approved by:

Dr. Frank I. Elijorde
Professor 6

West Visayas State University
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August 28, 2024

MR. ORLANDO C. CABILLOS
Part-time Lecturer
College of Information and Communications Technology
West Visayas State University
Luna St., La Paz, Iloilo City, 5000

Dear Mr. Cabillos,

We, a group of fourth-year Bachelor of Science in Computer Science (BSCS) students from West Visayas State University, are writing to formally request permission to conduct a study entitled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management" to your BSCS 4-A class in the subject, **Networks and Communications (CCS 234), Operating Systems, (CCS 236) & Programming Languages (CCS 238)**. The goal of this research is to monitor the stress levels of college students using facial expression recognition. We believe that by understanding the stress levels of students, we can develop strategies to improve their overall well-being and academic performance.

We understand the importance of your class schedule and assure you that the conduct of our study will not disrupt your regular class activities. We plan to implement the study within the duration of your class session.

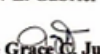
We believe that this study will be beneficial to both the participants and the academic community. Your cooperation in this matter is highly appreciated.

Thank you for considering our request. We look forward to your positive response.

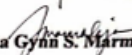
Yours sincerely,


Mathew E. Gabriel


Hazel B. Petate


Phoebe Grace C. Juayong


EJ Prince D. Sevileno


Glaiza Gynn S. Marmolejo

Noted:


Dr. Frank I. Elijorde
Research Adviser

Recommending Approval:


Dr. Ma. Beth S. Concepcion
Dean, CICT

Approved:

Mr. Orlando C. Cabillos
Part-time Lecturer

West Visayas State University
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August 28, 2024

DR. ARNEL N. SECONDES
Associate Professor 5
College of Information and Communications Technology
West Visayas State University
Luna St., La Paz, Iloilo City, 5000

Dear Dr. Secondes,

We, a group of fourth-year Bachelor of Science in Computer Science (BSCS) students from West Visayas State University, are writing to formally request permission to conduct a study entitled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management" to your BSCS 4-A in the subject, **Seminars and Field Study (CCS 235)**. The goal of this research is to monitor the stress levels of college students using facial expression recognition. We believe that by understanding the stress levels of students, we can develop strategies to improve their overall well-being and academic performance.

We understand the importance of your class schedule and assure you that the conduct of our study will not disrupt your regular class activities. We plan to implement the study within the duration of your class session.

We believe that this study will be beneficial to both the participants and the academic community. Your cooperation in this matter is highly appreciated.

Thank you for considering our request. We look forward to your positive response.

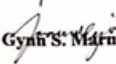
Yours sincerely,


Mathew E. Gabriel


Hazel B. Petate


Phoebe Grace C. Juayong


EJ Prince D. Sevileno


Glaiza Gynh S. Marmolejo

Noted:


Dr. Frank I. Elijorde
Research Adviser

Recommending Approval:


Dr. Ma. Beth S. Concepcion
Dean, CICT

Approved by:

Dr. Arnel N. Secondes
Associate Professor 5

West Visayas State University
COLLEGE OF INFORMATION AND COMMUNICATIONS TECHNOLOGY
Ia Paz, Iloilo City, Philippines

Appendix T Disclaimer

This research project and its corresponding documentation entitled "ZenLens: Image-Based Stress Detection and Tailored Recommendations for Enhanced Student Stress Management" is submitted to the College of Information and Communications Technology, West Visayas State University, in partial fulfillment of the requirements for the degree, Bachelor of Science in Computer Science. It is the product of our own work, except where indicated text.

We hereby grant the College of Information and Communications Technology permission to freely use, publish in local or international journal/conferences, reproduce, or distribute publicly the paper and electronic copies of this software project and its corresponding documentation in whole or in part, provided that we are acknowledged.

Mathew E. Gabriel

Phoebe Grace C. Juayong

Glaiza Gynn S. Marmolejo

Hazel B. Petate

EJ Prince D. Sevileno

June 2025